LECTURE NOTES

ON

BIOORGANIC CHEMISTRY

BACHELOR OF TECHNOLOGY

III SEMESTER

PREPARED BY

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&

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SEC : A&B

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SUBJECT IN CHARGE

SYLLABUS

SUBJECT : BIOORGANIC CHEMISTRY

SUBJECT CODE : BT2202

YEAR / SEM : II YEAR B.Tech (BIOTECH)/ III SEMESTER

MAIN AIM(S) OF THE COURSE

The course aims to develop skills of Students in the area of Organic Chemistry and its applications in Biology. This will be a prerequisite to courses like Molecular Modelling, Bioseparations etc.

MAIN TOPICS OF STUDY

UNIT I INTRODUCTION TO ENZYMES 9

Classification of enzymes. Mechanisms of enzyme action; concept of active site and energetics of enzyme substrate complex formation; specificity of enzyme action; Stereochemistry – R,S notation – re-si faces – e,z isomerism- conformers- ethane – cyclopean - reactivates- mechanisms of sn1 sn2 reactions, e1 e2 reactions – ester formation and hydrolysis, reaction rates - hammond's postulate – h/d effects. Catalysis – general acid – base and covalent catalysis.

UNIT II KINETICS OF ENZYME ACTION 9

Allosteric regulation of enzymes, Monod changeux wyman model, ph and temperature effect on enzymes & deactivation kinetics - Stereospecific enzymatic reactions – Stereochemistry of nucleophilic reactions – chiral methyl group – chiral phosphate.

UNIT III ENZYME IMMOBILIZATION & CASE STUDIES OF ENZYME STRUCTURE AND MECHANISM 9

Physical and chemical techniques for enzyme immobilization – adsorption, matrix entrapment, encapsulation, cross-linking, covalent binding etc., - examples, advantages and disadvantages. Case studies include dehydrogenases, proteases – - lysozyme-stability of proteins

UNIT IV KINETICS OF PROTEIN FOLDING 9

Kinetics of single substrate reactions; estimation of Michelis – Menten parameters, multi substrate reactions- mechanisms and kinetics; turnover number; types of inhibition & models –substrate, product - folding of peptides.

UNIT V FOLDING PATHWAYS & ENERGY LANDSCAPES 9

Folding of ci2 – nucleation condensation mechanism – folding of barnase – time resolution – insights from theory – optimization of folding rates – molecular chaperones. Production and purification of crude enzyme extracts from plant, animal and microbial sources; methods of characterization of enzymes; development of enzymatic assays .

TOTAL PERIOD: 45

TEXTS PRESCRIBED:

1. Harvey W. Blanch, Douglas S. Clark, “Biochemical Engineering”, Marcel Dekker, Inc.
2. James M. Lee, “Biochemical Engineering”, PHI, USA.

REFERENCES:

1. Structure and Mechanism In Protein Science: A Guide To Enzyme Catalysis and Protein Folding; A. R. Fersht, W.H. Freeman, 1999.
2. Bioorganic Chemistry; H. Dugas, Springer Verlag, 1999.
3. James. E. Bailey & David F. Ollis, “Biochemical Engineering Fundamentals”, McGraw-Hill.
4. Wiseman, “Enzyme Biotechnology”, Ellis Horwood Pub

UNIT I INTRODUCTION TO ENZYMES

Classification of enzymes. Mechanisms of enzyme action; concept of active site and energetics of enzyme substrate complex formation; specificity of enzyme action; Stereochemistry – R,S notation – re-si faces – e,z isomerism- conformers- ethane – cyclopean - reactivates- mechanisms of sn1 sn2 reactions, e1 e2 reactions – ester formation and hydrolysis, reaction rates - hammond's postulate – h/d effects. Catalysis – general acid – base and covalent catalysis.

Stereochemistry

Stereochemistry is defined as a branch of science that deals with molecular structure and properties dealing with structure in three dimensions.

Plane polarized light

When a polychromatic light is passed through a polarizer (Nicol prism) produces a monochromatic light which is having directions in a single direction ie in one direction is known as plane polarized light.

Optical rotation

The number of rotations made by the substance per second at a particular concentration and at one dm tube when a plane polarized light is passed through it is known as optical rotation (α).

$\alpha = \alpha / lx C$

where α = observed rotation in degrees

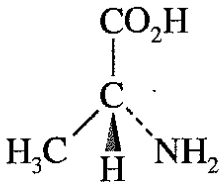
l = length of the cuvette (cm)

C= Concentration of the substance (g/ml)

Chiral molecule

A chiral molecule is defined as a molecule that has non-superimposable mirror image and which is bonded to four different atoms with absence of plane of symmetry. The asymmetric carbon atom attached to four different groups is known as a chiral atom and the molecule is referred to as a chiral molecule.

Eg. In alanine, the centre carbon atoms is attached to four different groups making the carbon centre as an asymmetric chiral molecule.



D-Alanine

Isomers

Isomers are compounds with same molecular formula but with different physical and chemical properties. The phenomenon exhibited by these compounds is referred to as isomerism. Depending on the type of this difference in arrangement of atoms, there can be two types of isomerism,

a) Structural Isomerism

b) Stereoisomerism.

Structural isomerism

Structural isomers are isomers that have the same molecular formula, but their atoms are connected in a different order eg. C_2H_6O can be either ethanol or dimethyl ether. Tautomers are structural isomers which interconvert via migration of a proton as in the case of keto and enol forms of a same compound.

Types of structural isomerism

There are six types of structural isomerism

a) Skeletal (or) chain isomerism
b) Position isomerism
c) Functional isomerism
d) Metamerism
e) Ring-chain isomerism
f) Tautomerism

Chain or Nuclear or Skeletal isomerism

Chain isomerism is due to the difference in the nature of the carbon chain which forms the nucleus of the molecule.

Eg. n-butane and Isobutane

Position isomerism

Position isomerism is due to the difference in the position of the substituted atom or group or an unsaturated linkage in the same carbon.

Eg. n-propanol and Isopropanol

Ring chain isomerism

Ring chain isomerism is due to the difference in the mode of linkage of carbon atoms and isomers may have either open chain or closed structures.

Eg. Propene and Cyclopropane

Metamerism

Metamerism is due to difference in the nature of alkyl groups attached to the same functional group. This type of isomerism is shown by compounds of same homologous series.

Eg. Methyl propyl ether and Diethyl ether.

Functional isomerism

Functional isomerism is due to the mode of attachment of functional groups. Functional isomerism is more or less similar to position isomerism

Tautomerism

Tautomerism is defined as a phenomenon in which a single compound exists in two readily interconvertible structures that differ markedly in the relative position of atleast one atomic nucleus, generally a proton atom. There are three types of tautomerism such as

a) Open chain tautomerism
b) Ring chain tautomerism
c) Valence tautomerism

a) Open chain tautomerism or ionotropy

In this type , both the tautomers are open-chain compounds sine the movement of ions is involved, it is referred to as ionotropy and is further classified as;

- Cationotropy
- Anionotropy

In cationotropy, the atoms or group of atoms shift as cation whereas in anionotropy, the phenomenon involves shift in anion.

Cationotropy

The mobile atoms is, in most cases is hydrogen atom and the phenomenon is then referred to as prototropy.

The prototropic system of tautomerism is divided into two types based on the number of atoms which are involved in the shift or change such as;

- Diad system.
- Triad system

Diad system

In the Diad system, the proton shifts from atom number one to atom number two.

Eg. Cyanide-isocyanide system

Triad system

In the Triad system, the proton shifts from atom number one to atom number three.

Eg. Keto-Enol system

Mechanism of prototropy

Prototropic inversions take place in the following two ways;

- By stepwise mechanism
- By concerted mechanism

Aniotropy

In anisotropy, involves movement of atoms or groups as anions.

b) Ring-chain tautomerism

Ring-chain tautomerism is a phenomenon in which one form is cyclic and otherform is acyclic. The best example of ring-chain tautomerism is found in carbohydrates.

c) Valence tautomerism

Valence tautomerism involves a shift in interatomic substances within the molecule due to the movement of valence electrons.

Stereoisomerism

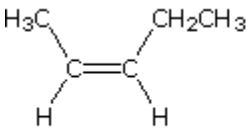
Compounds having the same atoms (or) groups with a different spatial arrangement are called stereoisomers. This type of isomerism is called stereoisomerism and is of two kinds.

- Geometrical isomerism
- Optical isomerism.

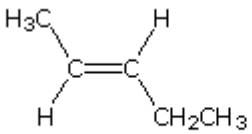
Geometrical isomerism

Geometrical isomers possess the same structural formula but differ in spatial arrangement of the groups around the double bond are known as geometrical isomers and the phenomenon is known as geometrical isomerism.

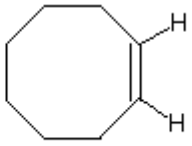
Eg. Cis-trans isomerism



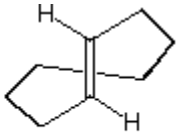
cis-2-pentene



trans-2-pentene



cis-cyclooctene



trans-cyclooctene

Optical isomerism

Optical isomers are configurational isomers that differ in the 3D relationship of the substituents about one or more atoms. The phenomenon exhibited by optical isomers are referred isomerism.

There are two types of optical isomers such as;

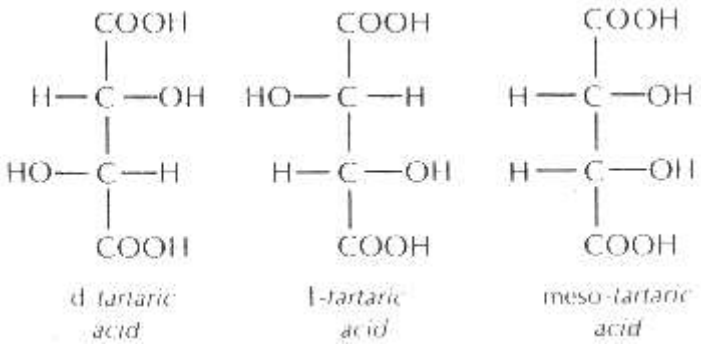
- Enantiomers
- Diastereomers

Enantiomers

Enantiomers are two molecules which are non-superimposable mirror images of each other. The phenomenon exhibited by these two non-superimposable molecules are referred to as enantiomerism.

Diastereomers

Diastereomers are compounds having two chiral centres ie has two stereocentres. Tartaric acid (2,3, dihydroxy butane dioic acid) is a compound with two stereocentres. It has three stereoisomers. Two isomers are pair of enantiomers, while the third isomer is the molecule which has a plane of symmetry. This isomer shows no optical activity and is therefore achiral.



"Diastereomers are stereoisomes which are not mirror images of each other. Diastereomers have different physical properties.

Resolution

Resolution is the term used to describe the separation of a racemic mixture into its enantiomers. There are three methods of resolution such as;

- Mechanical method
- Chemical method
- Biological method

Mechanical method

In 1848, Louis Pasteur discovered that sodium ammonium salt of racemic tartaric acid crystallized into a right handed and left handed crystals. He separated the crystals with a pair of tweezers. This was the earliest mechanical method of separation.

Chemical method

Racemic mixture of lactic acid can be separated by chemical method. The two components of the mixture are enantiomers. Enantiomers have identical physical properties and cannot be separated by the methods based on physical methods. Several naturally occurring alkaloids are used for the resolution of acids.

R,S system

Chang, Ingold and Prelog system is used along with IUPAC rules to solve the difficulty in naming the enantiomers. RS system has four sequene rules to name the enantiomers.

Step 1

- Each of the four groups attached to the stereocentre is assigned a priority.
- The directly attached atom having the highest atomic number has the highest priority. The other groups or atoms follow in order of decreasing atomic number.
- Eg. In lactic acid, oxygen has highest priority whereas hydrogen has lowest priority because the atomic number of oxygen is 16 whereas for hydrogen is one.

Step 2

- When the priority cannot be decided on the basis of the atomic number of the atoms that are directly attached to the stereocentre then it is determined by considering the next atom of the group.
- Eg. Thus from $–CH_3$ and –COOH, the COOH is of higher priority than $–CH_3$ because the next atom to the C has to be considered. The next atom is oxygen in–COOH whereas it is hydrogen in $–CH_3$, so obviously oxygen gets priority over the hydrogen therefore –COOH is considered.

Step 3

- Make a model. As a general example, put four different groups on the tetrahedral carbon atom with priorities: 1>2>3>4.

- Orient the model with the group of lowest priority away from you. Look dwon the bond from the stereocentre to the substituent at the lowest priority.
- Then tracing a path of remaining three groups on the stereocentre ie 1-2-3. If the trace back path is clockwise then the configuration is clockwise and if the trace back path is anti-clockwise then the configuration is S.

Step 4

E-Z isomerism

E-Z isomerism is a method of naming the geometrical isomers with cis-trans orientation. There are two rules to name such isomers such as;

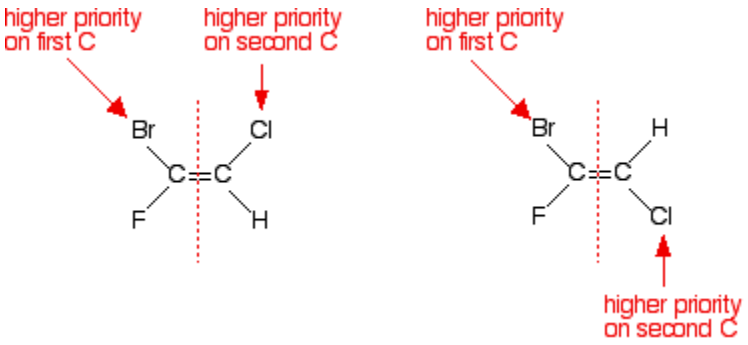
Step 1

If the two groups with the higher priorities are on the same side of the double bond, then it is described as the (Z) - isomer. The symbol Z (Zusammen) means together.

Step 2

If the two groups with the higher priorities are on the opposite side of the double bond diagonally, then it is described as (E) – isomer. The symbol E (Entgegen) means opposite

If the two



In the example above, at the left-hand end of the bond, it turns out that bromine has a higher priority than fluorine. And on the right-hand end, it turns out that chlorine has a higher priority than hydrogen. The two high priority atoms are on the same side so the compound is named as Z-1,1 bromo, fluoro,2 chloro ethene whereas the next compound is named as E-1,1 bromo, fluoro,2 chloro ethane.

Summary

- (E)- : the higher priority groups are on opposite sides of the double bond.
- (Z)- : the higher priority groups are on the same side of the double bond.

Organic reactions

Organic reactions depend on the nature of the reacting species. The molecules can be cleaved as;

- Homolytic cleavage
- Heterolytic cleavage

Hemolytic cleavage

In hemolytic cleavage, the compound splits by sharing the electrons forming free radicals.

Heterolytic cleavage

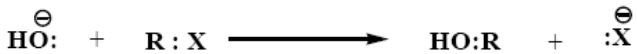
In heterolytic cleavage, there is a complete sharing of electrons resulting in the production of either electron rich species known as nucleophiles or electron deficient species known as electrophiles.

Nucleophiles and electrophiles are involved in various organic reactions such as

- Substitution reactions
- Elimination reactions
- Addition reactions
- Reduction reactions

Substitution reactions

Nucleophilic substitution reaction involves the displacement of a nucleophile by another These reactions have great synthetic importance. A classic example is the hydrolysis of alkyl halides.



There are three types of nucleophilic substitution reaction

- S_N1- Substitution nucleophilic unimolecular reaction
- S_N2- Substitution nucleophilic bimolecular reaction
- S_Ni - Substitution nucleophilic intramolecular molecular reaction

S_N1- Substitution nucleophilic unimolecular reaction

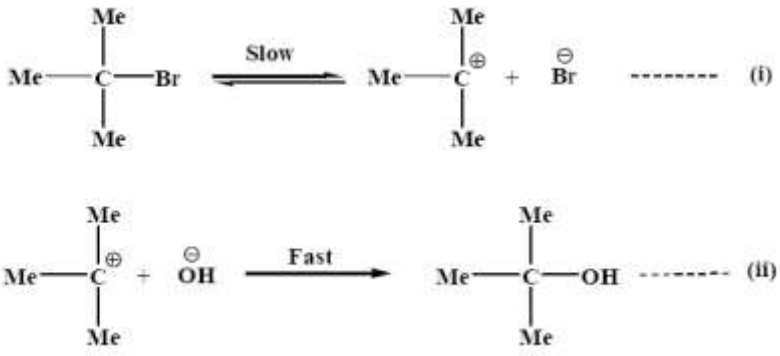
Characteristics of S_N1 reactions

- Two step unimolecular reactions
- First order kinetics
- Reactivity of RX ie hydrolysis of alkyl halides 3>2°>1° > MeX
- Rearrangement is involved
- Inversion-Walden inversion is observed
- The rate of S_N1 depends upon various factors such as polarity, solvent, nature of nucleophile, neigbouring group and ambient substrates

Mechanism

Kinetic studies of the hydrolysis of t-Butyl bromide indicate that the rate of the reaction is proportional to the concentration of the alkyl halide. (i.e.) rate $\propto[R_3CX]$. Since the rate of the reaction is dependent on one of the reactants, the reaction is a first order reaction. Nucleophilic substitution reaction which follows first order kinetics is designated S_N1(substitution nucleophilic unimolecular).

As the rate of the reaction is independent of ^-OH, it is interpreted that the halide undergoes slow ionization in the first step producing carbocation intermediate. In the second step a rapid attack of ^-OH on the carbocation completes the hydrolysis.

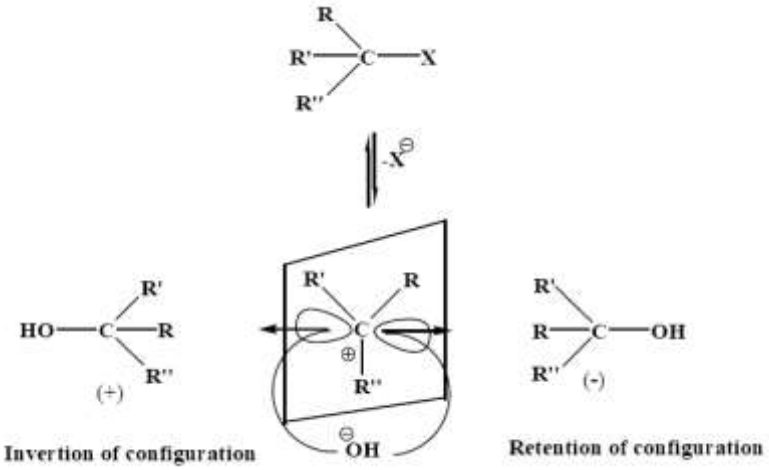


The energy required for the ionization of the halide is supplied by the energy of solvation of the ions. Since a carbocation is formed in the slowest step, the alkyl halide which can most easily form a stable carbocation will favor hydrolysis by SN1 path. Hence, the order of hydrolysis of alkyl halides by SN1 path is:

Allyl, Benzyl > 3° > 2>° 1 >° CH_3X

Stereochemistry of SN1 reaction

The carbocation is flat (Sp^2, trigonal planar) with the vacant 2P orbital vertical to the plane bearing the three groups, the attack o the reagent can occur from either side of the plane with equal probability, i.e. a racemic product should result if the alkyl halide is chiral.



Pure racemisation (50/50 mixture) is rarely observed. This is because the leaving group lies close to the carbocation shielding the side from the attack till it has sufficiently moved away.Thus, more attack of the reagent occurs from the opposite side to the leaving group. This causes more inversion than retention of configuration. Stable carbocations have longer life to permit salvation from either side of the plane of the carbocation

resulting in greater proportion of racemisation. Greater proportion of inversion is observed with more nucleophilic solvent due to faster attack from the opposite side to the leaving group.

S_N2 - Substitution nucleophilic bimolecular reaction

Characteristics of S_N2 reaction

- Single step process
- Bimolecular reaction
- Second order kinetics
- Reactivity of carbocation Me>1° > 2°> 3°
- Elemental effect RI>RBr>Rcl>RF
- Simultaneous reactions, no rearrangement is involved
- Complete inversion is observed
- The non-polar solvent favours the reaction when compared to the polar solvent

Mechanism

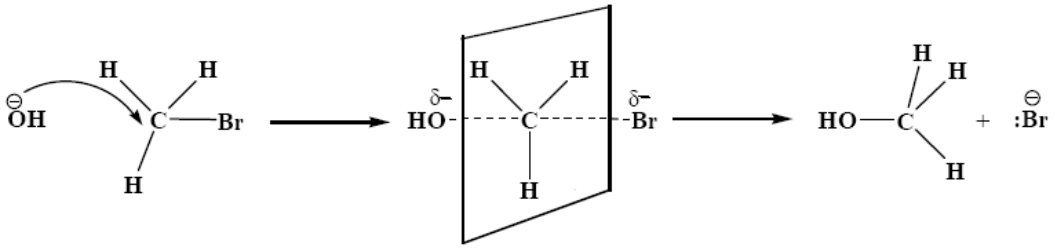
Nucleophilic substitution reactions which follow second-order kinetics are called S_N2 (substitution nucleophilic reactions which follow second-order kinetics depending upon theconcentrations of both the reactants. Thus, the rate of hydrolysis of methyl bromide with NaOH has been found to be of second order, i.e.,

$$\text{the rate} \propto [CH_3Br]\,[OH^-]$$

Since the rate determining step involves both CH_3Br and ^-OH, a collision between the two reactants resulting in the direct displacement of Br^- by ^-OH occurs in such a way that while a new C-OH bond is being formed, the C-Br bond starts breaking, i.e. the bond formation and the bond breaking are simultaneous. Hence, the reaction is a concerted one step reaction without any intermediate.

During the collision an energetic hydroxide ion approaches the methyl bromide molecule from the side opposite to bromine to avoid repulsion, i.e. at 180° to the leaving group- a backside attack. When the ^-OH is sufficiently near the electron-deficient carbon of the substrate, it begins to form a bond with it and the C-Br bond starts stretching. At one stage of the reaction, a state is reached when the OH and Br are partially bonded to

the central carbon and the nonparticipating groups lying in a plane perpendicular to the line HO….C….Br. This state is called the transition state. In the transition state partial negative charge of the hydroxide ion is transferred to bromine via the carbon atom. With further approach of hydroxide, a complete C-OH bond is formed and bromine departs with the bonding pair of electrons.

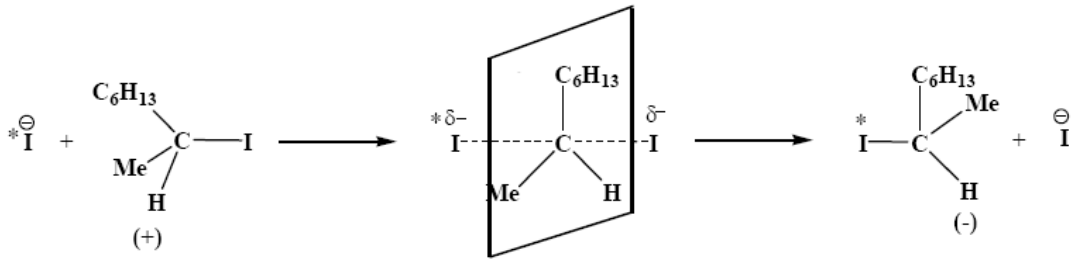


In the transition state five groups or atoms are bonded to the α-carbon. As we go along the series from methyl bromide to t-butyl bromide, the increasing crowding of the carbon bearing the bromine atom progressively decreases the nucleophilic attack. Also, the increasing +1 effect along the series makes the carbon bearing the bromine progressively less positively polarized and consequently less readily attacked by the nucleophile. The steric factor is, however, more important than the electronic factor. Hence, the E_{act} for the formation of the transition state will be highest for 3° halides and least for methyl halides. Therefore, the rate of hydrolysis of alkyl halides by S_N2 path is CH_3X >1° > 2° > 3°, reverse of S_N1 path.

Stereochemistry of S_N2 reaction

From the course of the direct displacement reaction as shown above, it is seen that the molecule is turned inside out. A Walden inversion is therefore expected to take place. An optically active halide on hydrolysis by S_N2 path, therefore, should give an alcohol with inversion of configuration. The change of configuration can be established by observing the directions of optical rotation. In this case, however the substrate (bromide) and the product (alcohol) are two different compounds. The directions of rotation and the configurations of two different compounds are not usually related. Hence, the configurations of the substrate and the product should be related to arrive at the

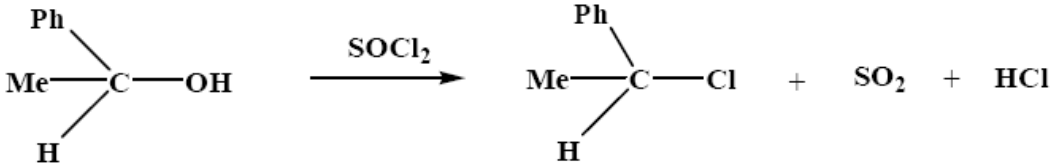
conclusion. An elegant method has been suggested (Huges and Ingold) to establish the inversion of configuration in S_N2 reaction. The method involves the conversion of (+2) iodooctane with potassium radioiodide ($K^{128}I$) in acetone to (-) 2 -iodooctane. The reaction was found to be bimolecular (S_N2), i.e. rate $\propto [C_6H_{13}CHICH_3]\ [I^*]$.The exchange of ordinary iodide with the radioactive iodide was accompanied by the loss of optical activity. This indicates the formation of (-) isomer from the (+) isomer to result in racemisation.



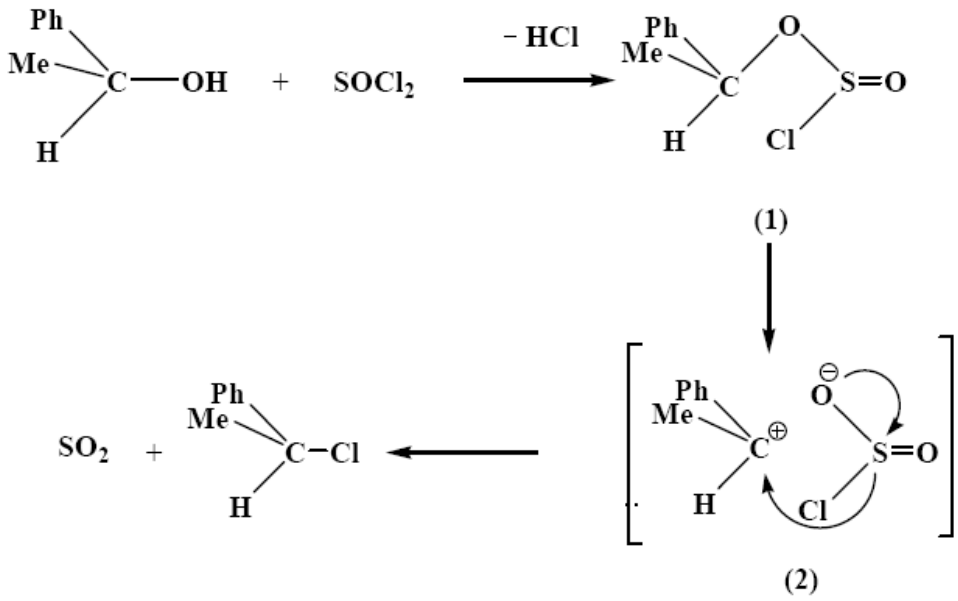
The rate of loss of optical activity (i.e. racemisation) was found to be twice the rate of iodine exchange (i.e. inversion) - one (+) molecule is inverted and another (+) molecule pairs with it to form a (±) modification. The above formulated mechanism of S_N2 reaction is therefore established. Inversion of configuration is always indicative of S_N2 reaction.

S_Ni - Substitution nucleophilic intramolecular molecular reaction

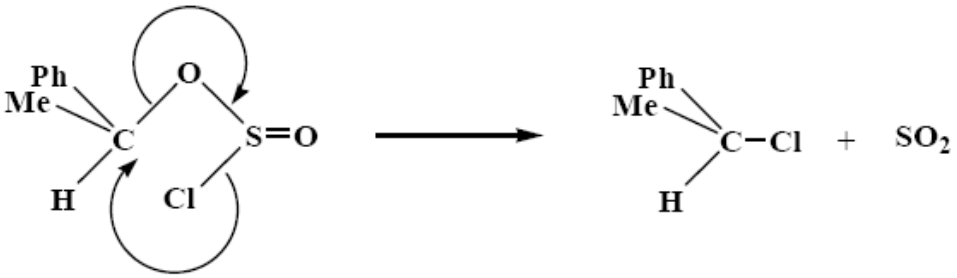
It means Substitution Nucleophilic Internal mechanism. This follows second-order kinetics and yet with no change in the configuration of the product is identified as S_Ni. Thus, the esterification of chiral alcohols with thionyl chloride results in the retention of configuration of the product.



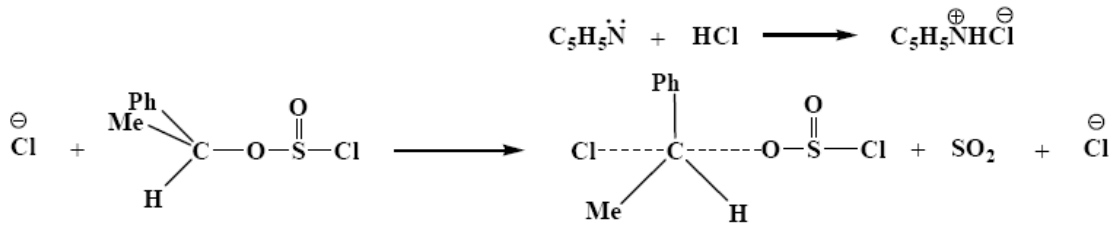
The rate of the reaction is found to be dependent on both the reactants, i.e.,rate ∞ [PhCH(Me)OH] [$SOCl_2$]. Thionyl chloride reacts with alcohol to furnish alkylchlorosulphite (1) which has the same configuration as the alcohol. There is evidence to show that the transformation of (1) to product involves an ion pair (2). The chloride ion is then supplied by the chlorosulphite anion for attack. The attack occurs from the front side since the chlorosulphite anion is held from the front side. Hence, no inversion of configuration is observed.



The overall reaction may be considered as an internal attack

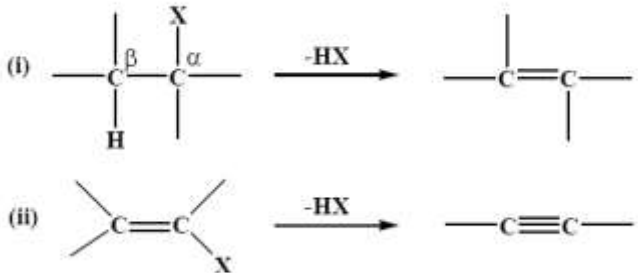


Hence the reaction is designated S_Ni as to distinguish it from S_N2.When the reaction is carried out in the presence of pyridine, the pyridine hydroxide formed in the reaction supplies the effective nucleophile, Cl-, for a back-side attack as in normal S_N2 reaction with inversion of configuration.



Elimination reactions

When two groups or atoms from adjacent carbons are eliminated with the formation of unsaturated compounds the reaction is called elimination reaction. Most commonly a nucleophile and a proton from the α, β or even from higher carbon atoms are eliminated. Hence the reaction is known as 1,2- or α,βelimination.



There are four types of elimination reactions

- α-elimination
- β-elimination
- γ-elimination
- Higher elimination

α-elimination

In α-elimination, there is a simultaneous abstraction of the proton and the halide from the same carbon atom leaving behind the lone pair of electrons forming a carbine.

β-elimination

In β-elimination, there is simultaneous abstraction of the proton from the β-carbon and the halide from the α-carbon atom resulting in the formation of an alkene

γ-elimination

In γ-elimination, the halide ion is abstracted by a metal ion leading to the formation of a cyclic compound. Here the metal ion acts as an electrophilic catalysts.

There are three types of elimination reaction such as

- E_1 = Elimination Unimolecular reactions
- E_2 = Elimination bimolecular reactions
- E_{1CB} = Elimination using conjugate base

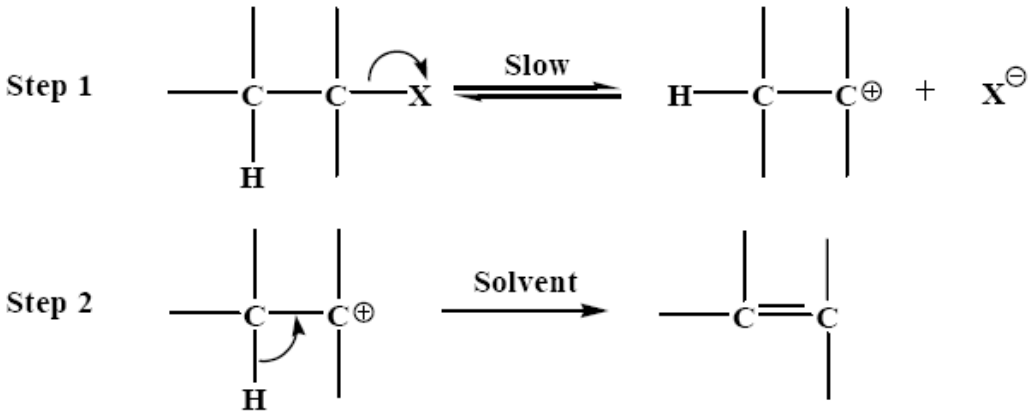
E_1-Elimination unimolecular reactions

Characteristics of E_1 reaction

- Unimolecular reaction
- Follows first order kinetics
- Rearrangement are not observed or accompanied
- Primary hydrogen isotope effects are not found
- Effect of reactivity is same as substitution unimolecular reactions

Mechanism

The E1 mechanism is a two step process in which the rate determining step is ionization of the substrate to give a carbocation that rapidly loses a β-proton to a base.

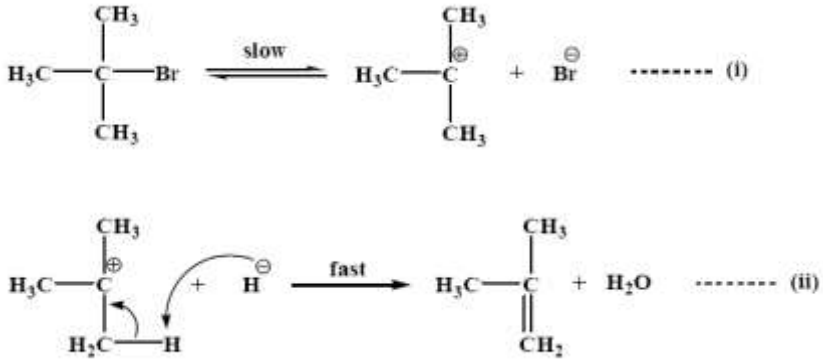


E1 mechanism reaction the product should be completely nonstereospecific, since the carbocation is free to adopt its most stable conformation.

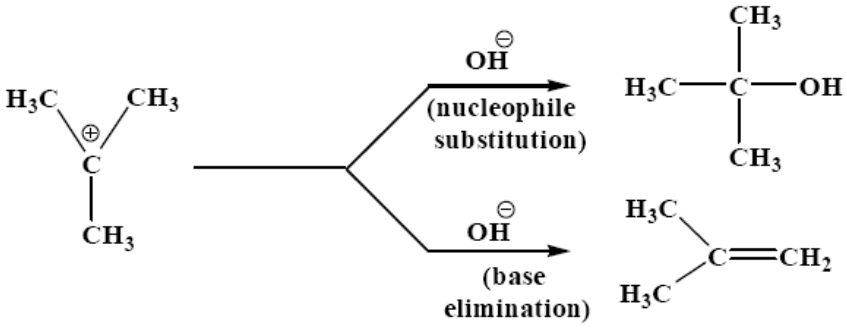
Dehydrohalogenation of alkyl halides

The rate of elimination of a halo acid from t-butyl bromide in basic medium is found to be proportional to [Me_3CBr]. Therefore, the halide undergoes slow ionization in

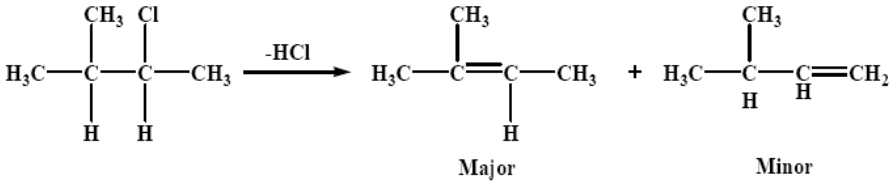
the first step. This is followed by a rapid extraction of a proton from the carbocation by the base or solvent in the second step.



The carbocation is formed in the first step in E_1 reaction. Hence the reagent can attack the carbon to give substitution product and also can accept a proton to give elimination product. In practice both alcohol and alkene are obtained on hydrolysis of Me_3CBr.



When more than one alkene can be formed, the alkene will predominate which has large number of alkyl groups on the double-bonded carbons – this is Saytzev's rule. This is understandable since the substituted alkyl groups will stabilize the alkene by hyperconjugation.



Saytzeff rule

In dehydrohalogenation, the preferred product is the alkene that has the greater number of alkyl groups attached to the double bonded carbon atom.

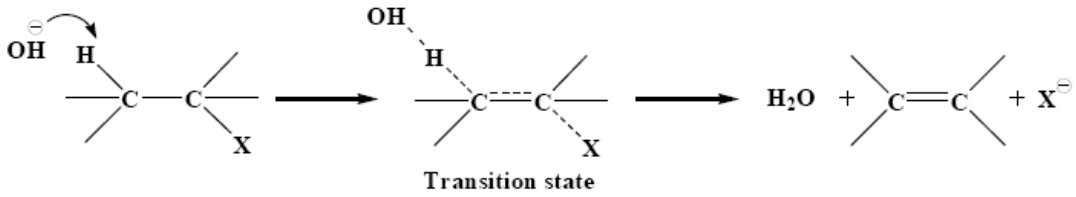
E_2-Elimination bimolecular reactions

Characteristics of E_2 reactions

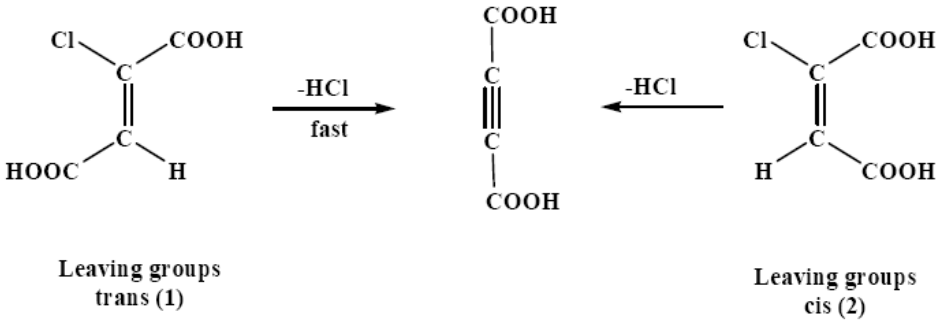
- Bimolecular reaction
- Follows second order kinetics
- Rearrangements are not observed or accompanied
- Does not show a large hydrogen isotope effect
- Show a large elemental effect
- More substituted alkene is formed ie preferred product following saytzeff's rule
- E_2 is stereospecific by nature

Mechanism

When the rate of elimination reaction is dependent on both the substrate and the reagent the reaction is kinetically second order or bimolecular.

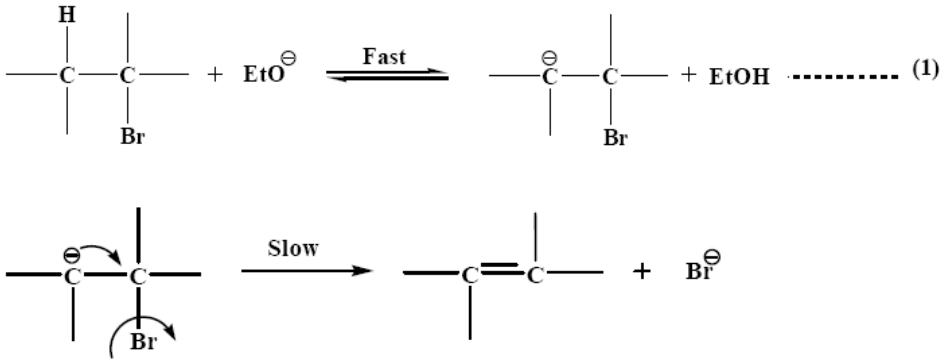


The base abstracts a proton from the β-carbon; simultaneous departure of the nucleophile takes place from the α-carbon. In the transition state the β-C-H and α–C-X bonds are stretched on the attack of the reagent with incipient π-bond formation. The energy of the transition state will be least when two leaving groups, the α and β carbons and the attacking base are coplanar in the transition state. Also, the two leaving groups (H and X) should be trans to each other to effect π bond.The two leaving groups orient themselves in the trans position when a σ bond exists between the α and β carbons. However, free rotation is not allowed as in the case of double bond, the elimination is difficult when two leaving groups are cis to each other. Thus, acetylene dicarboxylic acid is more easily formed from chlorofumaric acid (1) than from chloromaleic acid (2).



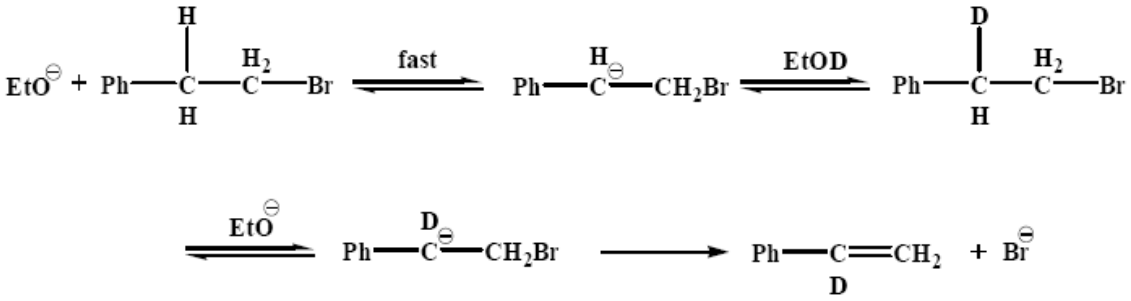
E_{1CB} –Elimination using conjugate base

The second-order elimination reaction may as well proceed in two steps as on E1 reaction. The first step involved a fast and reversible removal of a proton from the β-carbon with the formation of a carbon ion which then loses the leaving grouping in the second slow rate determining step.



The overall rate of this reaction is thus dependent on the concentration of the conjugate base of the substrate. Hence this designated as E1CB.

To distinguish between E_2 and E1CB mechanism, deuterium exchange experiment was performed. For this 2-phenyl ethyl bromide was treated with sodium ethoxide in EtOD. This substrate was selected because the Ph group is expected to increase the acidity of β– hydrogen and also to stabilize the carbanion to exist long enough for the incorporation of deuterium in the starting from the solvent EtOD.



The reaction was interrupted before completion and analysed for deuterium content. No deuterium incorporation was found either in the substrate or in the styrene. Hence no reversible carbanion was formed. The reaction followed E_2 path. However, the E1CB mechanism does operate in substrates having strong electron- withdrawing groups, e.g., chlorine on β carbon and poor leaving groups e.g., fluorine as in $Cl_2CH\text{-}CF_3$.

ELIMINATION *VS* SUBSTITUTION

Elimination reactions are usually accompanied by substitution reaction. When the reagent is a good base it accepts a proton to yield elimination product and if it is a good nucleophile then it attacks the carbon to give substitution product.

The proportion of elimination and substitution depends upon the following:

Structure of the substrate

The proportion of elimination increases from 1°→2°→3° substrates. The reason is that alkenes formed on elimination are stabilized by hyperconjucation. The steric strain is relieved on the formation of alkene, whereas on substitution the strain is reintroduced.

Nature of the base

Strong base promotes elimination over substitution and in particular E2 over E1. Alcoholic *KOH* favors elimination and aqueous *KOH* favors substitution. Strong nucleophiles but weak bases promote substitution over elimination whereas strong base but weak nucleophile promotes elimination over substitution. Though pyridine and R_3N are not strong bases they are poor nucleophile because the branching at the nitrogen atom causes steric hindrance to nucleophilic attack on carbon. Hence, they act as base to accept

the more exposed hydrogens of the substituent groups to afford alkene. A similar steric effect is observed with the size of the base or nucleophile. Elimination increases with increase in the size of the nucleophile.

Nature of the solvent

A less polar solvent not only favors bimolecular reaction but also E2 over SN2. Change of hydroxylic solvents to aprotic solvents increases the base strength as the solvents layer around the base by hydrogen bonding is absent. Thus Cl^-,OH^-,OR^- etc., are very strong bases in DMF or DMSO. The use of aprotic solvent may change the pathway from E1 to E2.

Effect of temperature

In elimination reaction a strong C-H bond has to break, hence a high activation energy is required for elimination reaction rather than for substitution reaction. In general, the proportion of elimination increases on using a strong base of high concentration and a solvent of low polarity. On the other hand the proportional substitution increases by using a weak base of low concentration and a solvent of high polarity.

Esterification and hydrolysis (Please refer Organic reaction mechanism by Khasi)

The plurality of mechanism is due to three reasons;

- Esterification and hydrolysis may proceed either through acyl-oxygen bond cleavage or an alkyl-oxygen bond cleavage
- Hydrolysis of an ester may be accompanished either by acids or alkalis and depending upon the reagent, the species undergoes/undergoing hydrolysis may be neutral ester molecule or an ionic conjugate acid.
- The esterification and hydrolysis may proceed either by a unimolecular or bimolecular mechanism

Esterification in basic medium is not possible because of the conversion of carboxylic acid to a resonance-stabilized carboxylate anion depending upon the above

three factors, there are four possible mechanisms by which esterification proceeds such as;

- Basic hydrolysis
- Acidic hydrolysis and esterification
- Acylium ion mechanism
- Carbonium ion mechanism

Basic hydrolysis ($B_{AC}2$)

The hydrolysis of esters by hydroxide ion in aqueous solution is kinetically a second order reaction, i.e., the reaction is a bimolecular reaction. The hydrolysis can be depicted as involving two mechanism which are consistent with each other

Mechanism 1

Mechanism 2

In the first mechanism, there is a TS with an energy maximum whereas in the second mechanism a true intermediate with minimum energy in the reaction co-ordinates is observed.

Acidic hydrolysis and esterification ($A_{AC}2$)

Acid catalysed hydrolysis of esters is a reversible reaction. The principle of microscopic reversibility tells the mechanism of reverse reaction.

The rate of acid-catalyzed hydrolysis of common esters is found to be proportional to both proton and the ester where the hydrolysis is carried out in the presence of hydronium ion (H_3O^+), the product alcohol has no O*. thus, it is a bimolecular reaction.

The acylium ion mechanism ($A_{AC}1$)

Hydrolysis and esterification of sterically hindered compounds such as derivatives of trialkyl acetic acids and ortho disubstituted benzoic acids, are carried out by dissolving them in concentrated sulphuric acid and then pouring this solution into cold water (hydrolysis) or into an alcohol (esterification).

Eg.Mesitoic acid provides a good example of this type whose solution in conc. Sulphuric acid indicates the presence of four particles. The available evidence indicates that there is an initial formation of an oxonium ion by the addition of a proton to the substrate which then undergoes a rate-controlling heterolytic fission to yield an acylium ion. The acylium ions are linear and their formation is favoured primarily because of relief in the steric strains of the sterically hindered compounds.

Another factor that favours their formation is the conjugation of the carbonyl group with the unsaturated system thus delocalizing the positive charge over the ion. A rapid attack by water or alcohol on this ion results in the formation of an acid or ester. The esterification of mesitoic acid proceeds as follows;

The carbonium ion mechanism

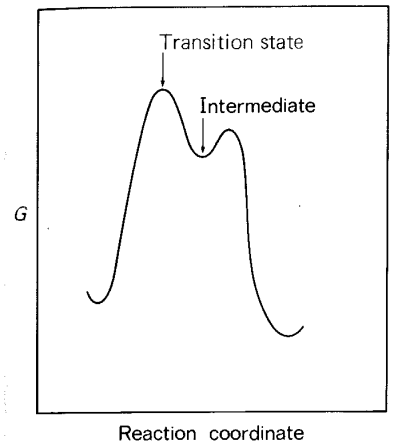
When groups such as tert-alkyl, benzyl etc capable of forming stable carbonium ions are alkyl moiety in the substrate where hydrolysis and esterification may take place by an alkyl-oxygen bond cleavage.

A mechanism of this type occurs in the solvolysis of tert-butyl benzoate which on treatment with methanol and acid gives tert-butyl ether and benzoic acid, rather than tert-butyl alcohol and methyl benzoate, the products of normal solvolysis. The following mechanism is consistent with the product analysis.

Transition state theory

Chemical kinetics can be explained by various theories. The simplest is the collision theory, which is used to calculate the rate constants for the collision of molecules in solution.

The transition theory is useful for analyzing structure-reactivity relationships, is the transition state theory. The processes by which the reagents collide are ignored whereas the physical entities considered are the reagents, or ground state, and the most unstable species on the reaction pathway, the transition state. The transition state occurs at the peak in the reaction coordinate diagram (Refer Figure 1), in which the energy of the reagents is plotted as the reaction proceeds.



In the transition state, chemical bonds are in the process of being made and broken whereas in the intermediates states, bonds are fully formed by occupying the troughs in the diagram. A simple way of deriving the rate of the reaction is to consider that the transition state and the ground state are in thermodynamic equilibrium, so that the concentration of the transition state may be calculated from the difference in their energies. The overall reaction rate is then obtained by multiplying the concentration of the transition state by the rate constant for its decomposition.

The analysis for a unimolecular reaction is as follows. Suppose that the difference in Gibbs free energy between the transition state, X*, and the ground state, X, is AG*. Then, from a well-known relationship from equilibrium thermodynamics:*

$$[X^{\ddagger}] = [X] \exp\left(-\frac{\Delta G^{\ddagger}}{RT}\right)$$

The frequency at which the transition state decomposes is the same as the vibrational frequency v of the bond that is breaking. This frequency is obtained from the equivalence of the energies of an excited oscillator calculated from quantum theory (E = hv) and classical physics (E = kT); that is,

$$\nu = \frac{kT}{h}$$

where k is the Boltzmann constant and h is the Planck constant. At 25°C, v= 6.212 X $10^{12}s^{-1}$ .The rate of decomposition of X is thus given by

The first-order rate constant for the decomposition of X is given by

$$\frac{-d[\mathrm{X}]}{dt} = \nu[\mathrm{X}^{\ddagger}]$$

$$= [\mathrm{X}]\left(\frac{kT}{h}\right)\exp\left(\frac{-\Delta G^{\ddagger}}{RT}\right)$$

The Gibbs free energy of activation, AG*, may be separated into enthalpic and entropic terms, if required, by using another relationship from equilibrium thermodynamics,

$$k_1 = \left(\frac{kT}{h}\right)\exp\left(\frac{-\Delta G^{\ddagger}}{RT}\right)$$

A more rigorous approach includes a transmission coefficient, k, which is a multiplication factor equal to the fraction of the transition state that proceeds to products.

$$k_1 = \left(\frac{kT}{h}\right)\exp\left(\frac{\Delta S^{\ddagger}}{R}\right)\exp\left(\frac{-\Delta H^{\ddagger}}{RT}\right)$$

The transmission coefficient is generally close to 1.0 for simple reactions. There can be a large attenuation factor for complex reactions because of viscous drag of the medium. Transition state theory is only qualitatively by looking at changes in rates where the transmission coefficient and other effects cancel out.

Significance of transition state theory

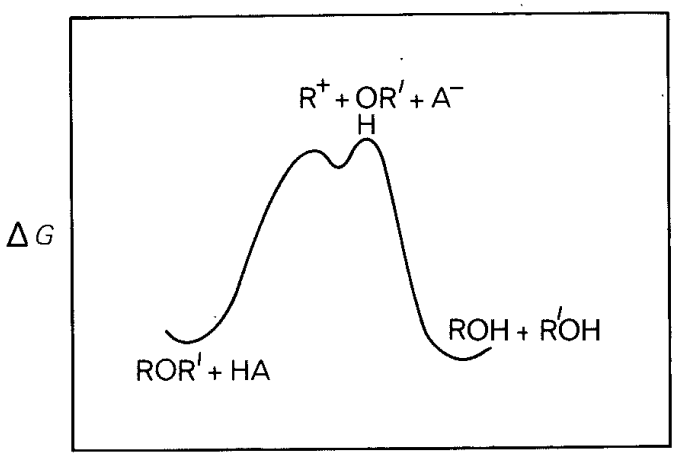
The importance of transition state theory is that

- it relates the rate of a reaction to the difference in Gibbs energy between the transition state and the ground state.

- This is especially important for comparing the relative reactivities of pairs of substrates, or the rates of a given reaction under different sets of conditions.
- Under some circumstances changes in rates may estimated be quantitatively or qualitatively. For example, the alkaline hydrolysis of an ester, such as phenyl acetate, involves the attack of the negatively charged hydroxide ion on the neutral ground state. This means that in the transition state of the reaction some negative charge must be transferred to the ester.
- It is assumed that p-nitro phenyl acetate to be more reactive than p-nitrophenol due to the presence of the electron withdrawing group and will eventually stabilize the negatively charged transition state with respect to the neutral ground state.
- TS can be used quantitatively in enzymatic reactions to analyze structure reactivity and specificity relationships involving discrete changes in the structure of the substrate.

Hammond's postulate

Hammond's postulate states that if there is an unstable intermediate on the reaction pathway, the TS for the reaction will resemble the structure of this intermediate. The unstable intermediate will be in a small dip at the top of the reaction coordinate diagram which is useful way of guessing the structure of the TS for predicting the types of stabilization it requires. Hammond's postulate is a useful guide in the application of the TS theory or in the analysis of structure-reactivity data. (Refer figure 1.2)

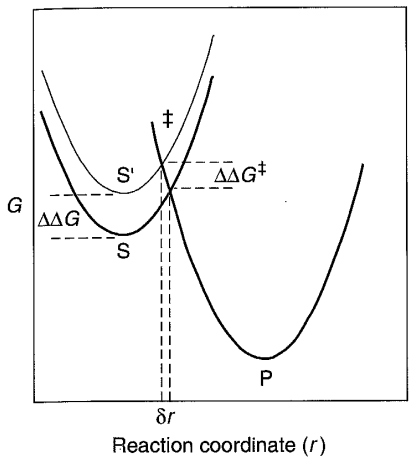


For example, an oxocarbenium ion may be an intermediate in the reaction catalyzed by lysozyme. Since oxocarbenium ion is known to be unstable high-energy compounds, the transition state is assumed to resemble the structure of the oxocarbenium ion.

At this juncture, one cannot apply the Hammond postulate to bimolecular reactions. Because these involve two molecules condensing to form one transition state, a large part of the Gibbs free energy change is caused by the loss in entropy. The Hammond postulate applies mainly to energy differences best fitting in unimolecular reactions. The postulate is sometimes extended to where there is a large energy difference between the reagents and products.

Chemical basis of the Hammond postulate

Generally substrates, intermediates and products sit at the bottom of the U-shaped energy wells. The shapes at the very bottom are parabolic for small changes in the reaction coordinate. The intersection of the two parabolic curves is the basis for Hammond's postulate. The energy curves of the substrate ad product intersect at the position of the transition state on the reaction coordinate. Suppose we make a small change in the structure of the substrate it raises its energy by $\Delta\Delta G$, subsequently raises its curve so that it intersects with the product curve at a higher energy. This causes the point of intersection t move closer to the bottom of the energy well of the substrate so that the transition state comes earlier in the reaction. This clearly indicates that as the substrate becomes more unstable, the TS approach it in structure. The movement along the reaction coordinate is known as ***Hammond effect***,



The figure also explains the basis of many linear free energy relationships. The change in equilibrium stability leads to a changes in the Gibbs free energy of activation $\Delta\Delta G^{\#}$. This value of $\Delta\Delta G^{\#}$ depends on where the curves intersect and on their shape. The ratio of $\Delta\Delta G^{\#}/ \Delta\Delta G$ is called Bronsted value.

Bronsted value is approximately constant for small changes in $\Delta\Delta G$. There is a linear relationship between changes in free energy of activation and equilibrium free energies. The value is between 0 and 1.0 for the simple reaction. Bronsted value is used as an indication of the extent of bond making and breaking in the transition state.

Principles of catalysis

The phenomenon by which a substance catalyses the rate of a chemical reaction without changing chemically the reactants is known as catalysis. There are two types of catalysis such as

- Positive catalysis
- Negative catalysis

Positive catalysis

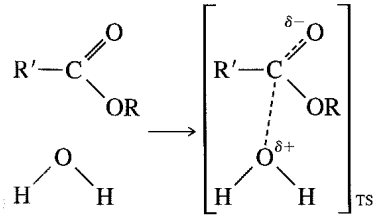
Catalyst which accelerates the rate of the chemical reaction is termed as positive catalyst and the phenomenon is known as positive catalysis.

Negative catalysis

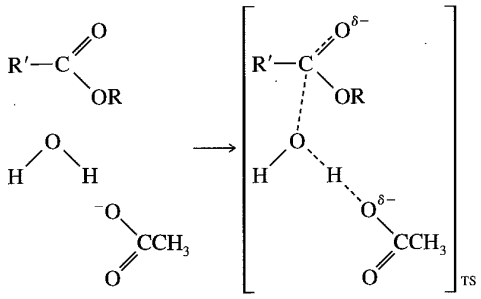
Catalyst which decelerates the rate of the chemical reaction is termed as negative catalyst and the phenomenon is known as negative catalysis.

1. General base catalysis

The uncatalyzed attack of water on an ester leads to a transition state in which a positive charge develops on the attacking water molecule, and a negative charge on the carbonyl oxygen.



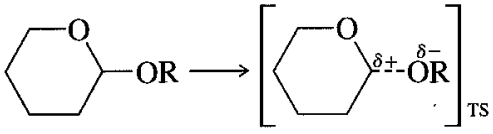
The transition state is very unfavorable because of the unstable positive and negative charges that are developed. Stabilization of these charges catalyzes the reaction by lowering the energy of the transition state. Such stabilization can be achieved in the case of the positive charge developing on the attacking water molecule by transferring one of the protons to a base during the reaction. This is known as general-base catalysis



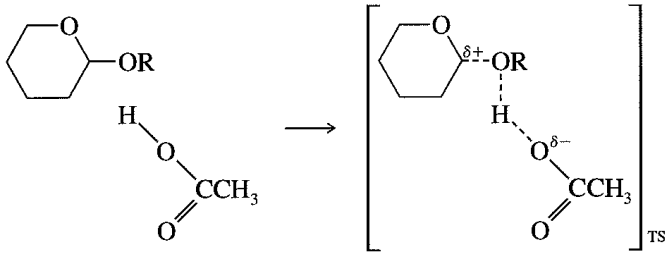
General-base catalysis by acetate ion

General acid catalysis

The uncatalyzed hydrolysis of an acetal involves a transition state that is close in structure to an oxocarbenium ion and an alkoxide ion.



The negative charge developing on the alcohol expelled from the acetal can be stabilized by proton transfer from an acid. This is known as general-acid catalysis.

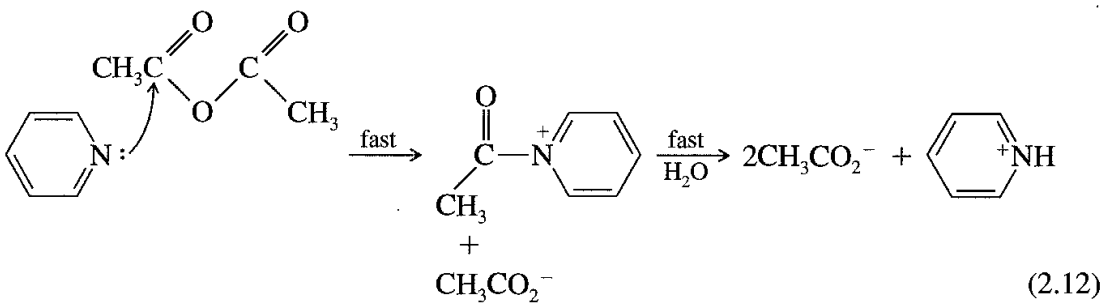


General-acid catalysis by acetic acid

Positive and negative charges may also be stabilized by electrostatic catalysis. The positively charged oxocarbenium ion cannot be stabilized by general-base catalysis because it does not ionize. But it can be stabilized by the electric field from a negatively charged carboxylate ion. The negative charge on an oxyanion may also be stabilized by a positively charged metal ion, such as Zn^{2+} or Mg^{2+}. The stabilization of a negative charge, i.e., an electron, is known as electrophilic catalysis.

Nucleophilic catalysis

Catalysts may also involve a different reaction pathway. A typical example is nucleophilic catalysis in an acyl transfer or hydrolytic reaction. The hydrolysis of acetic anhydride is greatly enhanced by pyridine because of the rapid formation of the highly reactive acetylpyridinium ion. For nucleophilic catalysis to be efficient, the nucleophile must be more nucleophilic than the one it replaces, and the intermediate must be more reactive than the parent compound.



(2.12)

Nucleophilic catalysis is a specific example of covalent catalysis; the substrate is transiently modified by formation of a covalent bond with the catalyst to give a reactive intermediate.

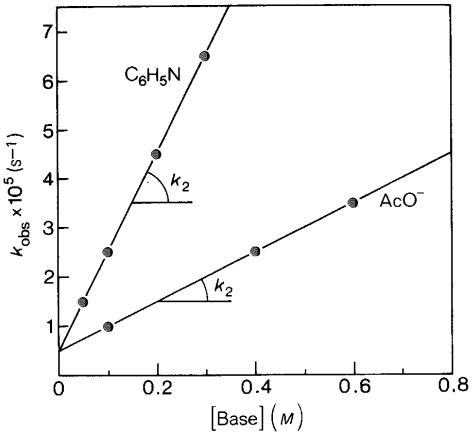
2. General acid-base catalysis

- Detection and measurement
- The efficiency of acid-base cataysis
- The efficiency of acid-base catalysis-The ionization state of the catalyst.

Detection and measurement

The general-acid-base catalysis of the hydrolysis of an ester is measured from the increase in the hydrolytic rate constant with increasing concentration of the acid or base. This is usually done at constant pH by maintaining a constant ratio of the acidic and basic forms of the catalyst by keeping the ionic strength of the reaction medium constant because many reactions are sensitive to changes in salt concentration.

The experiments are repeated at different buffer ratio whether the reaction is catalyzed by an acidic or basic form of catalyst. For ester hydrolysis, it is found that the increase in rate is generally proportional to the concentration of the basic form, so the catalysis is general-base. The slope of the plot of the rate constant against the concentration of base gives the second-order rate constant k2 for the general-base catalysis.

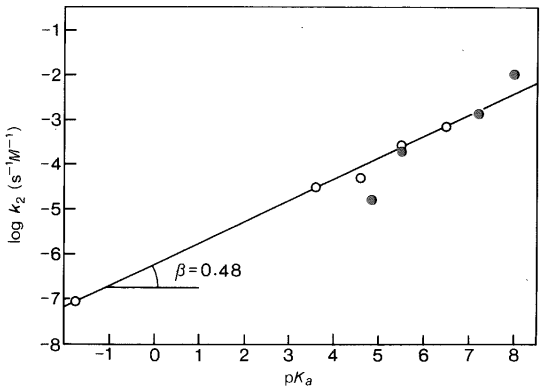


The efficiency of acid-base catalysis- The Bronsted equation

It is found experimentally that the general-base catalysis of the hydrolysis of an ester is proportional to the basic strength of the catalyst. The second-order rate constant k_2 for the dependence of the rate of hydrolysis on the concentration of the base is given by the equation

$$\log k_2 = A + \beta \, pK_a$$

The above equation is an example of the Br0nsted equation. The Bronsted β value measures the sensitivity of the reaction to the pKa of the conjugate acid of the base. The A is a constant for the particular reaction.



Bronsted equations are also common in general-acid catalysis, as, for example, in the hydrolysis of certain acetals. In acid catalysis, α rather than β is used:

$$\log k_2 = A - \alpha \, pK_a$$

The values of α and β are always between 0 and 1 for acid-base catalysis, because complete transfer of a proton gives a value of 1, and no transfer a value of 0. The usual values for ester hydrolysis are 0.3 to 0.5, and for acetal hydrolysis about 0.6.

The magnitude of the catalysis depends strongly on α and β and the pKa of the catalyst. In simple terms: The stronger the base, the better the general-base catalysis; the stronger the acid, the better the general-acid catalysis.

The efficiency of acid-base catalysis; The ionization state of the catalyst

A crucial factor in whether an acid-base catalyst is effective is whether it is in the correct ionization state under the reaction conditions: an acid has to be in acidic form to be an acid catalyst, and a base in its basic form.

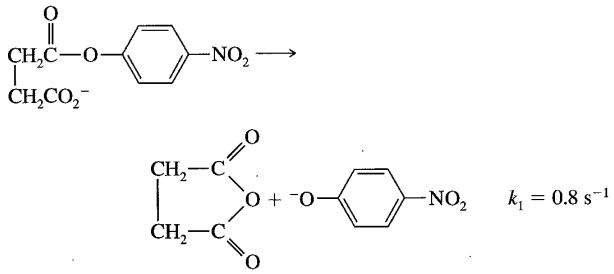
3. Intramolecular catalysis

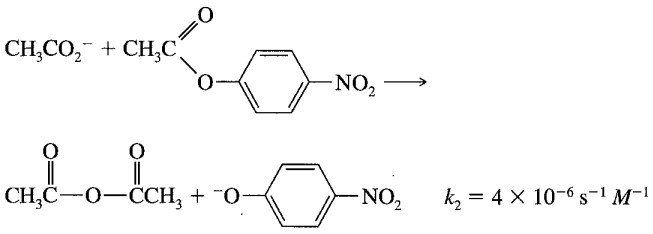
Acid-base catalysis being an integral part of the molecule is seems to be an effective way of catalyzing reactions. The crux of the matter is that the rate constants for the solution catalysis are second-order, the rate increasing with increasing concentration of catalyst; whereas reactions in an enzyme-substrate complex are first-order.

A typical example of an intramolecularly catalyzed reaction is the hydrolysis of aspirin. The hydrolysis of the ester bond is achieved by intramolecular general-base catalysis.

The effective concentrations of nucleophiles in intramolecular reactions are often for higher than intermolecular reactions. In the succinate and aspirin derivatives, the attacking nucleophile can rotate away from the ester bond to relieve any strain. The observed rate enhancements are due entirely to the high effective concentration of the neighboring group;

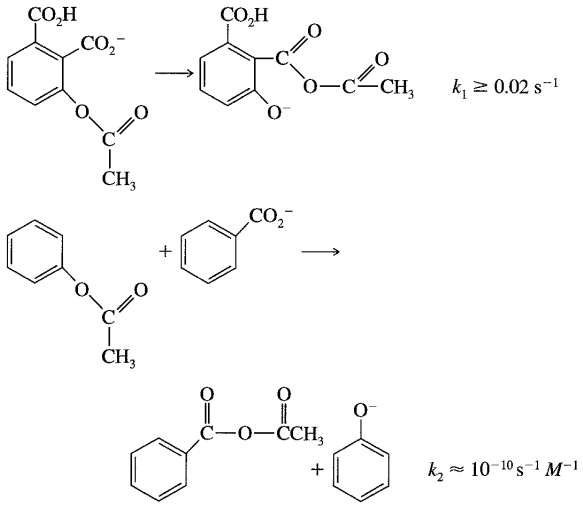
1. Rates of acyl transfer in succinates





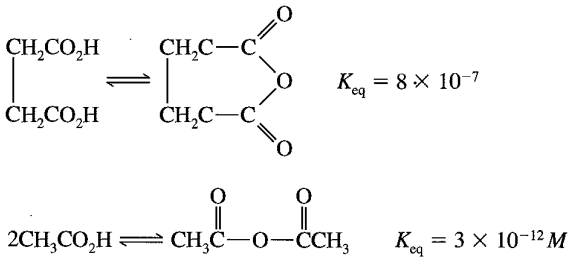
The effective concentration of $—CO_2^-$ is $k_1/k_2 = 2 \times 10^5\ M$.

2. Rates of acyl transfer in aspirin derivatives;



The effective concentration of $-CO_2^-$ is $k_1/k_2 > 2 \times 10^7\ M$.

3. Equilibria for acyl transfer in succinates



The effective concentration of $-CO_2H$ is $3 \times 10^5\ M$.

The above examples show that the rate enhancements come from intramolecuar nucleophilic catalysis.

4. Entropy: The theoretical basis of intramolecular catalysis and effective concentration

The high effective concentration of intramolecular groups is one of the most important reasons for the efficiency of enzyme catalysis which is explained theoretically by using transition state theory and examining the entropy term in the rate equation

$$k_1 = \left(\frac{kT}{h}\right) \exp\left(\frac{\Delta S^{\ddagger}}{R}\right) \exp\left(\frac{-\Delta H^{\ddagger}}{RT}\right)$$

The effective concentrations may be calculated by substituting certain entropy contributions into the exp ($AS^{\#}/R$) term of the above equation.

Before understanding the theoretical basis of entropy, one has to investigate the entropy concept in the form of magnitude, influence of entropy in unimolecular and bimolecular reactions, its dependence of effective concentration and orbital steering.

A. The meaning of entropy

The most naive explanation of entropy is that entropy is a measure of the degree of randomness or disorder of a system. The more disordered the system, the more it is favored and the higher its entropy. Entropy is similarly associated with the spatial freedom of atoms and molecules.

The catalytic advantage of an intramolecular reaction over its intermolecular counterpart is due to entropy. The intermolecular reaction involves two or more molecules associating to form one. This leads to an increase in "order" and a consequent loss of entropy. An effective concentration may be calculated from the entropy loss.

B. The magnitude of entropy

The entropy of a molecule is composed of the sum of its translational, rotational, and internal entropies. The translational and rotational entropies may be precisely calculated for the molecule in the gas phase from its mass and geometry. The entropy of the vibrations may be calculated from their frequencies, and the entropy of the internal rotations from the energy barriers to rotation.

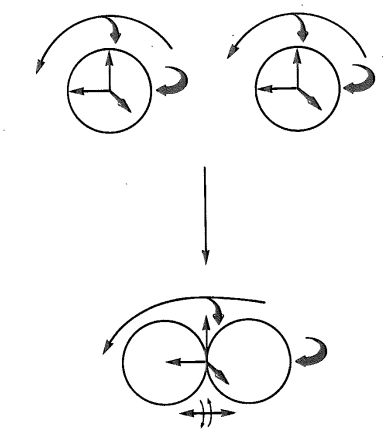
The translational entropy is proportional to the volume occupied by the molecule; the smaller the volume, the more the molecule is restricted and the lower the entropy. Similarly, the entropy decreases with increasing concentration, since the average volume occupied by a molecule is inversely proportional to its concentration. The dependence of mass is low.

The rotational entropy is also high and it increases only slowly with increasing mass, but it is independent of concentration. Low-frequency vibrations, where the atoms are less constrained, contribute a few entropy units.

C. The loss of entropy when two molecules condense to form one

The combining of two molecules to form one leads to the loss of one set of rotational and translational entropies. The rotational and translational entropies of the

adduct of the two molecules are only slightly larger than those of one of the original molecules, since these entropies increase only slightly with size.



D. The entropic advantage of a unimolecular over a bimolecular reaction

Consider the reaction of two molecules A and B combining together to form a third, with its intramolecular counterpart:

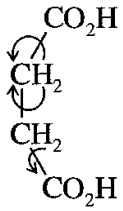
$$A + B \longrightarrow AB^{\ddagger} \longrightarrow AB$$

$$\overset{\frown}{A\ B} \longrightarrow \overset{\frown}{AB}^{\ddagger} \longrightarrow \overset{\frown}{AB}$$

The formation of the transition state AB* leads to the loss of translational and rotational entropy, although there are some compensating gains in internal rotation and vibration. The intramolecular cyclization in the below equation involves the loss of only some entropy of internal rotation. Depending on the relative gains and losses in internal rotation, the intramolecular reaction is favored entropically by up to 190 J/deg/mol (45 cal/deg/mol) or 55 to 59 kJ/mol (13 to 14 kcal/mol) at 25°C. Substituting 190 J/deg/mol (45 cal/deg/mol) into the exp (AS*/R) term of entropy equation gives a factor of 6 X 109.

Taking into account the difference in molecularity between the second-order and first-order reactions, this may be considered as the maximum effective concentration of a neighboring group, i.e., 6 X 109 M. In other words, for B in equation 2.22 to react with the same first-order rate constant as A_B in the above two equation, the concentration of A would have to be 6 X 109M.

The loss of internal rotation lowers the effective concentration quite considerably. Three internal rotations are lost for succinate:



E. Dependence of effective concentration on the tightness of the transition state.

The lower effective concentrations found in intramolecular base catalysis are due to the loose transition states of these reactions. In nucleophilic reactions, the nucleophile and the electrophile are fairly rigidly aligned so that there is a large entropy loss. In general-base or -acid catalysis, there is considerable spatial freedom in the transition state. The position of the catalyst is not as closely defined as in nucleophilic catalysis. There is consequently a smaller loss in entropy in general-base catalysis, so that the intramolecular reactions are not favored as much as their nucleophilic counterparts.

In summary, one of the most important factors in enzyme catalysis is entropy. Catalyzed reactions in solution are slow because the bringing together of the catalysts and the substrate involves a considerable loss of entropy. Enzymatic reactions take place in the confines of the enzyme-substrate complex. The catalytic groups are part of the same molecule as the substrate, so there is no loss of translational or rotational entropy in the transition state.

The catalytic groups have very high effective concentrations compared with bimolecular reactions in solution. This advantage in entropy is "paid for" by the enzyme-substrate binding energy; the rotational and translational entropies of the substrate are lost on formation of the enzyme-substrate complex, and not during the chemical steps.

F. Orbital steering

The lower effective concentrations found in intramolecular base catalysis are due to the loose transition states of these reactions. In nucleophilic reactions, the nucleophile and the electrophile are fairly rigidly aligned so that there is a large entropy loss. In general-base or -acid catalysis, there is considerable spatial freedom in the transition state. The position of the catalyst is not as closely defined as in nucleophilic catalysis.

There is consequently a smaller loss in entropy in general-base catalysis, so that the intramolecular reactions are not favored as much as their nucleophilic counterparts. An early treatment of the problem of calculating effective concentrations was to consider the concentration of an intramolecular group to be approximately the same as that of water in aqueous solution, since a molecule in solution is completely surrounded by water.22 This gives an upper limit of 55 M for effective concentration, equivalent to 34 J/deg/mol (8 cal/deg/mol) of entropy. That figure does represent the probability of two molecules being next to each other in solution. But, as soon as the two molecules are tightly linked, there is a large loss of entropy. A loose transition state may, perhaps, be interpreted as two molecules that are in close juxtaposition but that retain considerable entropic freedom. In summary, one of the most important factors in enzyme catalysis is entropy.

Catalyzed reactions in solution are slow because the bringing together of the catalysts and the substrate involve a considerable loss of entropy. Enzymatic reactions take place in the confines of the enzyme-substrate complex. The catalytic groups are part of the same molecule as the substrate, so there is no loss of translational or rotational entropy in the transition state. One way of looking at this is that the catalytic groups have very high effective concentrations compared with bimolecular reactions in solution. This advantage in entropy is "paid for" by the enzyme-substrate binding energy; the rotational and translational entropies of the substrate are lost on formation of the enzyme-substrate complex, and not during the chemical steps.

G. Electrostatic catalysis

The electrostatic interaction energy between two point charges e_1 and e_2 separated by a distance r in a medium of dielectric constant D is given by

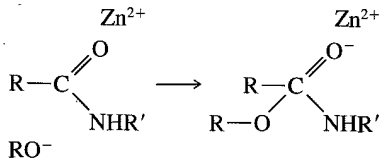
$$E = \frac{e_1 e_2}{Dr}$$

F. Metal ion catalysis

1. Electrophilic catalysis

Metals in the metalloenzymes functions as a electrophilic catalysts, stabilizing the negative charges that are formed in the reaction. In carboxy peptidase, the carbonyl oxygen of the amide substrate is coordinated to the Zn^{2+} of the enzyme. The coordination

polarizes the amide to nucleophilic attack and strongly stabilizes the tetrahedral intermediate.



This type of complex formation has been mimicked in model compounds to give rate enhancements of 10^4 to 10^6.

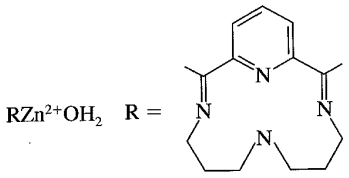
B. Sources of hydroxyl ions at neutral pH

Metal-bound hydroxyl ions are potent nucleophiles. The cobalt-bound water molecule in the below equation ionizes with a pKa of 6.6,

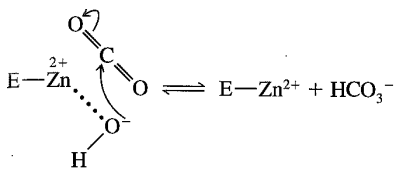
$$(NH_3)_5Co^{3+}\ OH_2 \rightleftharpoons (NH_3)_5Co^{2+}\ OH + H^+$$

a value 9 units below the pKa of free H_2O; yet the cobalt-bound hydroxyl group is only 40 times less reactive than the free hydroxide ion in catalyzing the hydration of carbon dioxide.This insensitivity of the high reactivity to pKa is quite general and independent of the metal involved. Thus, metal-bound water molecules provide a source of nucleophilic hydroxyl groups at neutral pH.

The mechanism of carbonic anhydrase takes place at neutral pH acting as a source of hydroxyl ions, which catalyzes the hydration of CO_2, has at its active site a Zn^{2+} ion ligated to the imidazole rings of three of its histidines. The classic mechanism for the reaction is that the fourth ligand is a water molecule which ionizes with a pKa of 7.37 The reactive species is considered to be the zinc-bound hydroxyl. Chemical studies show that zinc-bound hydroxyls are no exception to the rule of high reactivity. The H_2O in ionizes with a pKa of 8.7 and catalyzes the hydration of carbon dioxide and acetaldehyde.



The carbonic anhydrase mechanism probably involves the step in the below equation.



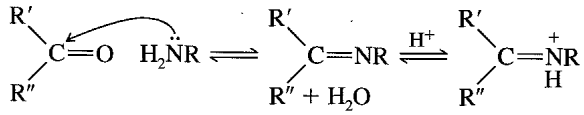
Covalent catalysis

Covalent catalysis can be studied using the following examples

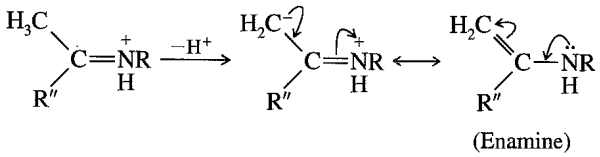
- Electrophilic catalysis by Schiff base formation
- Pyridoxal phosphate-electrophilic catalysis
- Thiamine pyrophosphate –electrophilic catalysis
- Nucleophilic catalysis

Electrophilic catalysis by Schiff base formation

A good example of how transient chemical modification can activate a substrate is Schiff base formation from the condensation of an amine with a carbonyl compound. The Schiff base may be protonated at neutral pH.



This acts as an electron sink to stabilize the formation of a negative charge on one of the carbons. After tautomerization to form the enamine, the methylene carbon is activated as a nucleophile.

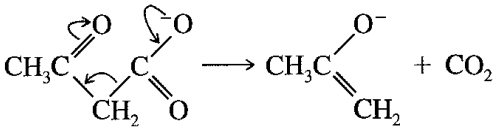


(Enamine)

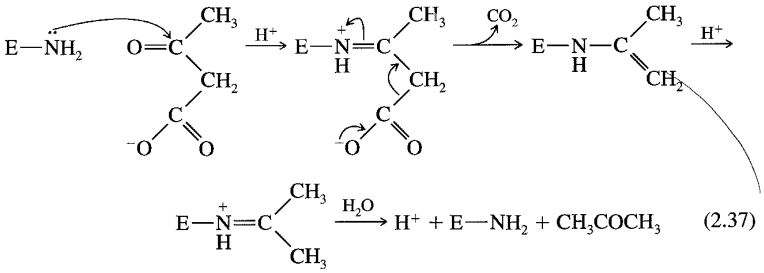
Another consequence of Schiff base formation is that the carbonyl carbon is activated toward nucleophilic attack because of the strong electron withdrawal of the protonated nitrogen.

a. Acetoacetate decarboxylase

This enzyme catalyzes the decarboxylation of acetoacetate to acetone and carbon dioxide. The nonenzymatic reaction involves the expulsion of a highly basic enolate ion at neutral pH (equation 2.36), but the enzymatic reaction circumvents this by the prior formation of a Schiff base with a lysine residue. The protonated imine is then readily expelled.

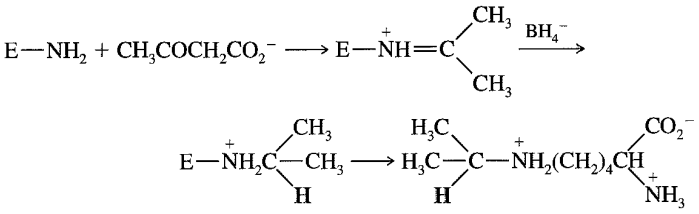


This process may be mimicked in solution by using aniline as a catalyst

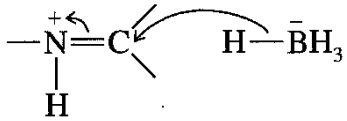


(2.37)

The evidence for the intermediate is that the enzyme is irreversibly inhibited when sodium borohydride is added to the complex with the substrate. Borohydride is known to reduce Schiff bases, and the hydrolysate of the inhibited protein is found to contain isopropyl-lysine.



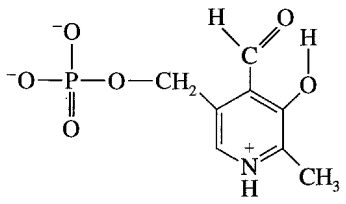
The carbon in the Schiff base is activated to the attack of an H^- ion from the borohydride:

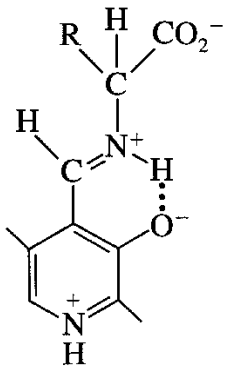


The aldol condensation and the reverse cleavage reaction catalyzed by these enzymes both involve a Schiff base. The cleavage reaction is similar to the acetoacetate decarboxylase mechanism, with the protonated imine being expelled. The condensation reaction illustrates the other function of a Schiff base, the activation of carbon via an enamine.

2. Pyridoxal phosphate—Electrophilic catalysis

The principles of the above reactions form the basis of a series of important metabolic interconversions involving the coenzyme pyridoxal phosphate. This condenses with amino acids to form a Schiff base. The pyridine ring in the Schiff base acts as an "electron sink" which very effectively stabilizes a negative charge.

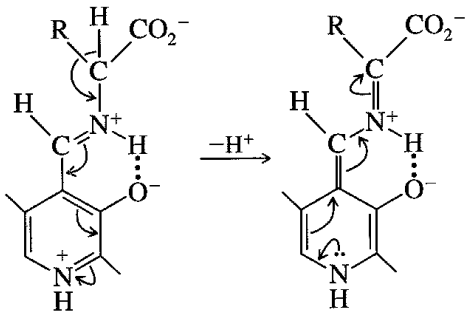




Each one of the groups around the chiral carbon of the amino acid may be cleaved, forming an anion that is stabilized by the Schiff base with the pyridine ring.

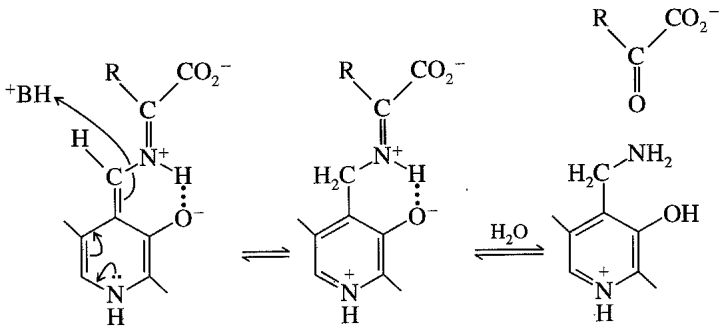
a. Removal of the a hydrogen

The removal of the a hydrogen gives a key intermediate that may react in several different ways:



1. Racemization. Addition of the proton back to the amino acid will lead to racemization unless it is done stereospecifically.

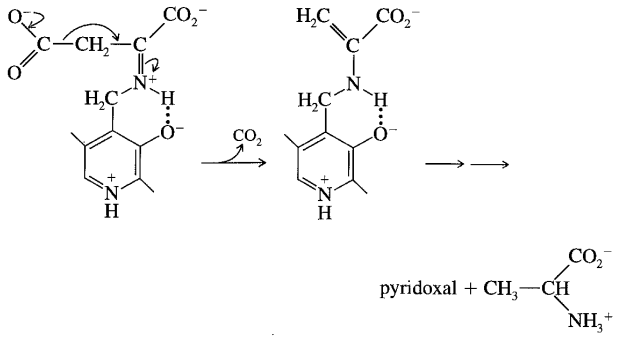
2. Transamination. Addition of a proton to the carbonyl carbon of the pyridoxal leads to a compound that is the Schiff base of an α-keto acid and pyridoxamine. Hydrolysis of the Schiff base gives the a-keto acid and pyridoxamine, which may react with a different a-keto acid to reverse the sequence:



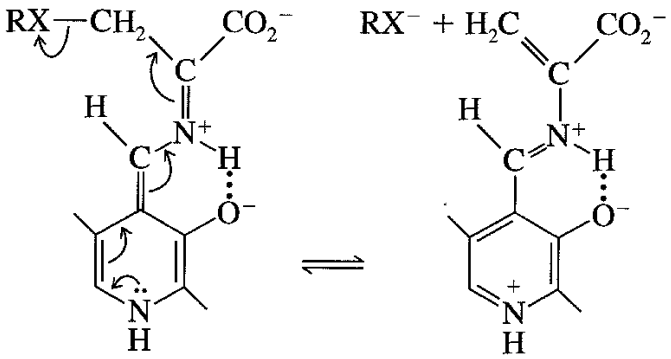
The overall reaction is as follows ;

$$R'CH(NH_3^+)CO_2^- + R''COCO_2^- \rightleftharpoons R'COCO_2^- + R''CH(NH_3^+)CO_2^-$$

3. β-Decarboxylation. When the amino acid is aspartate, the second compound in the below reaction is analogous to the Schiff base in the acetoacetate decarboxylase reaction and may readily decarboxylate:



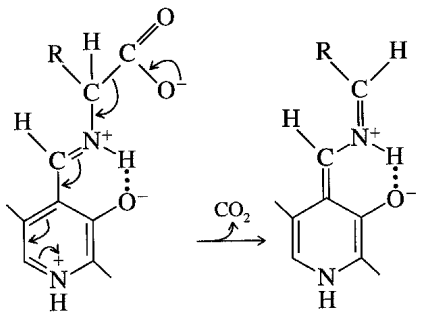
4. Interconversion of side chains. When RX— is a good leaving group, it may be expelled as follows,



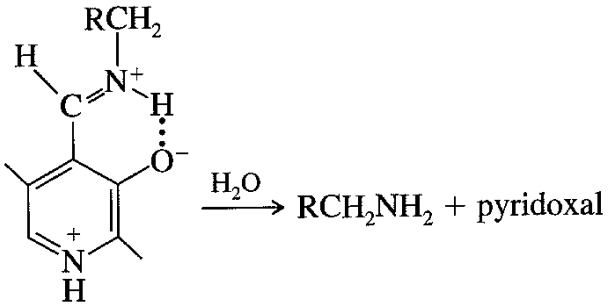
RX— may be a thiol, a hydroxyl, or an indole group. In this way, serine, threonine, cysteine, tryptophan, cystathionine, and serine and threonine phosphates may be interconverted or degraded.

b. α-Decarboxylation

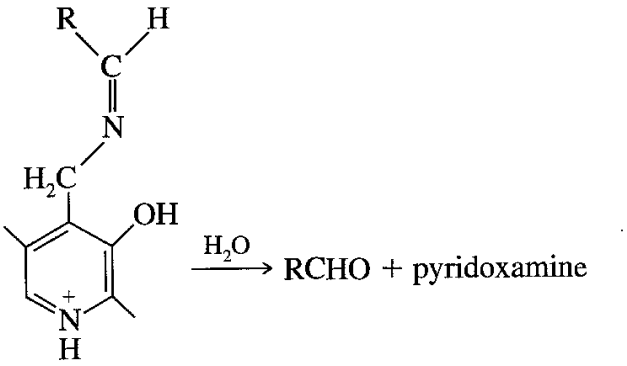
The "electron sink" allows facile decarboxylation. The decarboxylated adduct will add a proton to the amino acid carbonyl carbon and



Then hydrolyze to give the same amine and pyridoxal

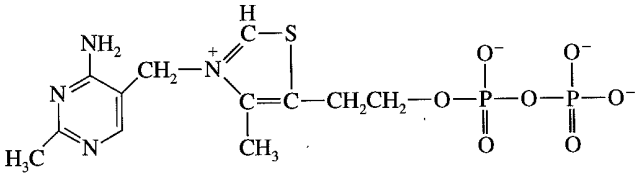


or else it will add the proton to the pyridoxal carbonyl carbon and then hydrolyze to give the aldehyde and pyridoxamine.

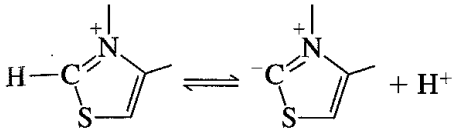


3. Thiamine pyrophosphate—Electrophilic catalysis

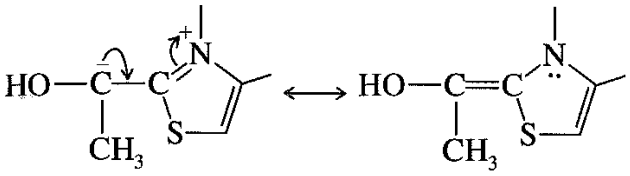
Thiamine pyrophosphate (structure 2.51) is another coenzyme that covalently bonds to a substrate and stabilizes a negative charge.



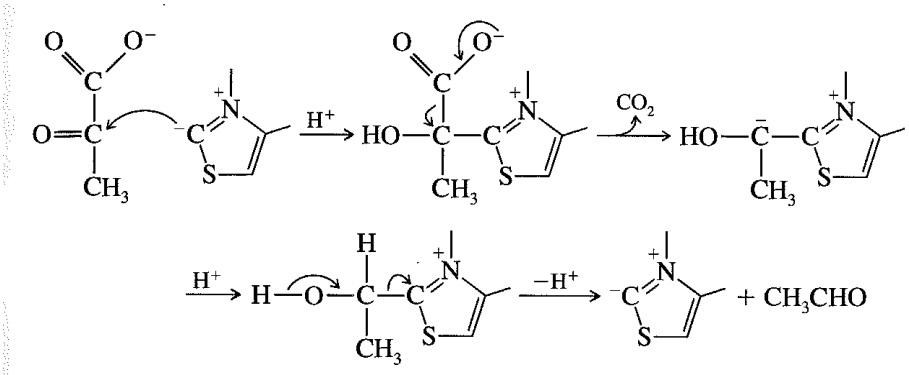
The positive charge on the nitrogen promotes the ionization of the C-2 carbon by electrostatic stabilization. The ionized carbon is a potent nucleophile.



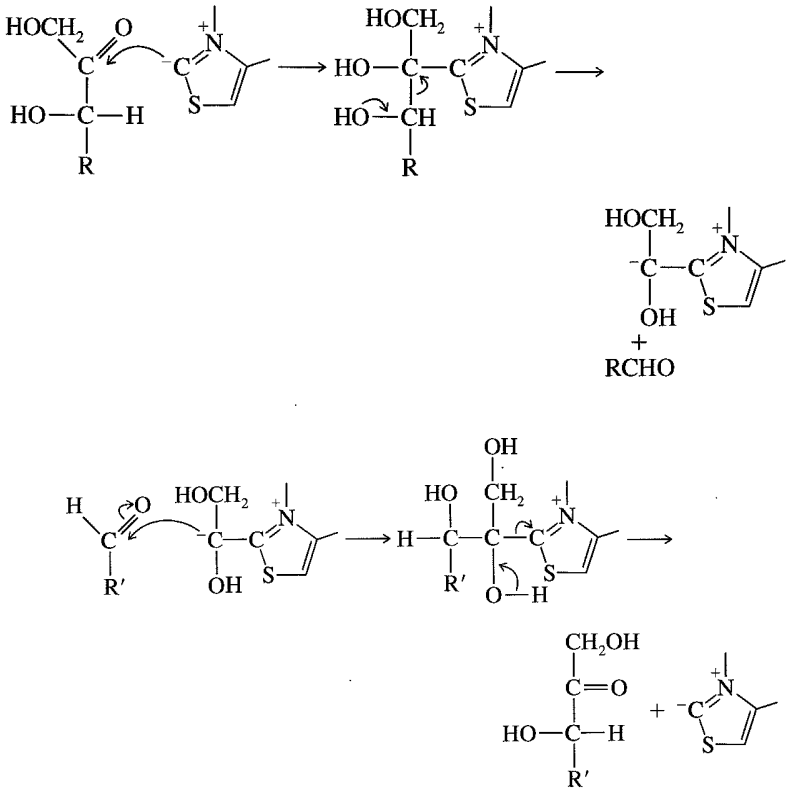
The nitrogen atom can also stabilize by delocalizing a negative charge on die adduct of thiamine with many compounds, as, for example, in hydroxyeihylthiamine pyrophosphate, & form in which much of the coenzyme is found in vivo (Refer equation below).



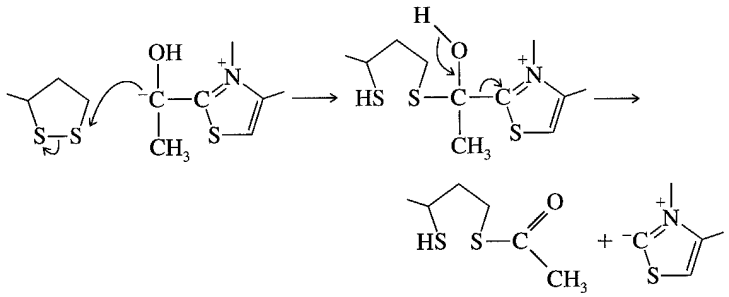
The combination of these reactions allows the decarboxylation of pyruvate by the route shown in below. Other carbon-carbon bonds adjacent to a carbonyl group may be cleaved in the same manner.



The hydroxyethylthiamine pyrophosphates are potent nucleophiles and may add to carbonyl compounds to form carbon-carbon bonds. A good illustration of carbon-carbon bond making and breaking occurs in the reactions of transketolase. The enzyme contains tightly bound thiamine pyrophosphate and shuttles a dihydroxyethyl group between D-xylulose 5-phosphate and D-ribose 5-phosphate to form D-sedoheptulose 7-phosphate and D-glyceraldehyde 3-phosphate (Refer the two equations below)



Hydroxyethylthiamine pyrophosphate is also nucleophilic toward a thiol of oxidized lipoic acid. A hemithioacetal is formed, and this decomposes to give a thioester:



Nucleophilic catalysis

In enzymes, the most common nucleophilic groups that are functional in catalysis are the serine hydroxyl—which occurs in the serine proteases, cholinesterases, esterases, lipases, and alkaline phosphatases—and the cysteine thiol—which occurs in the thiol

proteases (papain, ficin, and bromelain), in glyceraldehyde 3-phosphate dehydrogenase, etc. The imidazole of histidine usually functions as an acid-base catalyst and enhances the nucleophilicity of hydroxyl and thiol groups, but it sometimes acts as a nucleophile with the phosphoryl group in phosphate transfer.

The hydrolysis of peptides by these proteases represents classic nucleophilic catalysis. The relatively inert peptide is converted to the far more reactive ester or thioester acylenzyme, which is rapidly hydrolyzed. The use of the serine hydroxyl rather than the direct attack of a water molecule on the substrate is favored in several ways: alcohols are often better nucleophiles than the water molecule in both general-base-catalyzed and direct nucleophilic attack; the serine reaction is intramolecular and hence favored entropically; and the arrangement of groups is more "rigid" and defined for the serine hydroxyl compared with a bound water molecule.

Nucleophile	Enzyme	Intermediate
—OH (serine)	Serine proteases	Acylenzyme
	Alkaline phosphatases, phosphoglucomutase	Phosphorylenzyme
—OH (threonine)[a]	Proteasome, amidases	Acylenzyme
OH^- (zinc-bound)	Carbonic anhydrase, liver alcohol dehydrogenase	—
—SH (cysteine)	Thiol proteases, glyceraldehyde 3-phosphate dehydrogenase	Acylenzyme
$—CO_2^-$ (aspartate)	ATPase (K^+/Na^+, Ca^{2+})	Phosphorylenzyme
$—NH_2$ (lysine)	Acetoacetate decarboxylase, aldolase, transaldolase, pyridoxal enzymes	Schiff base
	DNA ligase	Adenylenzyme (phosphoamide)
Imidazole (histidine)	Phosphoglycerate mutase, succinyl-CoA synthetase, nucleoside diphosphokinase, histone phosphokinase, acid phosphokinase	Phosphorylenzyme
—OH (tyrosine)	Glutamine synthetase	Adenylenzyme
	Topoisomerases	Nucleotidylenzyme (phosphotyrosine)

UNIT – II STEREOCHEMISTRY OF ENZYMATIC REACTIONS

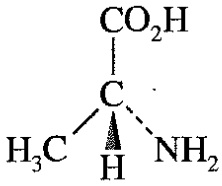
Stereospecific enzymatic reactions – fumarase catalysed reactions – NAD dependent oxidation and reduction reactions – Stereochemistry of nucleophillic reactions – chiral methyl group – chiral phosphate.

Introduction

Stereospecificity is a hallmark of enzyme catalysis, so a knowledge of the basic principles of stereochemistry is essential for appreciating enzyme mechanisms. Stereochemical evidence can provide important information about the topology of enzyme-substrate complexes. In particular, the positions of catalytic groups on the enzyme relative to the substrate may often be indicated, as may be the conformation or configuration of a substrate or intermediate during the reaction. Further, comparison of the stereochemistry of the substrates and products may reveal the likelihood of intermediates during the reaction.

Optical activity

A compound is optically active if it rotates the plane of polarization i.e., plane polarized light in right or left direction is said to be optically active. If it is not superimposable on its mirror image, it is referred to as an enantiomer. The presence of plane or centre of symmetry indicates the absence of optical activity whereas absence indicates presence of optical activity.



D-Alanine

Chiral carbon and chirality

The optical activity of many organic compounds is caused by the presence of an asymmetric carbon atom with four different groups attached to it. This type of carbon is now called a chiral carbon atom and the asymmetric centre is referred to as chiral centre.

A carbon atom of the form $CR_2R'R''$ is called prochiral. Although it is not optically active because it is bound to two identical groups, it is potentially chiral since it can be made chiral by any operation that produces a difference between the two R groups.

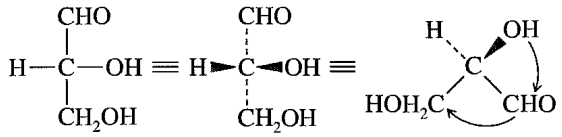
Notation

The letters d and l, which are often used to denote configuration, have the drawback that they are not absolute but are relative to a reference compound.

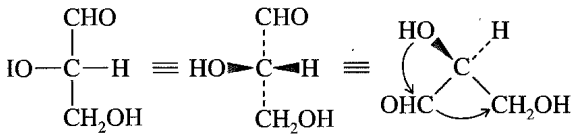
A more useful notation, denoting absolute configuration of a chiral center, is the RS convention. The groups around the chiral carbon are assigned an order of "priority" based on a series of rules that depend on atomic number and mass. The atom directly attached to the chiral carbon is considered first; the higher its atomic numer, the higher its priority. For isotopes, the higher mass number has priority. For groups that have the same type of atom attached to the chiral carbon, the atomic numbers of the next atoms out are considered.

This is best illustrated by the following list of the most commonly found groups: -SH > -OR > -OH >- NHCOR > $-NH_2$ > $-CO_2R$ > $-CO_2H$ > -CHO > $-CH_2OH$ > $-C_6H_5$ > $-CH_3$ > -T > -D >-—H.

A chiral carbon is designated as being R or S as follows. The carbon is viewed from the direction opposite to the ligand of lowest priority. If the priority order of the remaining three ligands decreases in a clockwise direction, the absolute configuration of the molecule is said to be R (Latin, rectus = right). If it decreases in the counterclockwise direction it is said to be S (Latin, sinister = left. For example:



D-Glyceraldehyde = *R*-glyceraldehyde



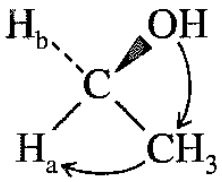
L-Glyceraldehyde = *S*-glyceraldehyde

If compounds contain different isotopes, e.g., hydrogen and deuterium, the priority rales are applied first to the atomic numbers. If this does not give an

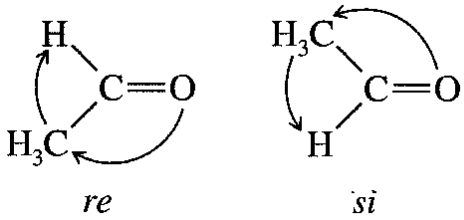
unambiguous assignment, the higher-mass-number isotope is given priority. There is also a prochirality.

Prochiral rule

There is also a prochirality rule. For example, in ethanol, if we label the two protons by the subscripts a and b, and arbitrarily give H_a priority over H_b (Refer below), H_a is said to be pro-R because of the clockwise order of priority. Conversely, H_b is pro-S*. If we repeat the treatment but give H_b priority over H_a, H_a is still found to be pro-R and Hb pro-S. Prochirality is thus absolute and does not depend on whether H_a or H_b is given priority.



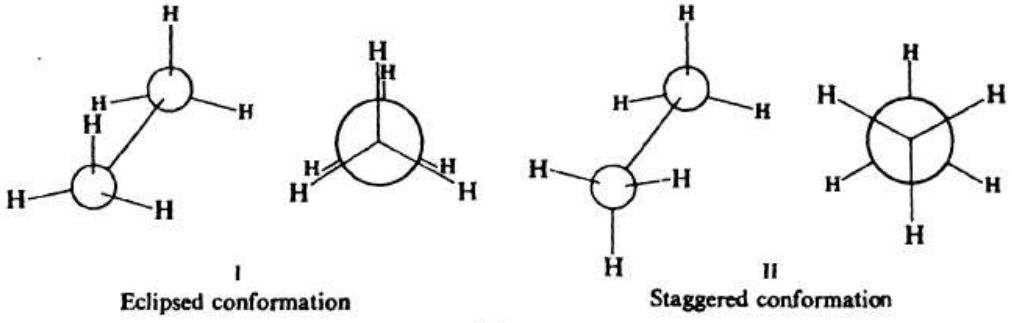
The two faces of a compound containing a trigonal carbon atom are described as re (rectus) and si (sinister) by a complicated set of rules. Two simple cases are illustrated by the faces presented below;



Conformation

Conformation refers to any one of the molecules instantaneous orientation in space caused by free rotation about its single bond.

The simple alkane ethane provides a good introduction to conformational analysis. The one carbon-carbon bond, and the rotational structures (rotamers) has two extremes, **staggered** and **eclipsed**.

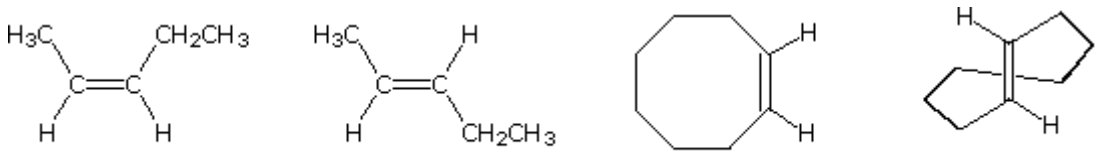


I
Eclipsed conformation

II
Staggered conformation

Configuration (Geometric isomerism)

Configuration refers to the geometry about a rigid or dissymmetric part of the molecule. Some examples of this configurational stereoisomerism (sometimes called geometric isomerism) are shown below. Cycloalkenes smaller than eight carbons cannot exist in a stable trans configuration due to ring strain. A similar restriction holds against cycloalkynes smaller than ten carbons. Since alkynes are linear, there is no stereoisomerism associated with the carbon-carbon triple bond.



cis-2-pentene

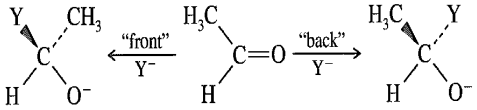
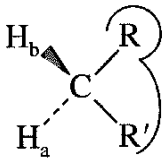
trans-2-pentene

cis-cyclooctene

trans-cyclooctene

Differences between the stereochemistry of enzymatic and non-enzymatic reactions

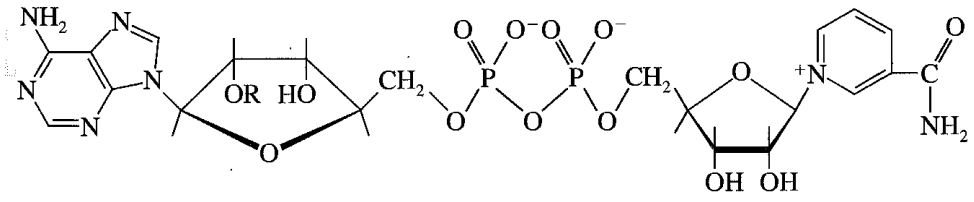
Non-enzymatic reactions	Enzymatic reactions
• Non-enzymatic reactions takes place in a homogenous solution	• Enzymatic reactions takes place on the surface of a protein
• Non-enzymatic reactions are mostly irreversible depends upon the catalyst	• Enzymatic reactions are autocatalysis
• Non-enzymatic reactions cannot be regulated	• Enzymatic reactions can be regulated.
• In chemical reactions, the equivalence	• In enzymatic reactions, the

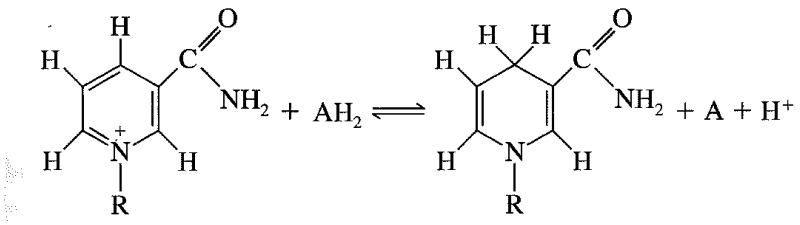
of the two hydrogen atoms of CH_2R'-R'' is lost during the course of the reaction	binding of the two R groups causes the two hydrogen to be exposed to different environments i.e., H_a binds to one position whereas H_b binds next to it, equivalence is not lost.
• In a chemical reaction, attack of the nucleophile causes a planar trigonal intermediate causes the attack of the second incoming group on the either side. Which results in racemic mixture of the product due to microscopic reversibility. 	• In an enzymatic reaction, since the enzyme is firmly bound to the enzyme active side and only one face is exposed, so only one type of product is only formed. 
• Chemical reactions have stereospecificity	• Enzymatic reactions also has stereospecificity.

Stereospecific enzymatic reactions

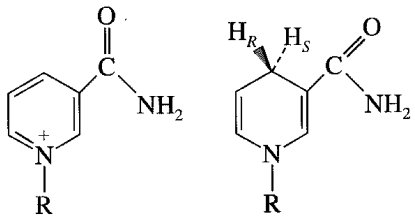
1. **NAD^+ and $NADP^+$ dependent oxidation and reduction**

NAD+ (In first figure below R = H) and NADP+ (In the second figure below R = PO3") function as coenzymes in redox reactions by reversibly accepting hydrogen at the 4 position of the nicotinamide ring.





The 4 position in the dihydronicotinamide ring is prochiral. The faces of the nicotinamide ring and the C-4 protons of the dihydronicotinamide rings may be labeled according to the RS convention by giving the portion of the ring containing the -$CONH_2$ group priority over the other portion:



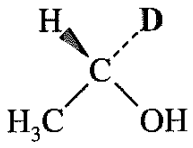
re

It was discovered by a series of experiments that there is direct and stereospecific transfer between the substrate and NAD^+.Yeast alcohol dehydrogenase transfers 1 mol of deuterium from CH_3CD_2OH to NAD^+.When the NADD formed with the enzyme and unlabeled acetaldehyde is incubated, all the deuterium is lost from the NADD and incorporated in the alcohol that is formed. The deuterium or hydrogen is transferred stereospecifically to one face of the NAD^+ and then transferred back from the same face. A nonspecific transfer to both faces would lead to a transfer of 50% of the deuterium back to the acetaldehyde (or considerably less than 50% because of the kinetic isotope effect slowing down deuterium transfer). Some dehydrogenases transfer to the same face as does the alcohol dehydrogenase (Class A dehydrogenases), others to the opposite face (Class B); these are listed in Chapter 16. It has subsequently been found that the Class A enzymes transfer to the re face of NAD^+ (or $NADP^+$) and use the pro-.R hydrogen of NADH:

$$NAD^+ + CH_3CD_2OH \longrightarrow NADD + CH_3CDO + H^+$$

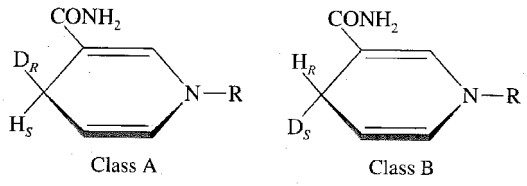
$$NADD + CH_3CHO + H^+ \longrightarrow NAD^+ + CH_3CDHOH$$

The transfer to the aldehyde is also stereospecific. The alcohol formed is the R enantiomer. The S enantiomer, formed from CH_3CDO and NADH, has a specific rotation of - 0.28 ± 0.03.



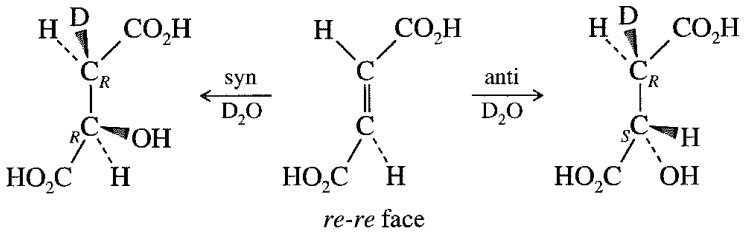
Types of dehydrogenases

There are two types of dehydrogenases such as class A and class B dehydrogenases. Class A dehydrogenases transfers the pro-R hydrogen from NADH, whereas class B transfers the pro-S. The crystal structures of dehydrogenases solved so far show that NAD^+ is bound to the Class A enzymes with the nicotinamide ring in the anti conformation about the glycosidic bond, as is consistent with NMR studies in solution, and that the Class B enzymes have the ring bound in the syn conformation.



2. Stereochemistry of fumarase catalysed reaction

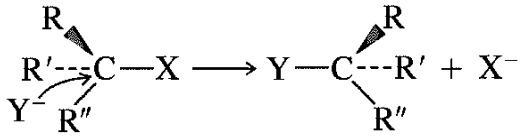
Fumarase catalyzes the addition of the elements of water across the double bond of fumarate to form malate. An NMR analysis of the stereochemistry of the malic acid formed shows that the addition of D_2O is anti (D^+ adding to the re face at the top and OD^- adding to the si face at the bottom) rather than syn (both adding to the re-re face). Similarly, other enzymes in the citric acid cycle catalyze anti hydrations of double bonds. From the principle of microscopic reversibility, the dehydration reaction of malate occurs in a conformation in which the two carboxyl groups are anti, as the elements of water that are to be eliminated.



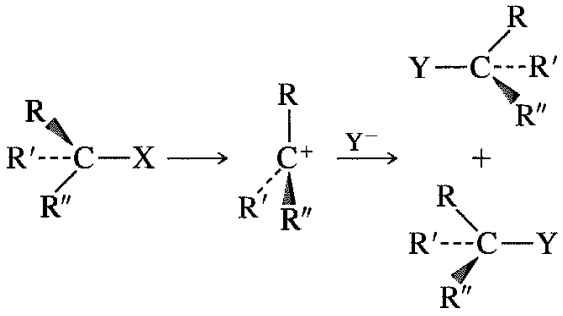
re-re face

Stereochemistry of nucleophilic reactions

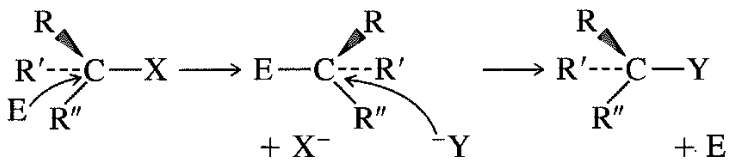
The reaction that usually begins most elementary courses on mechanistic organic chemistry is nucleophilic substitution at saturated carbon. There are two extreme forms of the mechanism. The biomolecular SN2 reaction of a nucleophile (Y) and an alkyl halide leads to inversion of the chiral carbon.



The unimolecular SN1 reaction, on the other hand, generates a planar carbonium ion that reacts randomly at each face to generate a racemic product.



The inversion of configuration during an SN2 reaction has led to certain criteria for the detection of intermediates in enzymatic reactions. For example, suppose that an enzyme catalyzes the nucleophilic reaction. Then a direct attack of Y" on the alkyl halide will lead to inversion. But if the substrate first reacts with a nucleophilic group on the enzyme to give an intermediate that then reacts with Y",



there will be two successive inversions, and hence an overall retention of configuration. Thus, retention in such reactions is generally taken as evidence for the presence of an intermediate on a reaction pathway, whereas inversion of configuration is interpreted as evidence for the direct reaction of one substrate with the other.

The chiral methyl group

Many enzymatic reactions involve the conversion of methylene groups (CH_2XY or CH_2=) to methyl (CH_3-), and vice versa. The stereochemistry of these reactions may be studied by synthesizing a methyl group that is chiral. This is possible by using all three isotopes of hydrogen to form a classical asymmetric carbon atom.

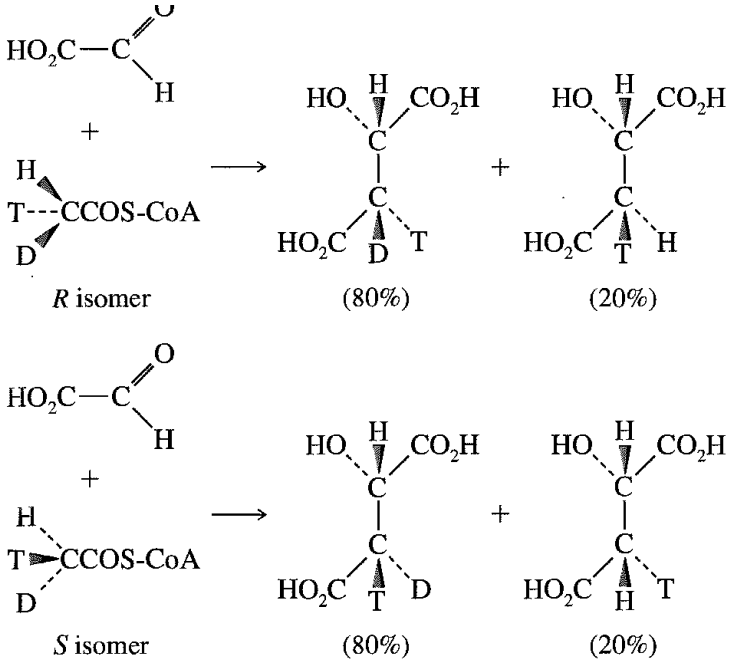
The fundamental difference between generating a chiral methyl group from a methylene group and converting a chiral methyl group into methylene

There is a crucial difference between generating a chiral methyl group and converting the same group to methylene. A methylene group CH_2XY is prochiral, and when it interacts with an enzyme by a three-point contact the two hydrogen atoms are non-equivalent. Their reactivity is thus governed by stereospecificity. A methyl group in CH_3X, on the other hand, rotates freely about the C—X bond, both in solution and when the group is bound to an enzyme. The reactivities of the hydrogen isotopes in CHDTX are thus governed by the kinetic-isotope effect on the reaction. Accordingly, whereas a prochiral methylene compound may be transformed by a stereospecific reaction into a homogeneous chiral product, a chiral methyl group reacts to give a mixture of products, the distribution of which depends on the magnitude of the kinetic isotope effect. Further, because of the possibility that a chiral methyl group is gradually exchanging isotopes by recycling in a reversible reaction, it is important that either the reaction being studied is irreversible, or that the products are rapidly converted into stable form by a subsequent

reaction so that the isotopes do not become scrambled by equilibration. The chirality assay. The assay depends on radioactivity and not on optical measurements It is clearly not practical to measure the optical activities of chiral methyl compounds because of the vast amounts of radioactive tritium that would have to be handled. It is sufficient that practically all tritium-containing methyl groups also contain one deuterium, and that all, or nearly all, such methyl groups have the same chirality. There is thus a small amount of chiral, but also radioactive, methyl in a sea of nonradioactive material. By using a radioactivity assay on a mixture of unlabeled and labeled reagents, the only compounds that are measured are those that are labeled and hence chiral.

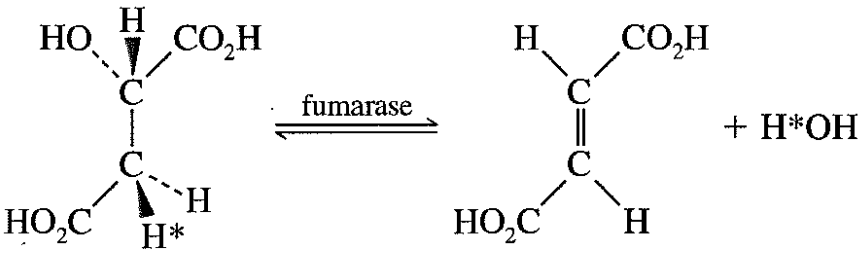
The assay uses the stereochemistry of known enzymatic reactions

R and S isomers of [HDT] acetic acid were synthesized by chemical and enzymatic methods that yield products of known stereochemistry. The two isomers were then distinguished by using the following ingenious enzymatic assays. The acetic acid was first converted to acetyl-coenzyme A. The acetyl-coenzyme A was then condensed with glyoxylate to form malate in an essentially irreversible reaction catalyzed by malate synthase.



The crucial feature of this reaction is that it is subject to a normal kinetic isotope effect, so that more H than D is lost on condensation. In fact, there is a k_H/k_D effect of 4

so that 80% of the radioactive product has lost H and 20% has lost D. The stereochemistry is resolved by using the enzyme fumarase. The crucial feature of the fumarase reaction is that, as described previously, the stereochemistry of dehydration of malate is controlled stereospecifically and is not influenced by a kinetic isotope effect: dehydration occurs from a malate conformation in which the two carboxyl groups are anti, as are the H and OH groups that are eliminated.

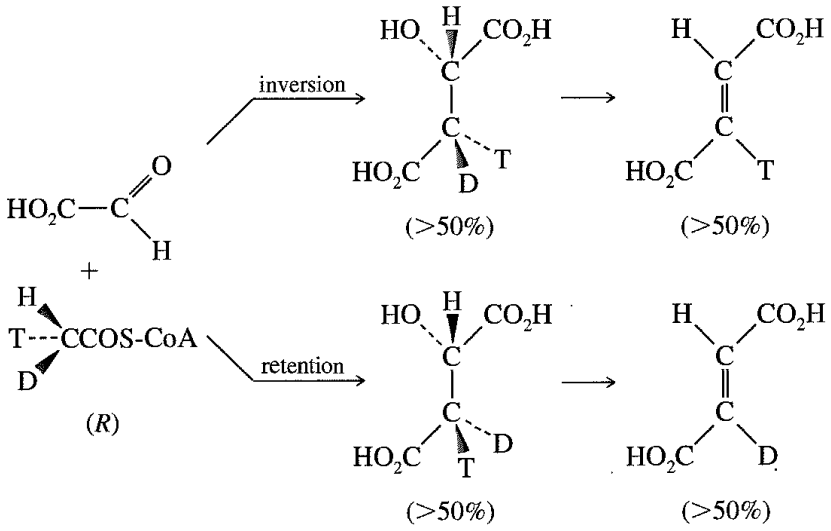


It is found that the fumarate eventually formed from the R isomer of acetic acid gives a product containing 80% of the tritium and that the fumarate from the S isomer contains 20%, compared with a control reaction of TCH2CO2H, which gives a product containing 50%. This clearly provides an assay for the chirality of unknown acetyl groups: R should retain 80% of T, and S 20%.

3. Stereochemistry of the malate synthase reaction

The stereospecificity of the malate synthase reaction was inferred from prior knowledge that the -C(H)(OH)(CO_2H) moiety of malic acid is S and the specificity of fumarase is anti, and assuming a normal k_H/k_D effect of greater than 1. The malic acid has lost more H than D, inversion of the configuration of acetate will yield the products shown above, where the D-containing form is greater than 50% and the H-containing form less than 50%. Retention of configuration will give products with the opposite configurations. Equations below predict that the fumarate formed from R-acetate in the condensation of glyoxylate with acetyl-coenzyme A will contain greater than 50% of the T if inversion has occurred, and less than 50% with retention. The observed figure of 80% is therefore consistent with inversion of configuration. Similarly, it is predicted, as found, that inversion of the configuration of the S isomer will give fumarate containing

less than 50% of the T. It is seen that even though the rotation of a chiral methyl group is not constrained, stereochemical information may be derived, provided that there is a kinetic isotope effect. The absence of an effect would give 50% of each product, and hence no information on stereospecificity.



It has since been proved that malate synthase proceeds with inversion independently of any assumption about isotope effect. This has been done via a lyase system that splits malate to acetate-a methylene-to-methyl conversion. By using methylene-labeled malate and generating chiral acetate, it could be shown unambiguously that this cleavage is an inversion. When the lyase reaction was applied to malate formed from chiral acetate by malate synthase, it was found that the acetate thus generated was of the same chirality as the starting material. This proved that there is also inversion in the reaction of malate synthase. The stereospecificity of other methyl-methylene transformations has been assigned by converting the reaction products to acetyl-coenzyme A and using the malate synthase/fumarase assay. Similarly, the stereochemistry of transfer of chiral methyl groups between two acceptors has also been studied by converting the methyl group into chiral acetate.

Chiral phosphate

Phosphate esters play a central and ubiquitous role in biochemistry: the genetic information of all living organisms is stored in the phosphodi ester polymers DNA and RNA; the genetic information is translated into protein via RNA; the di- and triphosphates of nucleosides are important energy carriers, as are other phosphorylated compounds; groups are often activated by phosphorylation; energy transduction is generally mediated via the hydrolysis of a nucleoside triphosphate; phosphorylation of proteins and enzymes is typically an important part of control mechanisms; phosphorylation of metabolites is often invoked to increase their solubility or prevent their passage across membranes. Phosphate transfer reactions are accordingly most common. The chemistry of phosphoryl transfer is far more complicated than that of acyl transfer. A very important piece of information in the elucidation of the mechanism of phosphoryl transfer is the stereochemistry.

1. A preview of phosphoryl transfer chemistry

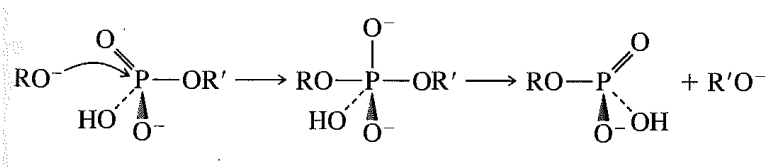
Unlike carbon, which forms only stable tetravalent compounds, phosphorus forms stable trivalent, tetravalent, and pentavalent compounds. A phosphoryl transfer reaction such as can proceed by either of two general types of mechanisms,

$$R'OPO_3^{2-} + ROH \longrightarrow ROPO_3^{2-} + R'OH$$

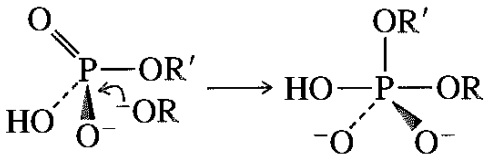
analogous to the SN1 and SN2 mechanisms of carbon chemistry. In a dissociative mechanism, there is first the elimination of the leaving group to produce an unstable metaphosphate intermediate, and then the rapid addition of the nucleophile:

$$R'OPO_3^{2-} \longrightarrow R'O^- + PO_3^- \xrightarrow{ROH} ROPO_3^{2-} + R'OH$$

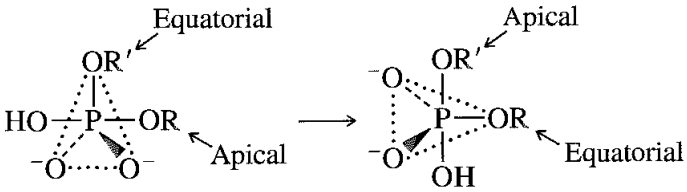
In an associative mechanism, there is first the addition of the nucleophile to give a pentacovalent intermediate, followed by the elimination of the leaving group. This mechanism is subdivided further. There is an in-line mechanism, in which the attacking nucleophile enters opposite the leaving group,



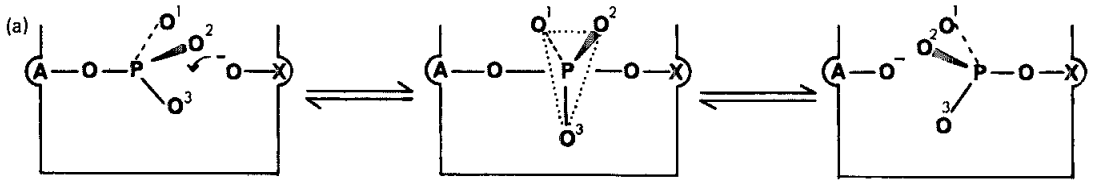
and there is an adjacent mechanism, in which the nucleophile enters on the same side as the leaving group.



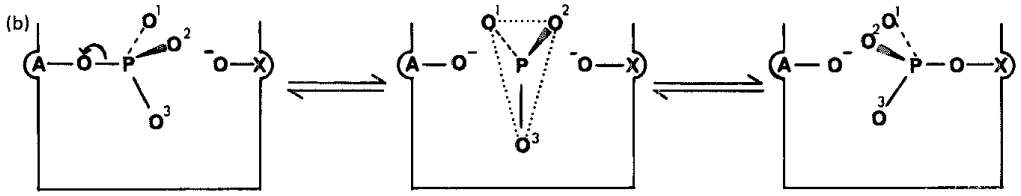
The adjacent mechanism involves an additional step that is a consequence of the symmetry of the intermediate. The pentacovalent intermediate is a trigonal bipyramid in which groups assume either of two topologically different positions, equatorial and apical. In the in-line mechanism, the entering and leaving groups naturally take up apical positions. In the adjacent mechanism, however, the entering nucleophile is apical whereas the leaving group is equatorial. For the leaving group to be expelled, it must move to an apical position. The movement occurs by a process termedpseudorotation.



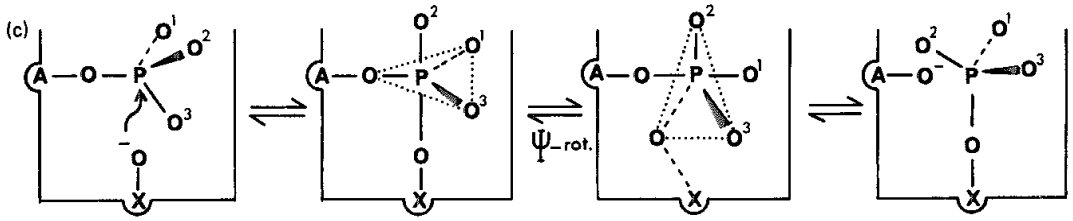
The stereochemical consequences of the three mechanisms in enzymatic reactions are illustrated below.The phosphoryl group has the same tetrahedral geometry as saturated carbon compounds. A direct in-line associative reaction between two substrates leads to inversion is illustrated below,



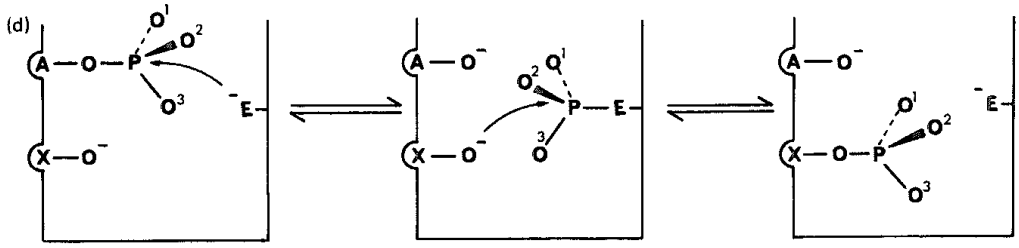
The phosphoryl transfer in in-line dissociative mechanism is illustrated below.



Although a dissociative reaction in solution would generally lead to racemization, the stereochemistry in an enzymatic reaction is governed by the spatial arrangements of the substrates. The adjacent associative mechanism leads to retention of configuration (Refer figure below).



In an enzymatic reaction there is the fourth possibility that a covalent intermediate is formed with a nucleophilic group on the enzyme (Refer figure below).



The stereochemistry should obey the rules that normally apply when such intermediates occur: an odd number of intermediates with inversion at each step will lead to retention of configuration, whereas an even number will lead to inversion. But, the adjacent mechanism always leads to retention, since configuration is retained at each step.

Chirality of phosphoryl derivatives

A phosphate diester of the form $ROPO_2OR'$ is prochiral, since one of the non-alkylated oxygen (^{16}O) atoms must be replaced to produce chirality. A monoester $ROPO_3^{2-}$ is pro-prochiral, since two oxygen atoms must be substituted to produce a chiral compound. Chiral phosphates have been synthesized de novo by using stereospecific chemical and enzymatic reactions with isotopic and or atomic substitutions. For example, a chiral phosphorothioate may be synthesized from a prochiral phosphate by replacing an oxygen atom with a sulfur atom. Similarly, what would otherwise be a pro-prochiral phosphate has been synthesized as a chiral product by replacing one oxygen atom with sulfur and another with $^{17}0$ or ^{18}O, or by replacing two oxygen atoms with both isotopes to give [$^{16}0$,$^{17}0$,^{18}O]phosphate.

Assays of isotopic and chiral phosphoryl compounds have been greatly facilitated by ^{31}P NMR: substitution with ^{18}O causes a shift to higher field, the magnitude of the isotope shift being related to the bond order; ^{17}O, with its spin off and nuclear electronic quadrupole moment, rapidly relaxes the ^{31}P resonance and hence washes out its signal.

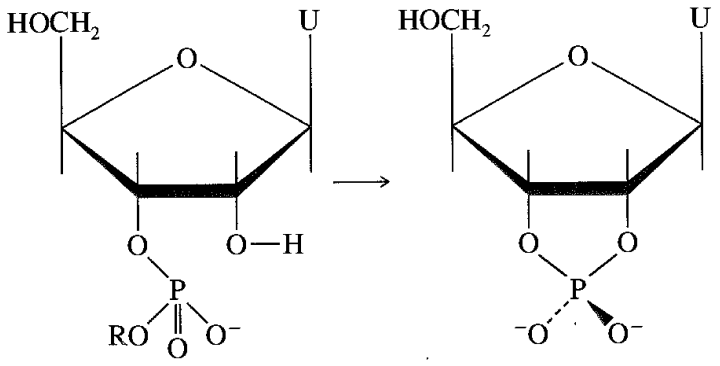
The different degrees of substitution required to generate chirality have led to different methods for studying the stereochemistry of the various reactions.

These reactions may be divided into four categories:

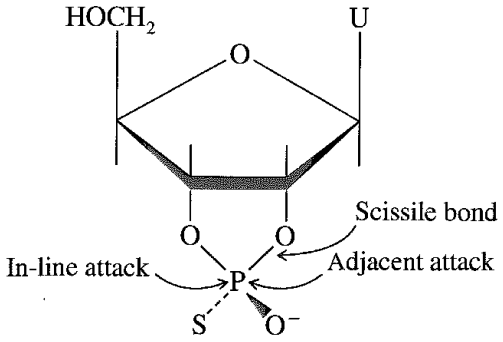
1. Prochiral substrate giving prochiral product (i.e., diester —» diester).
2. Prochiral substrate giving pro-prochiral product (i.e., diester —» monoester).
3. Pro-prochiral substrate giving pro-prochiral product (i.e., monoester —> monoester).
4. Pro-prochiral substrate giving pro-pro-prochiral product (i.e., monoester -¦»phosphate).

a. Prochiral substrate giving prochiral product

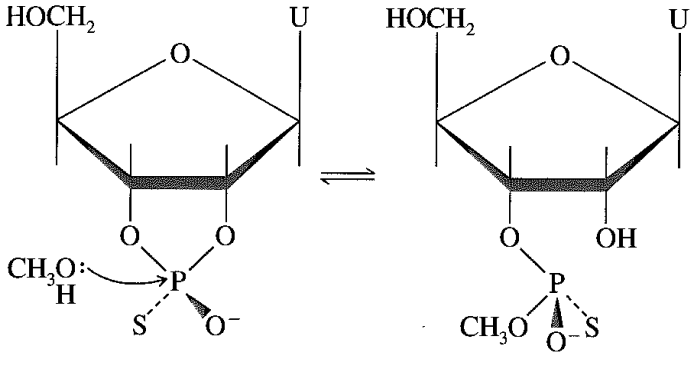
These reactions are the easiest to tackle, since they require only one phosphory] oxygen to be substituted in both the substrate and the product. The classic example of this experiment is the first step in the hydrolysis of RNA catalyzed by bovine pancreatic ribonuclease. As discussed in detail in Chapter 16, ribonuclease catalyzes the hydrolysis of RNA by a two-step reaction in which a cyclic intermediate is formed. The stereochemistry of the first step (cyclization) (Refer figure below)



was solved by some powerful, elegant experiments with the chiral substrate uridine 2', 3'-cyclic phosphorothioate (Refer figure below)



This compound was crystallized and its structure and absolute stereochemistry were determined by x-ray diffraction. Incubation with ribonuclease in aqueous methanol solution formed a methyl ester by the reverse of cyclization (Refer below).

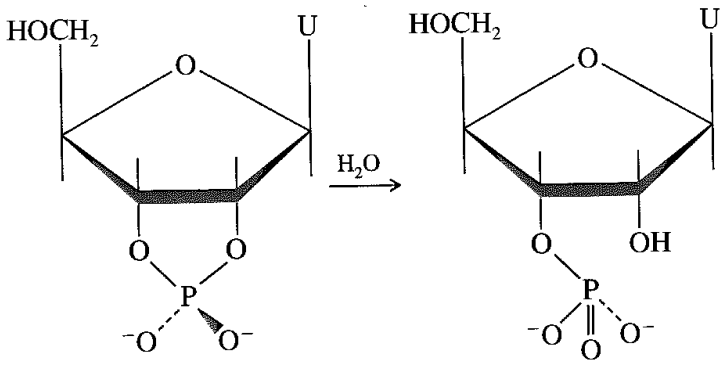


The methyl ester was crystallized and its absolute stereochemistry was determined by x-ray diffraction to be as in equation 8.37. This product corresponds to an in-line attack. When incubated with ribonuclease in aqueous solution, the methyl ester re-forms the original cyclic phosphorothioate. This result is expected from the principle of

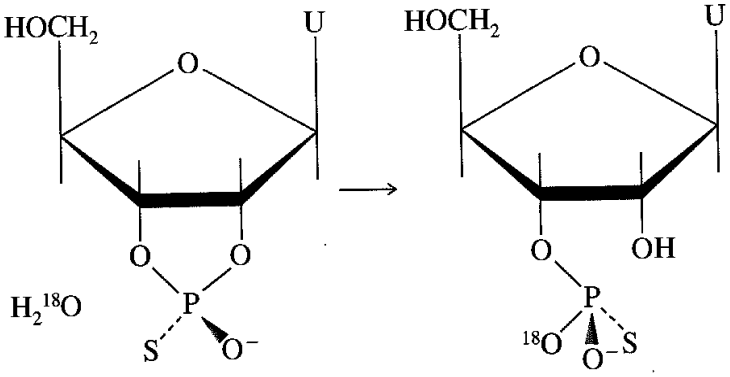
microscopic reversibility, since the forward and reverse reactions must go through the same transition state. But it does show directly that the cyclization step involves an in-line attack: an adjacent attack of the ribose hydroxyl in the cyclization of the methyl ester as in the right- hand structure (Refer figure below) would give the enantiomer of structure phosphorohionate.

b. Prochiral substrate giving pro-prochiral product

The second step of the ribonuclease reaction, ring opening (Refer figure below), is expected by analogy with the methanolysis step in the preceding section to be an in-line attack of water.



In this reaction, the chiral phosphorothioate was hydrolyzed in $H_2{}^{18}O$ to give a chiral product whose configuration could be assigned (Refer figure below).



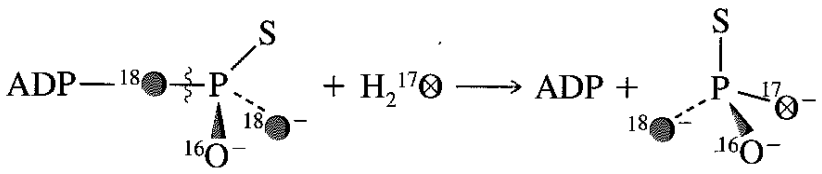
c. Pro-prochiral substrate giving pro-prochiral product

These reactions involve the transfer of the phosphoryl moiety itself and require that two of the three phosphoryl oxygen atoms be tagged in the substrate. This can be achieved by synthesizing an ^{18}O-substituted phosphorothioate or [^{16}O, ^{17}O, ^{18}O] phosphomonoester substrate. An example of the latter is the demonstration that the hexokinase reaction (Refer figure below) proceeds with inversion of configuration.25'40 The products were derivatized and their configurations were determined by NMR.

$$\beta\text{-D-glucose} + \text{ATP} \rightleftharpoons \text{glucose 6-phosphate} + \text{ADP}$$

d. Pro-prochiral substrate giving pro-pro-prochiral product

Experiments conducted so far for studying the stereochemistry of hydrolysis of phosphomonoesters to inorganic phosphate have used [^{16}O, ^{17}O, ^{18}O] thiophosphate. For example, the hydrolysis of ATP catalyzed by the myosin ATPase (Refer figure below) has been shown to proceed with inversion of configuration.

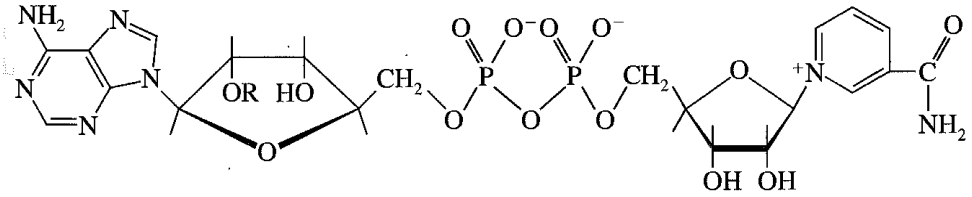


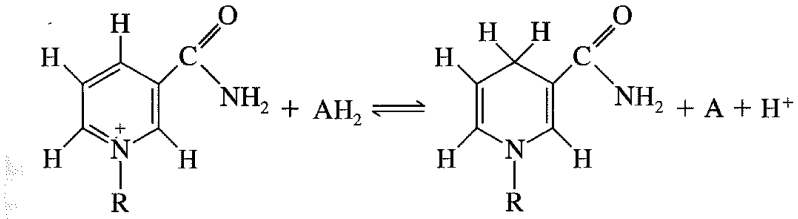
UNIT – III CASE STUDIES OF ENZYME STRUCTURE AND MECHANISM

The dehydrogenases – the proteases – ribonucleases – lysozyme – stability of proteins- stability – activity tradeoff.

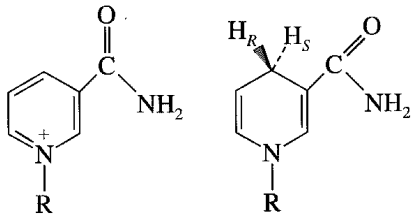
The dehydrogenases

NAD+ (In first figure below R = H) and NADP+ (In the second figure below R = PO_3^-) function as coenzymes in redox reactions by reversibly accepting hydrogen at the 4 position of the nicotinamide ring.





The 4 position in the dihydronicotinamide ring is prochiral. The faces of the nicotinamide ring and the C-4 protons of the dihydronicotinamide rings may be labeled according to the RS convention by giving the portion of the ring containing the -$CONH_2$ group priority over the other portion:



re

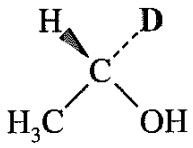
It was discovered by a series of experiments that there is direct and stereospecific transfer between the substrate and NAD^+.Yeast alcohol dehydrogenase transfers 1 mol of deuterium from CH_3CD_2OH to NAD^+.When the NADD formed with the enzyme and unlabeled acetaldehyde is incubated, all the deuterium is lost from the NADD and incorporated in the alcohol that is formed. The deuterium or hydrogen is transferred stereospecifically to one face of the NAD^+ and then transferred back from the same face. A nonspecific transfer to both faces would lead to a transfer of 50% of the deuterium back to the acetaldehyde (or considerably less than 50% because of the kinetic isotope

effect slowing down deuterium transfer). Some dehydrogenases transfer to the same face as does the alcohol dehydrogenase (Class A dehydrogenases), others to the opposite face (Class B); these are listed in Chapter 16. It has subsequently been found that the Class A enzymes transfer to the re face of NAD^+ (or $NADP^+$) and use the pro-.R hydrogen of NADH:

$$NAD^+ + CH_3CD_2OH \longrightarrow NADD + CH_3CDO + H^+$$

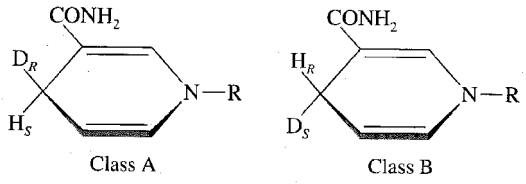
$$NADD + CH_3CHO + H^+ \longrightarrow NAD^+ + CH_3CDHOH$$

The transfer to the aldehyde is also stereospecific. The alcohol formed is the R enantiomer. The S enantiomer, formed from CH_3CDO and NADH, has a specific rotation of - 0.28 ± 0.03.



Types of dehydrogenases

There are two types of dehydrogenases such as class A and class B dehydrogenases. Class A dehydrogenases transfers the pro-R hydrogen from NADH, whereas class B transfers the pro-S. The crystal structures of dehydrogenases solved so far show that NAD^+ is bound to the Class A enzymes with the nicotinamide ring in the anti conformation about the glycosidic bond, as is consistent with NMR studies in solution, and that the Class B enzymes have the ring bound in the syn conformation.



Class A enzymes catalyzes the reduction of the more reactive carbonyls, while class B is associated with the reactive ones. NADH is a weaker reducing agent when it is in syn conformation than when it is anti.

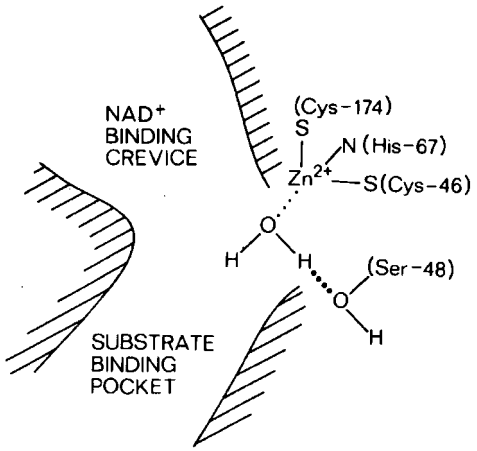
The alcohol dehydrogenases

The alcohol dehydrogenases are zinc metalloenzymes of broad specificity. They oxidize a wide range of aliphatic and aromatic alcohols to their corresponding aldehydes and ketones, using NAD^+ as a coenzyme. The two most studied enzymes are those from yeast and horse liver. The crystal structures of the horse liver apo- and holoenzymes were solved at 2.4 and 2.9 A respectively. The molecule is a symmetrical dimer, composed of two identical chains of Mr 40 000. Each chain contains one binding site for NAD^+ but two sites for Zn^{2+} with only one zinc ion is directly binds. The yeast enzyme is a tetramer of Mr 145 000, and each chain binds one NAD^+ and one Zn^{2+}. The overall reaction mechanism is same for both enzymes except the rate-determining step and the pH dependence, differ between the two.

The liver enzyme exists in two forms, E and S, which differ only by some six residues in their amino acid sequence.13 Only the S is active toward 3-β-hydroxysteroids, but both forms are active toward ethanol. E and S chains combine in statistical ratios to form EE, SS, and ES dimers. These different species are termed isozymes, which mean that they are multiple molecular forms of the same enzyme. When it is isolated from liver, the enzyme consists of about 40 to 60% of the EE dimer, the remainder being SS and ES.

A. Structure of the active site

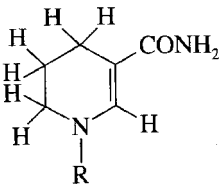
The Zn^{2+} ion sits at the bottom of a hydrophobic pocket formed at the junction of the catalytic and nucleotide-binding domains.



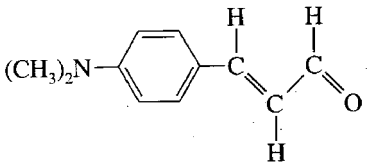
It is ligated by the sulfur atoms of Cys-46 and Cys-174, and by a nitrogen atom of His-67. The fourth ligand is an ionizable water molecule that is hydrogen-bonded to the hydroxyl group of Ser-48. It is known from the pH dependence of the binding of NAD^+, from the direct determination of the proton release on the binding of NAD^+,and from other kinetic procedures that the apoenzyme has a functional group of pKa -9.2 that is perturbed to a pKa of ~ 7.6 in the holoenzyme. Crystallographic studies suggest that this is the ionization of the zinc-bound water molecule. The nicotinamide ring of the NAD^+ is bound close to the zinc ion at the bottom of the pocket. The 2'-hydroxyl of the ribose ring is located between the hydroxyl of Ser-48 and the imidazole ring of His-51, within hydrogen-bonding distance of both.

B. The structure of the enzyme - substrate (ternary) complex

The structure of the reactive ternary complex composed of the enzyme NAD^+, and 4-bromobenzyl alcohol was solved at 2.9-A resolution. There is a favorable equilibrium between this complex and the enzyme-bound ternary complex reaction product of NADH and 4-bromo-benzaldehyde. The structure of the unreactive product like complex composed of the enzyme, the coenzyme analogue H_2NADH, and trans-4-(N,N-dimethylamino)-cinnamaldehyde (DACA) was also solved at the same resolution.

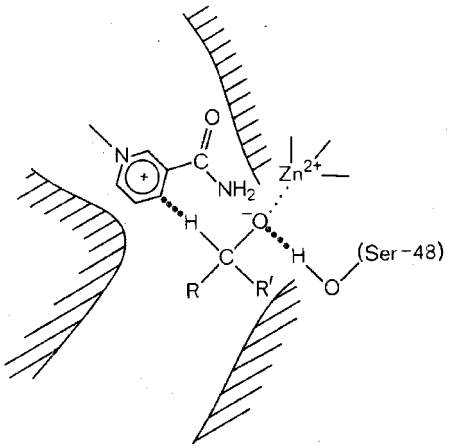


H_2NADH

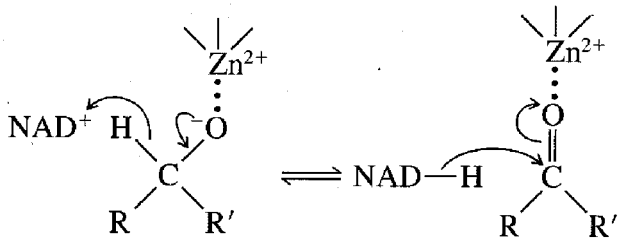


DACA
(chromophoric)

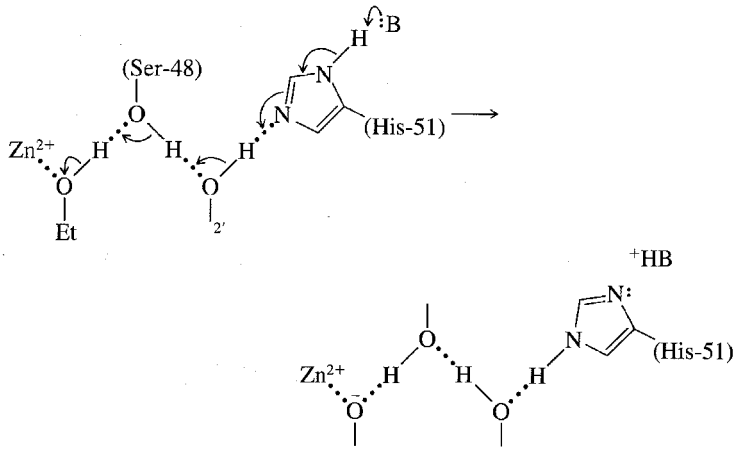
In both examples, the relevant oxygen atom of the substrate is directly coordinated to the zinc ion, displacing the bound water molecule so that the metal remains tetra coordinated. The hydrophobic side chain of the substrate binds in the hydrophobic "barrel" of the pocket. Both observations are consistent with earlier solution studies using model building, spectroscopy, and kinetics: the binding of the carbonyl oxygen of the chromophoric cinnamaldehyde (DACA) to a positively charged center in the enzyme was indicated by a characteristic shift in the spectrum.



It is difficult to predict whether the ionization state of the bound alcohol is bound as alcoholate anion or as neutral alcohol. Evidence in support of alcoholate anion comes from x-ray diffraction studies, pH studies and kinetic isotope effects. Kinetic isotopes effects is consistent with the hydride ion transfer taking place from an alcoholate ion.



Interestingly, although the substrate is bound in the pocket in a water-free environment, there is a system of hydrogen bonds that can shuttle the proton from the bound alcohol to the surface.



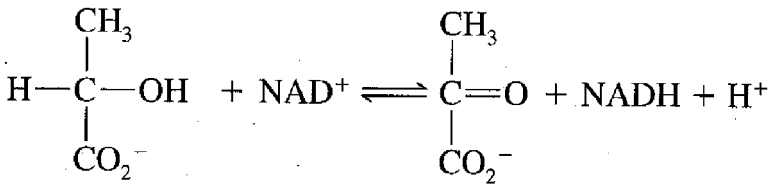
C. The Kinetic mechanism

Steady state and stopped-flow kinetic studies on the horse liver enzyme have shown that the oxidation of alcohols is an ordered mechanism, with the coenzyme binding first and the dissociation of the enzyme-NADH complex being rate determining step with enzyme-NAD^+ complex isomerizes to a second complex. In the reverse reaction, the reduction of aromatic aldehydes involves the rate determining dissociation step of the enzyme-alcohol complex whereas in the reduction of acetaldehyde, hydride transfer step is the rate determining step.

In the case of yeast enzymes, the enzyme-product complex dissociates rapidly such that the chemical steps are rate determining steps. The oxidation of deuterated alcohol RCD_2OH and the reduction of benzaldehydes by deuterated NADH are significantly slower than the reaction with normal isotopes indicating that the hydride transfer is the rate determining step.

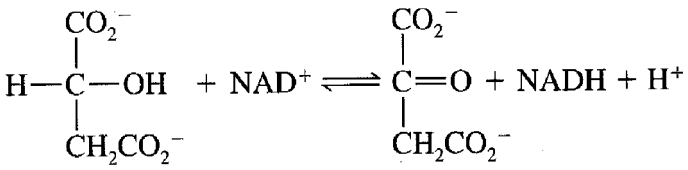
L-Lactate and L-malate dehydrogenases

L-Lactate dehydrogenase catalyzes the reversible oxidation of L-lactate to pyruvate, using NAD+ as a coenzyme.



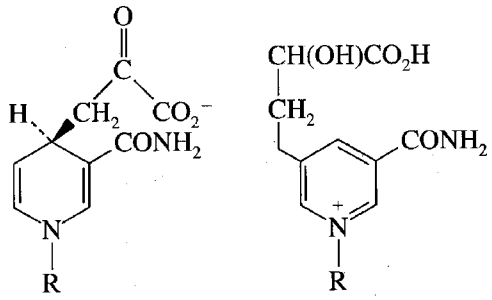
The isoenzyme is a tetramer of Mr 140 000 existing in two forms, the H4, predominating in heart muscle, and the M4, predominating in skeletal muscle. The amino acid composition of the polypeptide chain M4 form is significantly different from that of the H4 form, and the two have different kinetic properties. Despite this, the sites for the association of the subunits had to be very similar.

L-Malate dehydrogenase catalyzes the reversible oxidation of malate to oxaloacetate, using NAD+ as a coenzyme. Malate dehydrogenase is structurally and mechanistically similar to the lactate enzyme.

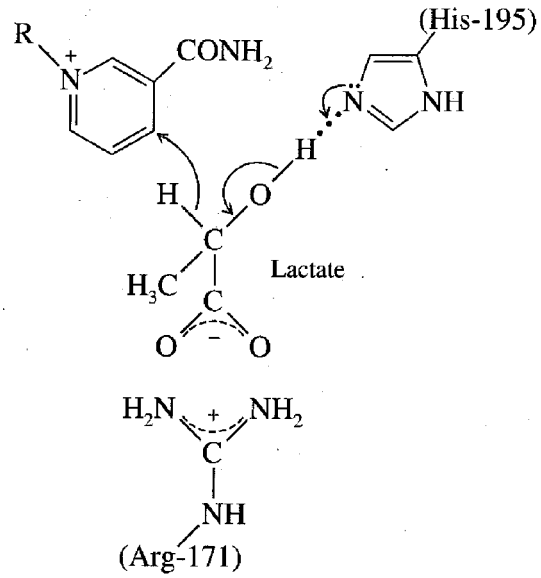


A. The structure of the enzyme-substrate complex

X-ray crystallographic studies reveal the structures and binding sites of ternary complexes of lactate dehydrogenase of NAD-pyruvate, covalent analogue of pyruvate plus NADH and the binding of the S-lac-NAD^+ of lactate dehydrogenases.



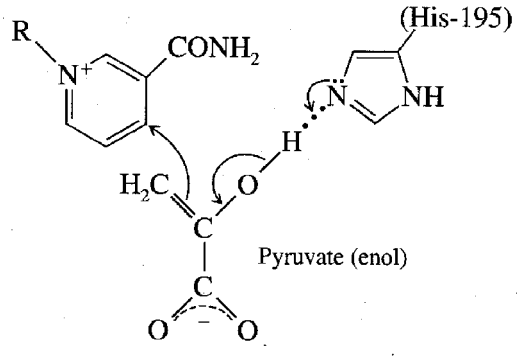
The active ternary complexes are shown below.



The carboxylate group of the substrate forms a salt bridge with the side chain of Arg-171. The hydroxyl group of lactate forms a hydrogen bond with the unprotonated imidazole ring of His-195. His-195 acts as an acid-base catalyst, removing the proton from lactate during oxidation.

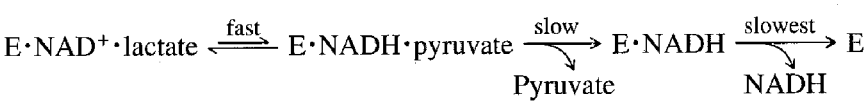
Further mutagenesis studies show that the hydrophobic nature of the nicotinamide-binding pocket causes tighter binding of the neutral NADH than of NADH+, a factor that helps to stabilize the ketoacid form of the ternary complex.

The NAD-pyruvate is formed by the enzyme from NAD+ and the enol form of pyruvate. The mechanism for the reaction is presumably catalyzed by His-195 in a manner similar to that of the oxidation of lactate.



b. The kinetic mechanism

Crystallographic studies reveal that the binding of the enzyme to the lactate or pyruvate occurs only in the presence of coenzyme revealing the ordered mechanism by inducing a conformational change in which residues 98 to 110 of the chain move through a relatively large distance to close over the active site in the ternary complex. Rapid reaction studies show that the mechanism differs from that of horse liver alcohol dehydrogenase, although the dissociation of the enzyme-NADH complex is rate-determining in both reactions. With alcohol dehydrogenase, the dissociation of the aldehyde from the ternary complex E-RCHO-NADH is faster than the rate of hydride transfer, so that the two ternary complexes do not equilibrate. However, the hydride transfer steps are very rapid in the reactions of lactate dehydrogenase and the, dissociation of pyruvate is slow, so that the two ternary complexes equilibrate. Further, the equilibrium position favors lactate and NAD+ at neutral pH. When lactate is mixed with the holoenzyme at pH 7, 20% of the bound NAD+ is reduced during the first millisecond of reaction, as the equilibrium between [E-NAD+-lactate] and [E-NADH-pyruvate] is rapidly attained. As the pyruvate dissociates, the equilibrium is displaced toward products, so that all four bound NAD+ molecules are reduced. There is then the slower dissociation of NADH, the rate-limiting step in the steady state.

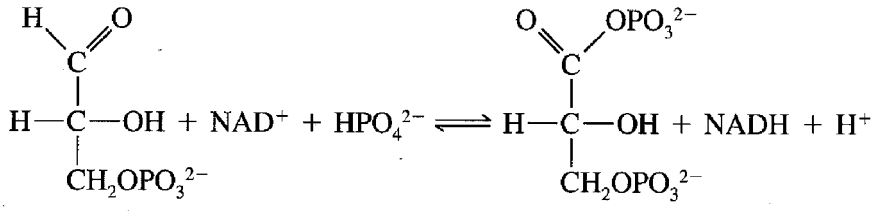


c. Changing specificity by protein engineering

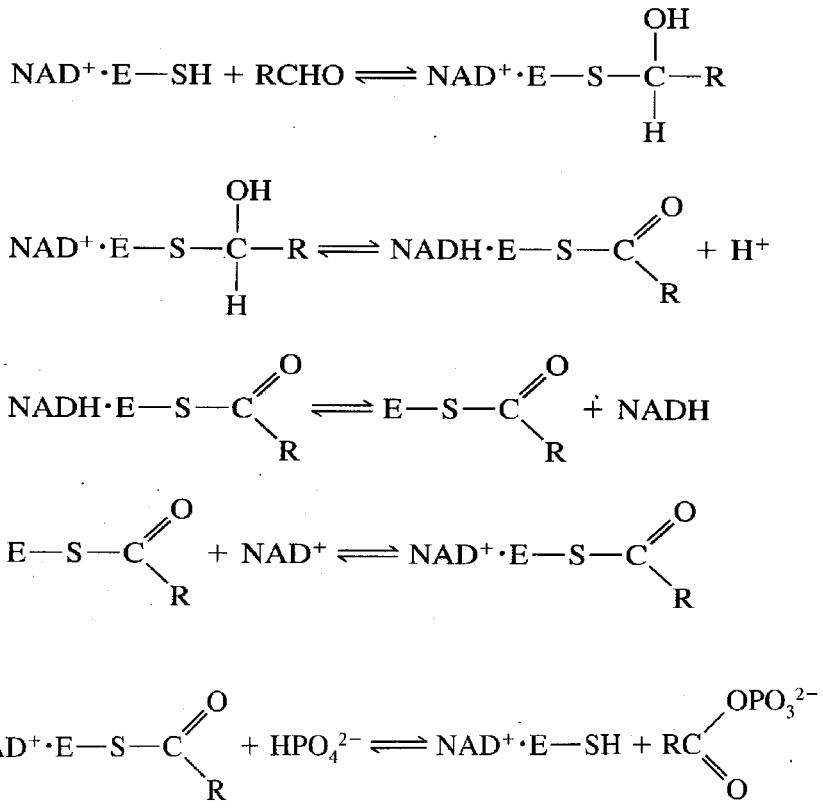
A key structural and mechanistic feature of lactate and malate dehydrogenases is the active site loop, residues 98-110 of the lactate enzyme. The loop has two functions: it carries Arg-109, which helps to stabilize the transition state during hydride transfer; and contacts around 101-103 are the main determinants of specificity. Tryptophan residues were placed in various parts of lactate dehydrogenase to monitor conformational changes during catalysis. Loop closure is the slowest of the motions. All malate dehydrogenases have an arginine residue at position 102, whereas most lactate dehydrogenases have a glutamine at this position. Mutation of Gin —> Arg-102 in lactate dehydrogenase increases the activity toward oxaloacetate by a factor of 103 and decreases the activity toward pyruvate by 104; i.e., a single mutation converts lactate into malate dehydrogenase.

3. Glyceraldehyde 3-phosphate dehydrogenase

Glyceraldehyde 3-phosphate dehydrogenase, a tetrameric enzyme of Mr 150 000 containing four identical chains, catalyzes the reversible oxidative phosphorylation of glyceraldehyde 3-phosphate to 1,3-diphosphoglycerate, using NAD+ as a coenzyme.



The reaction pathway consists of a series of reactions. The currently accepted mechanism (Refer the equation below) which is supported by extensive pre-steady state and steady state kinetic studies.



The enzyme has a reactive cysteine residue that is readily acylated by acyl phosphates to form a thioester (the reverse of the last reaction).

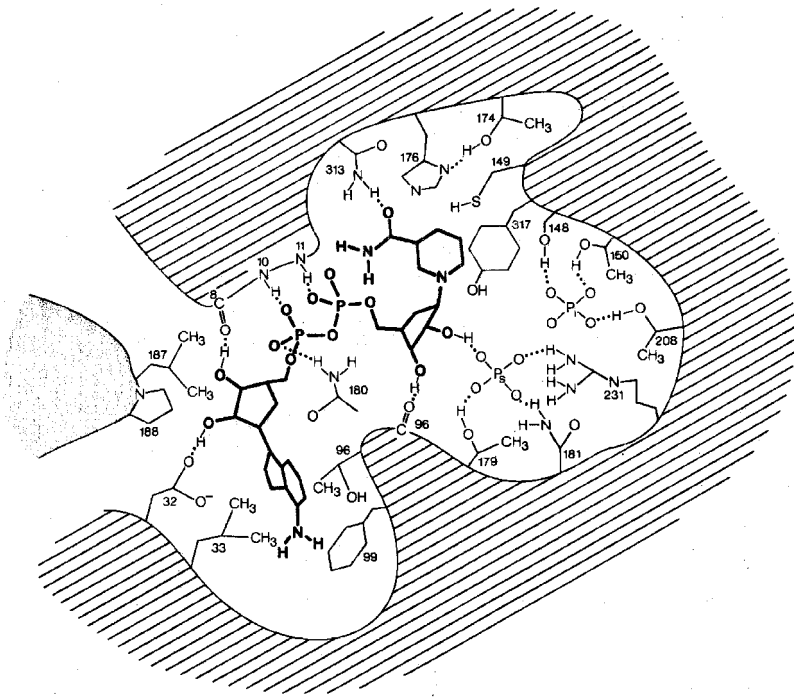
- The first step in the reaction sequence is the formation of a hemithioacetal between the cysteine and the substrate. This has the effect of converting the carbonyl group, which is not easy to oxidize directly, into an alcohol that is readily dehydrogenated by the usual procedure (second reaction).
- The thioester that is formed in the second reaction reacts with orthophosphate to give the acylphosphate (Last reaction).
- The acyl transfer is very slow unless NAD+ is bound to the enzyme. Therefore, the replacement of NADH by NAD+ in reactions fourth and fifth is a necessary part of the reaction sequence.
- The dissociation of the complex of the acylenzyme and NADH(Fourth reaction) is the rate-determining step in the sequence at saturating reagent concentrations at high pH. A consequence of this replacement of NADH by NAD+ before the release of acylphosphate is that the free apoenzyme does not take part in the

reaction. Also, because acylation of the enzyme by the diphosphate is activated by NAD+, the holoenzyme initiates the reductive dephosphorylation of 1,3-diphosphoglycerate.

a. The structures of the enzyme-substrate complexes

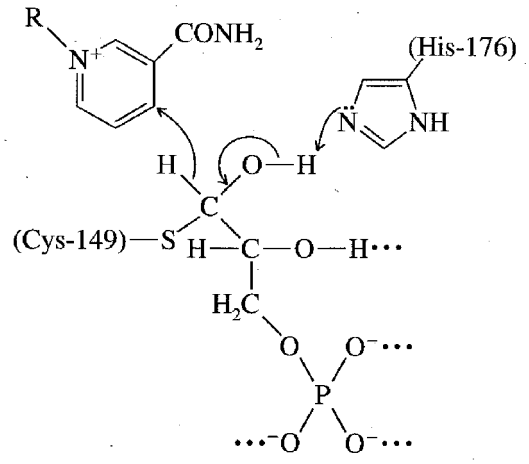
The crystal structures of the holoenzymes from lobster and Bacillus stearothermophilus were solved at 2.9 and 2.7 A, respectively; the human holoenzyme was solved at low resolution. The lobster apoenzyme was solved at 3.0-A resolution. The structures of the enzyme - substrate complexes were deduced from model-building experiments on the lobster and bacterial enzymes.

Two binding sites for sulfate ions were identified at the active site of the enzyme that had been crystallized from ammonium sulfate (Refer figure below).



A chemically and stereochemically reasonable model for the course of the reaction is constructed by assuming that these are the binding sites for the phosphate residue of the substrate and the nucleophilic phosphate in the deacylation reaction. The aldehyde group of the substrate can form a hemithioacetal with Cys-149 when the 3-phosphate is placed to make hydrogen bonds with the hydroxyl of Thr-179, the positively charged side chain of Arg-231, and the 2'-hydroxyl of the ribose ring that is attached to

the nicotinamide of NAD+. The C-2 hydroxyl of the substrate can then form a hydrogen bond with Ser-148, while the C-1 hydroxyl forms one with a nitrogen of His-176. These interactions orient the substrate so that the C-1 hydrogen points toward the C-4 position of the nicotinamide ring, less than 3 A away. In this mode of binding, the dehydrogenation reaction may take place as described earlier for lactate dehydrogenase, with His-176 as the general-base catalyst(Refer the figure below).



The transition state for the attack of orthophosphate on the thioester can be stabilized by hydrogen bonds to the attacking phosphate from the hydroxyl of Ser-148, the hydroxyl of Thr-150, and the C-2 hydroxyl of the substrate, and also from the amido nitrogens of Cys-149 and Thr-150. The presence of this specific binding site for phosphate explains phosphorylation of thioester rather than being hydrolyzed. The sulfur atom of the thioester is close enough to the C-4 carbon of the nicotinamide ring of NAD+ to be polarized by its positive charge. This perhaps explains the activation of the acyl transfer reactions on NAD+ binding.

b. The symmetry of the enzyme and the cooperativity ofligand binding

The binding of NAD+ to the enzyme is strongly cooperative and there is strong negative cooperativity in the binding to the rabbit muscle and bacterial enzymes, although there is positive cooperativity in the binding to the yeast enzyme at some temperatures. Glyceraldehyde 3-phosphate, on the other hand, binds independently to all four subunits.

The proteases

The proteases were the first enzymes to be discovered because of their abundance in the digestive system. But they have a very wide range of important biological activities because they are involved in processing proteins in general. For example,

- The activities of some hormones are modulated by proteolysis; e.g., the angiotensin converting enzyme, which is a metalloprotease.
- The blood clotting cascade involves a series of serine proteases. Activation of the immune system also uses serine proteases.
- Metalloproteases and thiol proteases are involved in the destruction of cells.
- Proteases are used to punch holes in cell walls in processes varying from fertilization to the formation of metastases.
- Viruses encode proteases that act in their maturation:
    - adenoviruses encode a cysteine protease;
    - picornaviruses, a serine protease; and
    - retroviruses, such as HIV, an aspartyl (acid protease).
- Proteases are the targets for many drugs, some of which are mechanism-based inhibitors.

The proteases may be conveniently classified according to their activities and functional groups. The serine proteases are endopeptidases that have a reactive serine residue and pH optima around neutrality. The carboxyl or aspartyl proteases are endopeptidases that have catalytically important carboxylates and pH optima at low pH. The thiol proteases are endopeptidases that differ from the serine proteases by having reactive cysteine residues. The zinc proteases are metalloenzymes that function at neutral pH. Many of these proteases are small monomeric enzymes of Mr 15 000 to 35 000, readily amenable to kinetic and structural study.

The different types of proteases are

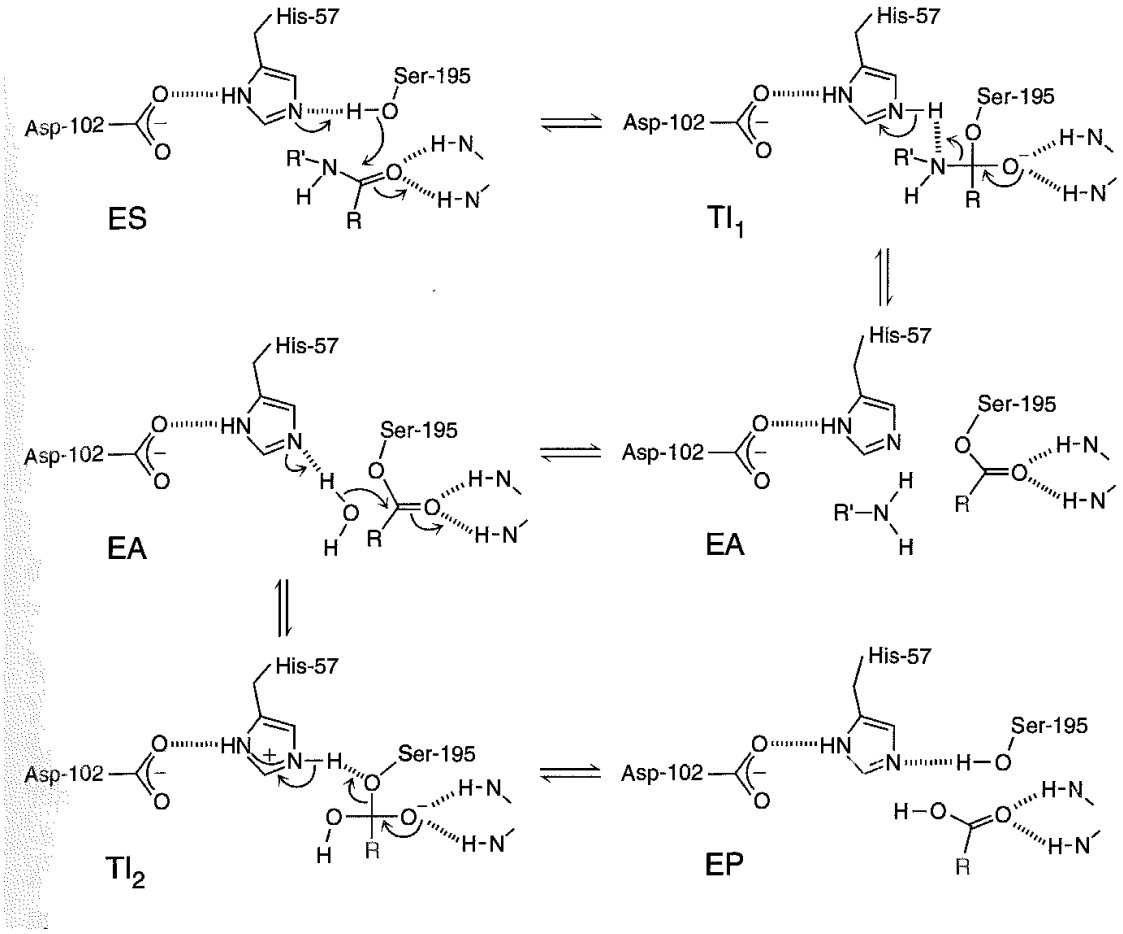
- Serine proteases
- Cysteine proteases
- Carboxyl proteases

Serine proteases

Peptide and synthetic ester substrates are hydrolyzed by the serine proteases by the acyl enzyme mechanism.

a. Mechanism of enzyme action-active site

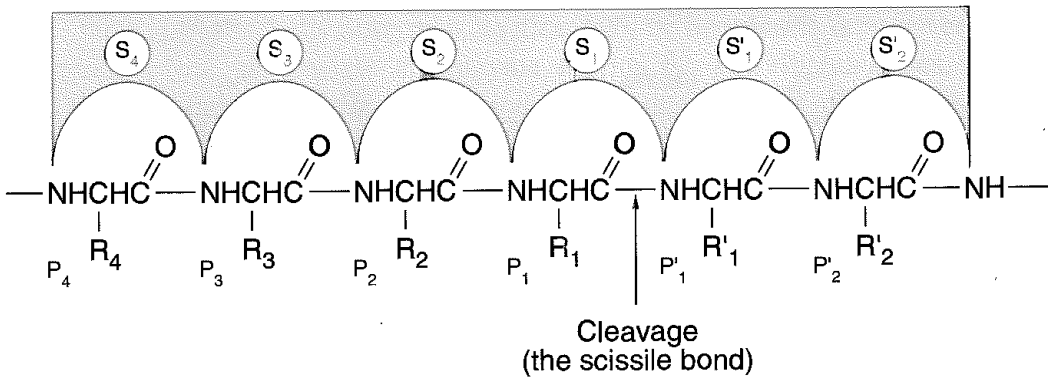
The enzyme and substrate first associate to form a non-covalent enzyme-substrate complex. This is followed by the attack of the hydroxyl of Ser-195 on the substrate to give the first tetrahedral intermediate. The intermediate then collapses to give the acylenzyme releasing the amine or alcohol. The acylenzyme then hydrolyzes to form the enzyme-product complex via a second tetrahedral intermediate.



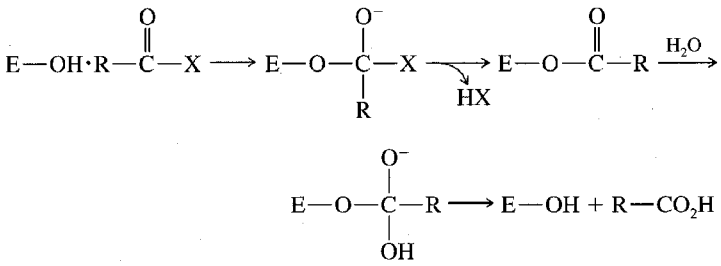
b. The structure of enzyme-substrate complex

The binding site for a polypeptide substrate consists of a series of subsites across the surface of the enzyme. By convention, they are labeled as below. The substrate residues are called P (for peptide); the subsites, S. Except at the primary binding site S_1

for the side chains of the aromatic substrates of chymotrypsin or the basic amino acid substrates of trypsin, there is no obvious, well-defined cleft or groove for substrate binding. The subsites run along the surface of the protein.



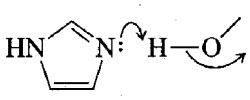
The hydrolysis of ester or amide substrates catalyzed by the serine proteases involves an acylenzyme intermediate in which the hydroxyl group of Ser-195 is acylated by the substrate. The formation of the acylenzyme is the slow step in the reaction of amide substrates at saturating concentrations, but the acylenzyme often accumulates in the hydrolysis of esters. The attack of Ser-195 on the carboxyl group of the substrate probably forms a high-energy tetrahedral intermediate (Refer equation below), but there is no direct evidence for it.



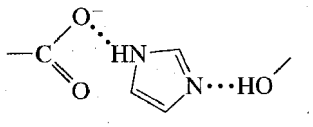
Crystallographic studies reveal that the active site of the enzyme is complementary in structure to the transition state of the reaction, a structure that is very close to the tetrahedral adduct of Ser-195 and the carbonyl carbon of the substrate.

a. The hydrogen-bond network at the active site

It has long been thought that the imidazole base of His-57 increases the nucleophilicity of the hydroxyl of Ser-195 by acting as a general-base catalyst (Refer mechanism below).



The carboxylate ion of the Asp 102 is also involved at the active site to give a catalytic traid.



The carboxyl group is buried inside the interior of the protein, it is surrounded by polar residues and water molecules. NMR and IR studies confirmed that the ionizing group is Histidine residue which facilitates the proton transfer between Asp 102 and His 57 while Asp-102 has an unusually low pK_a.

b. The contribution of the oxyonion binding site to catalysis

The protein engineering experiments on subtilisin indicated that, after mutagenesis of all three catalytic residues to alanine, there is a residual activity about 103 times higher than the spontaneous hydrolysis.This residual activity must result from enzyme-transition state complementarity. The complementarity is oxyanion of the tetrahedral intermediate that is formed by the attack of Ser-195 on the substrate and that is solvated by two backbone NH groups.

Protein engineering studies are difficult to study but the vibrational spectroscopy has given some clues such as the resonance spectra of acylenzymes of serine proteases have a signal at about 1700 cm^{-1} that is attributable to the stretching of its C=O group.

Mutagenesis of one of the residues in the oxyanion binding site of subtilisin, Asn-155 —» Leu removed the spectral features associated with the polarized C=O group of the acylenzyme. A component of the stretching mode is the contribution of the valence bond structures,

$$> C{=}O \longleftrightarrow > C^{+}{-}O^{-}$$

The Zinc proteases

The digestive enzyme bovine pancreatic carboxypeptidase A, which is a metalloenzyme - containing one atom of zinc bound to its single polypeptide chain of 307 amino acids and Mr 34 472. It is an exopeptidase, which catalyzes the hydrolysis of C-terminal amino acids from polypeptide substrates, and is specific for the large hydrophobic amino acids such as phenylalanine.

carboxypeptidase B catalyzes the hydrolysis of C-terminal lysine and arginine residues. The two enzymes are structurally almost identical, except that the B form has an aspartate residue that binds the positively charged side chain of the substrate.

Other classes of zinc proteases are

- Collagenase
- Angiotensin –converting enzyme
- Thermolysin

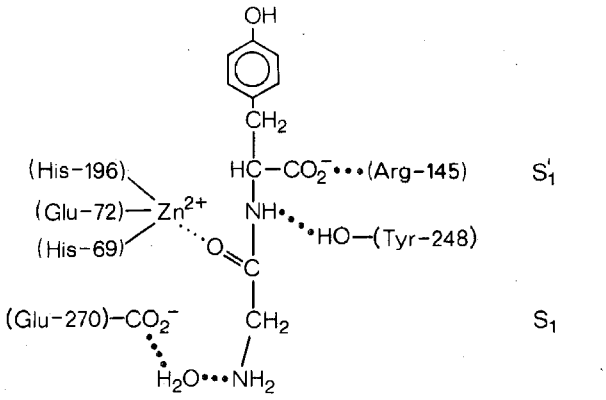
There is no direct evidence for the intermediates and chemical mechanism of the zinc proteases and the Zn^{2+} ion acts as electrophile to polarize > C=O group of a substrate and/or as a source of metal-bound nucleophilic ^{-}OH ions.

a. The structure of the active site of carboxypeptidase A

The crystal structure of the enzyme was solved very early at 2.0-A resolution. The active site consists of a shallow groove on the surface of the enzyme leading to a deep pocket, lined with aliphatic and polar side chains and parts of the polypeptide chain, for binding the C-terminal amino acid.

The catalytically important zinc ion is ligated by the basic side chains of Glu-72, His-196, and His-69. In few molecules, the phenolic oxygen of Tyr-248 is a fourth ligand. The side chain of this residue is conformationally very mobile. The phenolic side chain may rotate about its C_{α}—C_{β} bond. This residue is also so mobile in carboxypeptidase B that its position cannot be determined in the crystal structure. The fourth ligand of the zinc ion is normally a water molecule or a hydroxide ion.

A model was proposed for the structure of a productively bound enzyme-substrate complex, extrapolated from the structures of inert complexes (Refer figure below).



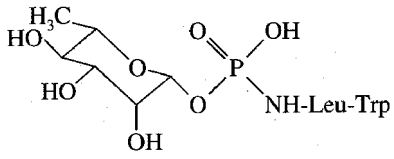
Tyr-248 had been implicated in catalysis from early chemical modification experiments. But one of the first protein engineering experiments, the mutation Tyr-248 to Phe, showed that it is not involved.

b. The structure of the active site of thermolysin

Thermolysin is an exopeptidase whereas the carboxypeptidases are an endopeptidases. The nature of the binding site instead of the deep pocket of the carboxypeptidases, thermolysin has an open expended cleft that can bind the polypeptide chain of the substrate on both sides of the scissile bond. There are also important differences in the catalytic apparatus;

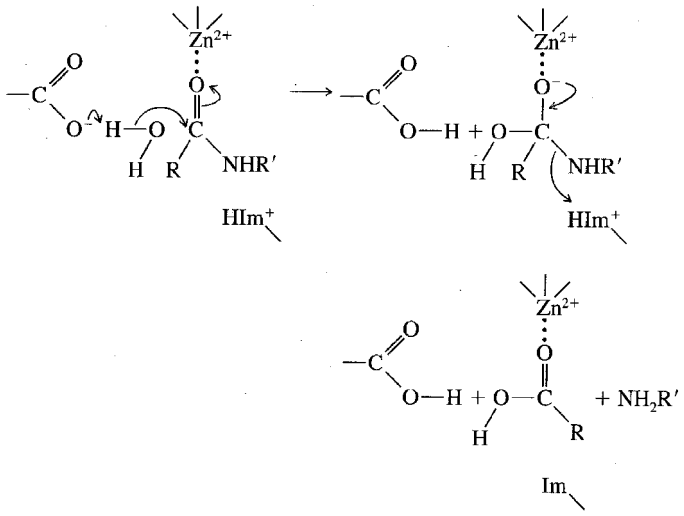
- The catalytic glutamate residue (Glu-143) is located at the bottom of a narrow cleft, where it is bound to a water molecule and cannot approach the substrate as readily as can Glu-270 in carboxypeptidase.
- There is no tyrosine residue corresponding to Tyr-248. Instead, a histidine residue, His-231, is located to be a general-acid-base catalyst in the reaction.

The crystal structure of the complex between the enzyme and a tight-binding inhibitor, phosphoramidon (Refer figure below), has been solved.

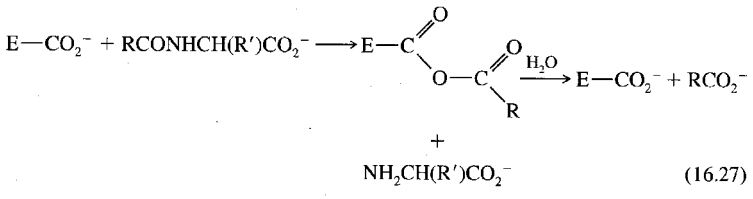


The phosphoryl group mimics a carbonyl group to which a water molecule is added to generate a tetrahedral intermediate. A phosphoryl oxygen displaces the molecule of water that is bound to the Zn2+ ion and becomes its fourth ligand, at a distance of ~ 2.0A°. The hydroxyl oxygen bonded to the phosphorus atom is 2.6A° from an oxygen atom of Glu-143, and is hydrogen-bonded to it. The phosphoramide nitrogen is 4.1A° from a nitrogen atom of His-231.

A likely reaction mechanism is immediately suggested by this structure, since it appears to be a transition state analogue for the Glu-143 general-base-catalyzed attack of a water molecule on a carbonyl group that has been polarized by binding to the zinc ion. Thus His-231 acts as an efficient proton donors.



An alternative catalytic role for. Glu-143 is that it acts as a nucleophile (Refer figure below). It is possible to alkylate it with active-site-directed a-chloro-ketone irreversible inhibitors. But the steric inaccessibility of the carboxylate group leads to such distortion of the enzyme on the formation of covalent bonds that nucleophilic attack on a substrate is most unlikely.



(16.27)

c. The reaction mechanism of carboxypeptidase

An important difference between thermolysin and carboxypeptidase leads to the major uncertainty in the mechanism of carboxypeptidase. This difference is that the catalytic carboxylate of carboxypeptidase is far more sterically accessible. It is unclear that the carboxypeptidase-catalyzed hydrolysis of peptides proceeds via general-base catalysis, or via nucleophilic catalysis. Early kinetic work concentrated on establishing the participation of the various groups in catalysis.

The ionized carboxylate of Glu-270 is implicated in catalysis from the pH-rate profile where the zinc ion acts as an electrophilic catalyst to polarize the carbonyl group and stabilize the negative charge that develops on the oxygen.

Evidence suggests the covalent mechanism ie general base catalysed mechanisms for thermolysin and carboxypeptidase A where His-231 is the general base for thermolysin and the carboxy-terminal carboxylate for carboxypeptidase A with common feature of Zn^{2+} acting as a lewis acid to polarize the substrate.

d. The zymogen

Procarboxypeptidase A is activated by the removal of a peptide of some 64 residues from the N-terminus by trypsin. This zymogen has significant catalytic activity. As well as catalyzing the hydrolysis of small esters and peptides, procarboxypeptidase removes the C-termmal leucine from lysozyme only seven times more slowly than does carboxypeptidase. In contrast to the situation in chymotrypsinogen, the binding site clearly pre-exists in procarboxypeptidase.

The carboxyl (aspartyl) proteases

The carboxyl proteases have two catalytically essential aspartate residues which are active at low pH. The best-known member of the family is pepsin. Other members are chymosin (rennin); cathepsin D; Rhizopus-pepsin (from Rhizopus chinensis); penicillinopepsin (from Penicillium janthinellum); the enzyme from Endothia parasitica; and renin, which is involved in the regulation of blood pressure. These constitute a homologous family, and all have an M_r of about 35 000. The aspartyl proteases have been thrown into prominence by the discovery of a retroviral subfamily, including one from

HIV that is the target of therapy for AIDS. These are homodimers of subunits of about 100 residues. All the aspartyl proteases contain the two essential aspartyl residues.

Pepsin

Pepsin consists of a single polypeptide chain of molecular weight 34 644 and 327 amino acid residues. Ser-68 is phosphorylated, but is removed without significantly altering the catalytic properties of the enzyme. The active site is an extended area that can accommodate at least four or five, and maybe as many as seven, substrate residues. The enzyme has a preference for hydrophobic amino acids on either side of the scissile bond.

The bond cleavages in proteins show that there is a specificity for leucine, phenylalanine, tryptophan, and glutamate (!) in the S_1 site, and for tryptophan, tyrosine, isoleucine, and phenylalanine in the S_J site. Pepsin rarely catalyzes the hydrolysis of esters, the exceptions being esters of L-β-phenyllactic acid and some sulfite esters.

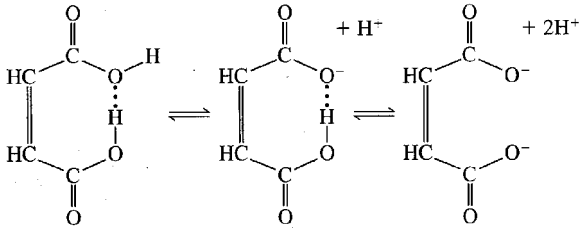
There are two catalytically active residues in pepsin: Asp-32 and Asp-215. Their ionizations are seen in the pH-activity profile with an optimum at pH 2 to 3, and depends upon the acidic form of a group of pKa — 4.5 and the basic form of a group of pKa ~ 1.1. The pKa values have been assigned from the reactions of irreversible inhibitors that are designed to react specifically with ionized or un-ionized carboxyl groups. Diazo compounds—such as N-diazoacetyl-L-phenylalanine methyl ester, which reacts with un-ionized carboxyls—react specifically with Asp-215 up to pH 5 or so (Refer figure below).

$$\text{(Asp-215)}\text{—}CO_2H + N_2CHCONHCH(Ph)CO_2CH_3 \longrightarrow$$
$$\text{(Asp-215)}\text{—}CO_2CH_2CONHCH(Ph)CO_2CH_3 + N_2$$

Epoxides, which react specifically with ionized carboxyls, modify Asp-32 (Refer figure below).

$$\text{(Asp-32)}\text{—}CO_2^- + CH_2\text{—}CHR \longrightarrow \text{(Asp-32)}\text{—}CO_2CH_2CHR \quad (OH)$$

The pH dependence of the rate of modification shows that the pKa of Asp-32 is less than 3. It is seen in the high-resolution crystal structures that the carboxyl groups of the two aspartate residues are hydrogen-bonded to each other. This is similar to the ionization of maleic acid, which has pKa values of 1.9 and 6.2 (Refer figure below).

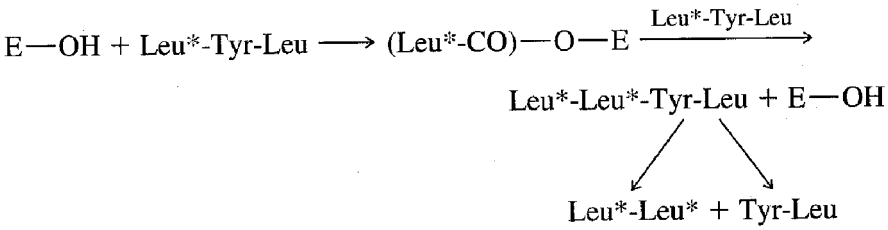


1. The chemical reaction mechanism of pepsin.

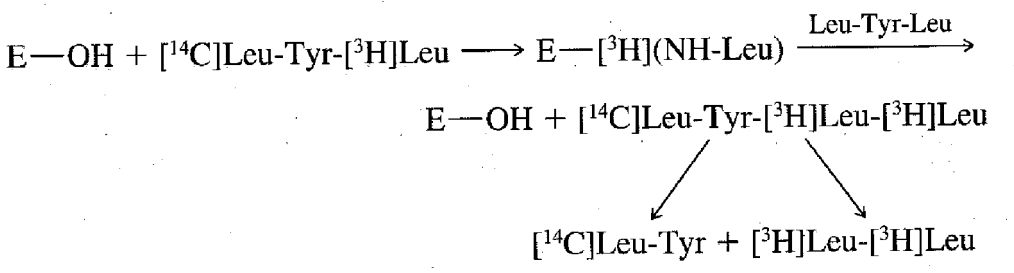
Evidences support the idea that both an acylenzyme and an aminoenzyme intermediates were formed during the reaction. Isotope studies are incorrect. Transpeptidation studies were found to be consistent with the absence of these intermediates. The interpretation of these experiments is quite instructive and will be discussed below.

Transpeptidation experiments.

The pepsin-catalyzed hydrolysis of Leu-Tyr-Leu gives the product Leu-Leu, which can be formed from the acyl transfer shown below



This experiment has been extended by using the double-labeled substrate [^{14}C] Leu-Tyr-[^{3}H]Leu to show that simultaneous amino and acyl transfer could take place. Both [^{3}H]Leu-[^{3}H]Leu and [^{14}C]Leu-[^{14}C]Leu are formed. The ^{14}C-labeled product comes from the acyl transfer route, whereas the ^{3}H-labeled product comes from the [^{14}C]Leu-Tyr-[^{3}H]Leu-[^{3}H]Leu produced from an aminoenzyme by mechanism (Refer the equation below), where E-[^{3}H](NH-Leu) is [^{3}H]Leu bound to E by the NH group of the amino acid.



There are various alternative possible explanations for the aminoenzyme and acyl enzyme intermediates such as

- The amino group in the "aminoenzyme" may not be covalently bound, but is merely be "activated" by the enzyme; and
- Similarly, the acyl transfer reaction of the above transpeptidation equation could occur by the direct attack of Leu-Tyr-Leu on the enzyme-bound Leu-Tyr-Leu.
- Other interpretations include small peptides stimulate the pepsin-catalyzed hydrolysis of other peptides by being first synthesized into larger peptides in a condensation reaction that is the reverse of the hydrolytic step; e.g., Refer equations below

$$\text{Z-Ala-Leu} + \text{Phe-Trp-NH}_2 \rightleftharpoons \text{Z-Ala-Leu-Phe-Trp-NH}_2$$

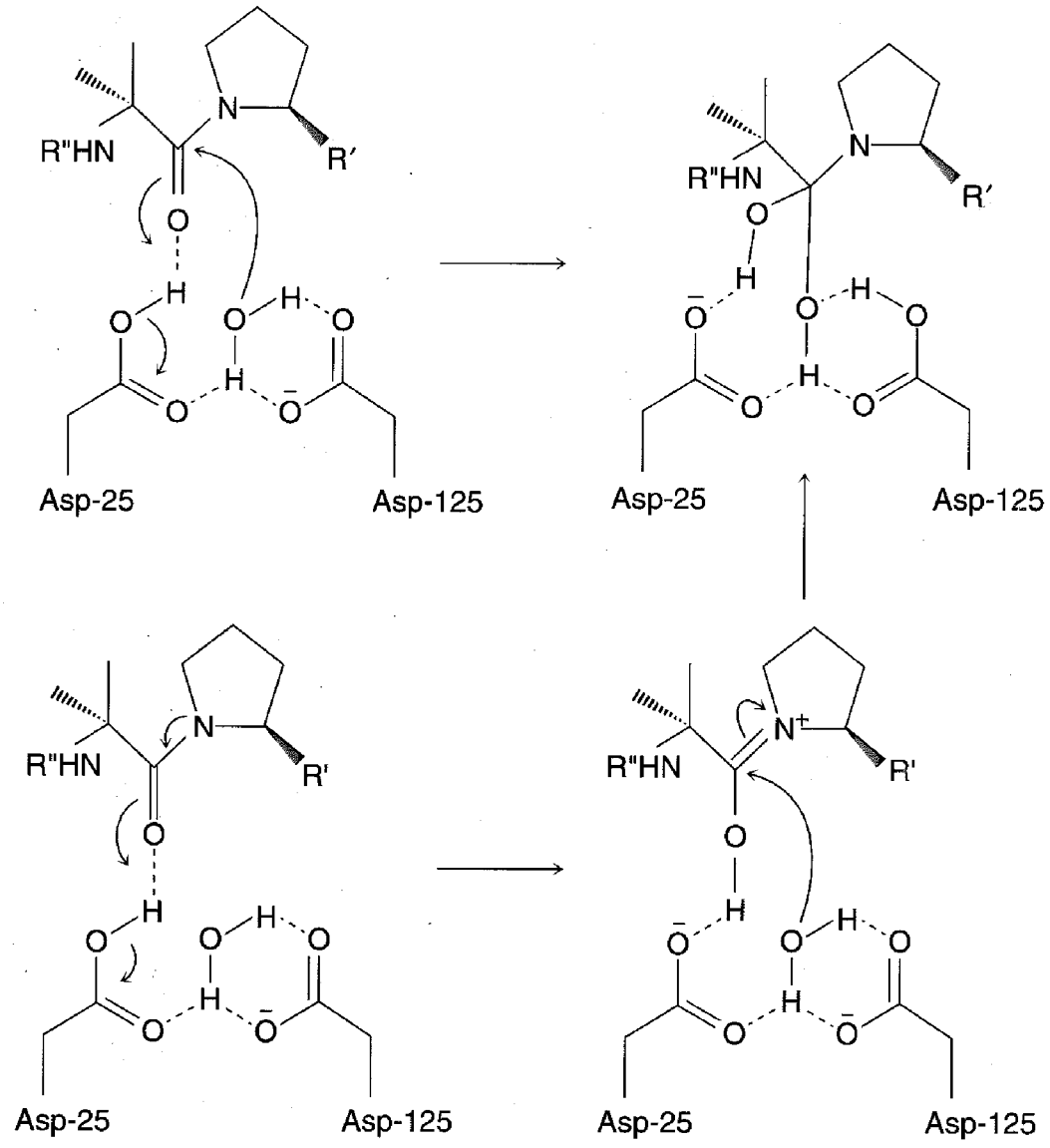
$$\text{Z-Ala-Leu-Phe-Trp-NH}_2 \longrightarrow \text{Trp-NH}_2 + \text{Z-Ala-Leu-Phe}$$

$$\text{Z-Ala-Leu-Phe} \longrightarrow \text{Z-Ala-Leu} + \text{Phe}$$

The rate of condensation of two small peptides to give a larger peptide is relatively faster compared with hydrolysis of the small peptide during conditions such as; pepsin has an extended active site so that long peptides react far more rapidly than short ones. Therefore, at high concentrations of small peptides, the reaction products of a reactive larger peptide, the rate of condensation of the small peptides should be comparable with the rate of hydrolysis of the larger peptide, and faster than the rate of hydrolysis of the small peptides. The reaction products in the transpeptidation equations may be generated by the prior condensation of two molecules of Leu-Tyr-Leu, followed by various modes of hydrolysis and condensation.

Mechanism for carboxy peptidase

The mechanism in Figure below fits the structural and kinetic data. There is formation of a terahedral intermediate in which both oxygen atoms become protonated. In the intermediate, two oxygen atoms are equivalent and sufficiently long-lived that exchange can take place in the solvent exchange experiments. Asp-25 acts as a general acid and Asp-125 as a general base. On the basis of ^{15}N-kinetic isotope effects on the hydrolysis of Ac-Ser-Gln-Asn-Tyr-Pro-Val-Val-NH2, it has been suggested that the first step in this reaction may be the isomerization of the peptide bond (Refer figure below, bottom).



The zymogen.

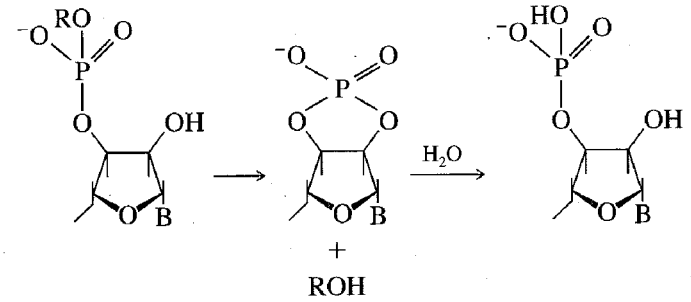
Pepsin is formed from pepsinogen by the proteolysis of 44 residues from the N-terminus. The zymogen is stable at neutral pH, but below pH 5 it rapidly and spontaneously activates. The activation process takes place by two separate routes,

- A pepsin-catalyzed and
- An intramolecularly catalyzed process.

There is much evidence that pepsinogen may activate itself in a unimolecular process, the active site cleaving the N-terminus of its own polypeptide chain. Exposure to pH 2, the pepsinogen spontaneously activates. The two routes for activation compete, with the bimolecular activation dominating at higher zymogen concentrations and above pH 2.5, and the intramolecular activation dominating at low pH. The result of this spontaneous activation of the pure zymogen is that the majority of pepsinogen molecules will be active 10 s after mixing with the hydrochloric acid in the stomach.

Ribonuclease

Bovine pancreatic ribonuclease catalyzes the hydrolysis of RNA by a two- step process in which a cyclic phosphate intermediate is formed (Refer figure below).



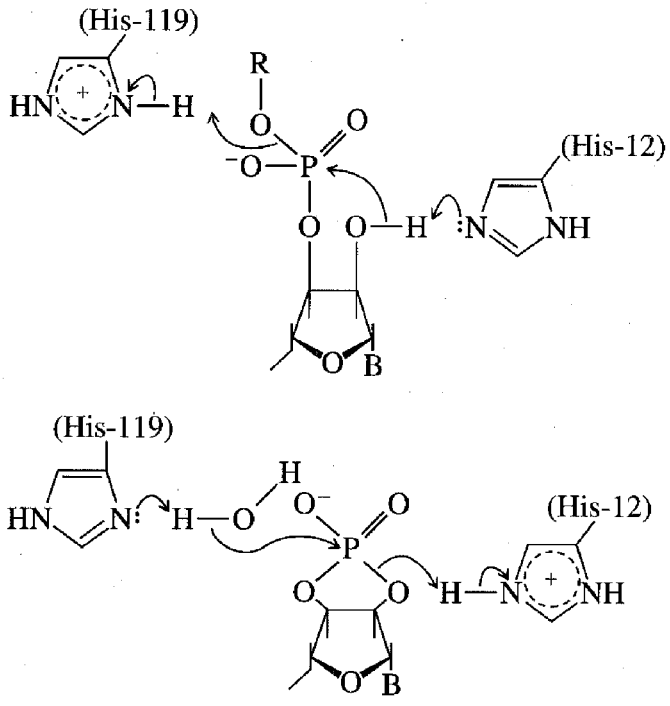
The cyclization step is usually much faster than the subsequent hydrolysis making the intermediate may be readily isolated. DNA is not hydrolyzed, as it lacks the 2'-hydroxyl group that is essential for this reaction. There is a strong specificity for the base B on the 3' side of the substrate to be a pyrimidine—uracil or cytosine.

The enzyme consists of a single polypeptide chain of M_r 13 680 and 124 ammo acid residues. The bond between Ala-20 and Ser-21 may be cleaved by subtilisin, but the peptide remains attached to the rest of the protein by noncovalent bonds. The modified protein, called ribonuclease S, and the native protein is now termed as ribonuclease A, have identical catalytic activities. Ribonuclease is very amenable to physical and chemical study and was the first enzyme to be sequenced. The crystal structures of both forms of the enzyme were solved at 2.0-A resolution several years ago.From the NMR studies, it has been interpreted that the catalytic activity depends on the ionizations of two histidine residues.

Bovine pancreatic RNase A is a member of a homologous superfamily. In addition, there is a separate family of guanine-specific microbial RNases that have evolved to have a similar active site. Ribonuclease Tl from Aspergillus oryzae and the 110-residue barnase from Bacillus amyloliquefaciens of Mr 12 392 are the best known examples.

The currently accepted chemical mechanism for the reaction of RNase A was deduced by an inspired piece of chemical intuition before the crystal structure was solved. It was found that the pH-activity curve is bell-shaped, with optimal rates around neutrality. The pH dependence of K_{cat}/K_M shows that the rate depends upon the ionization

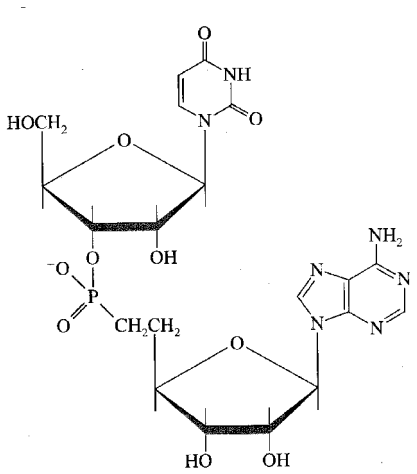
of a base of pKa 5.22 and an acid of pKa 6.78 in the free enzyme, whereas the pH dependence of K_{cat} shows that these are perturbed to pKa values of 6.3 and 8.1 in the enzyme-substrate complex. It was proposed that the reaction is catalyzed by concerted general-acid-general-base catalysis by two histidine residues, later identified as His-12 and His-119 (Refer the two figures below)



In the cyclization step, His-12 acts as a general-base catalyst and His-119 acts as a general acid to protonate the leaving group. Their catalytic roles are reversed in the hydrolysis step: His-119 activates the attack of water by general-base catalysis and His-12 is the acid catalyst, protonating the leaving group.

1. The structures of ribonuclease A and its complexes

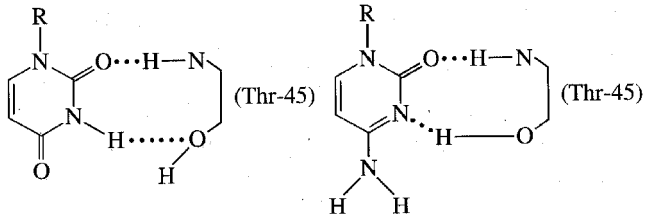
Ribonuclease has a well-defined binding cleft for the substrate. In it are located His-12, His-119, and the side chains of Lys-7, Lys-41, and Lys-66. The structure of the enzyme-substrate complex for the cyclization step was first deduced from the crystal structure of the enzyme and the substrate analogue UpcA, (Refer figure below)



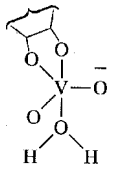
the phosphonate analogue of UpA. It is a very good analogue, differing from the real substrate only in that a —CH2— group replaces an oxygen atom, so that the structure of its complex with ribonuclease should be close to that of a productively bound enzyme-substrate complex. It was found that His-119 is within hydrogen-bonding distance of the leaving group, and His-12 within hydrogen-bonding distance of the 2'-hydroxyl of the pyrimidine ribose. The pKa values of His-12 and His-119 have been determined by NMR measurements to be 5.8 and 6.2, respectively.

The structures of the native enzyme and its complexes with several inhibitors have since been obtained at higher resolution in other laboratories, to deduce a more complete description of the enzyme - substrate interactions.

The lysine residues 7,41, and 66 play an important part of the catalytic machinery. Lys-41 is particularly important. The lysine side chains are very mobile in the free enzyme, but their mobilities are much decreased on the binding of nucleotide substrate analogues. Lys-41 interacts directly with the phosphate moiety and is thought to stabilize the pentacovalent intermediate. Another residue that has been conserved through evolution is Thr-45. This residue is responsible for the specificity of the enzyme for pyrimidines (on the 3' side). Its backbone NH group and -OH side chain are able to form complementary hydrogen bonds with both uracil and cytosine (Refer figure below).



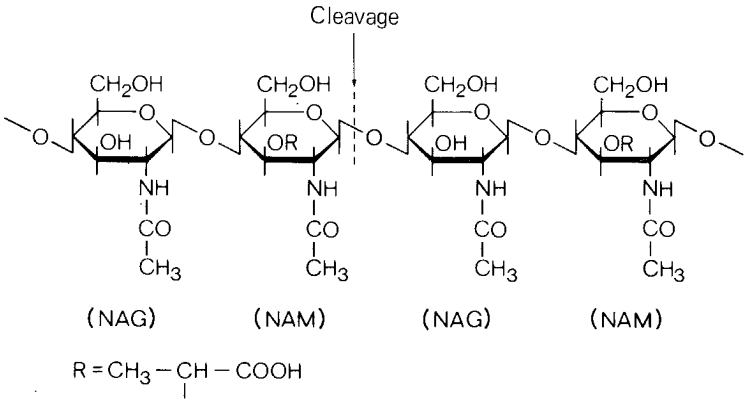
The most interesting crystal structure solved is that of the complex with uridine vanadate (Refer figure below),



by a combined neutron and x-ray diffraction study, and most recently refined to 1.3 A resolution. Uridine vanadate is a transition state analogue, mimicking the pentacovalent intermediate or transition state. An apical and an equatorial oxygen are donated by the ribose, and an apical water molecule is the sixth ligand. Lys-41 and His-12 are transposed from what is expected from the in-line mechanism of equation 16.36: His-12 is coordinated to an equatorial rather than an apical oxygen and Lys-41 to an apical rather than an axial. Some NMR evidence suggests that His-12 is in a different position in the solution structure from that in the crystal. A quantum mechanical analysis of the structure of the uridine vanadate complex suggests that His-12 may have an additional role in stabilizing the pentacovalent intermediate (or transition state) by acting as a conduit for proton transfers.

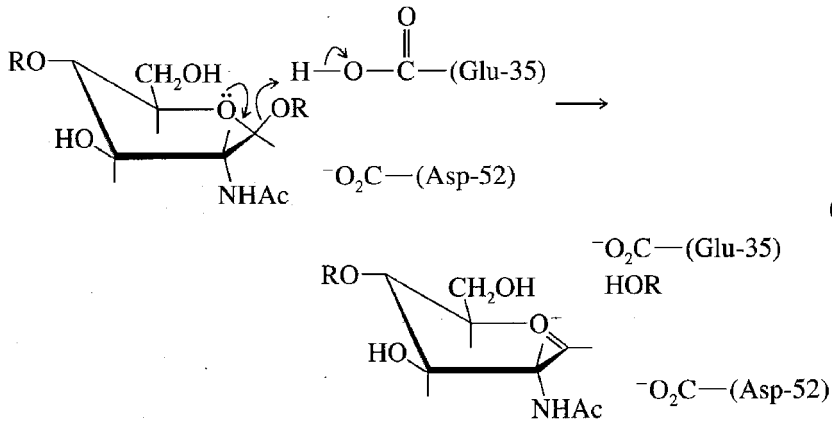
Lysozyme

Lysozyme catalyzes the hydrolysis of a polysaccharide that is the major constituent of the cell wall of certain bacteria. The polymer is formed from β(1 —»4)-linked alternating units of *N*-acetylglucosamine (NAG) and *N*-acetyl-muramic acid (NAM) (Refer figure below).



Hen egg white lysozyme is a small protein of M_r 14 500 and 129 amino acid residues. Hen egg white lysozyme has the distinction of being the first enzyme to have had its structure solved by x-ray crystallography. It is an atypical member of the hexosaminidase class of glycosyl transfer enzymes. It catalyzes the hydrolysis of substrates with retention of stereochemistry. T4 lysozyme was for many years thought to have the same fold and mechanism of lysozyme. But it has now been found that the T4 enzyme has inversion of configuration and so operates by a different mechanism. A mechanism proposed for the enzymatic reaction was based on the structure of the active site and on ideas from physical organic chemistry. This mechanism consisted of the following points:

- There are six subsites—labeled A, B, C, D, E, and F—for binding the glucopyranose rings of the substrate.
- The scissile bond lies between sites D and E, close to the carboxyl groups of Glu-35 and Asp-52.
- The reaction proceeds via an oxocarbenium ion intermediate that is stabilized by the ionized carboxylate of Asp-52 (Refer figure below).

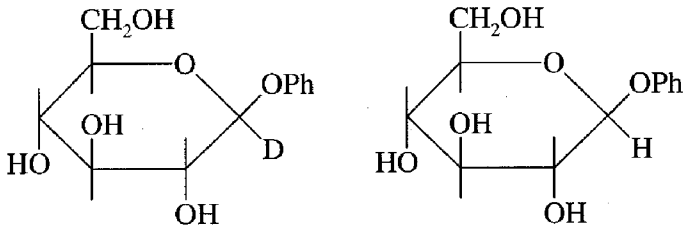


- The expulsion of the alcohol is general-acid-catalyzed by the un-ionized carboxyl of Glu-35.
- Further, as discussed in Chapter 12, sections B2a and D5c, the sugar ring in site D is distorted to the sofa conformation expected for a carbonium ion.

Small polysaccharides avoid the strain in the D subsite by binding the A, B, and C sites.

1. The oxocarbenium ion

Alternatives to the oxocarbenium ion mechanism are the direct attack of water on the substrate, and the nucleophilic attack of Asp-52 on the C-l carbon to give an ester intermediate. The single-displacement reaction has been ruled out by showing that the reaction proceeds with retention of configuration. The oxocarbenium ion or SN_1 mechanism has been substantiated by secondary isotope effects, using substrates containing either deuterium or tritium instead of hydrogen attached to the C-1 carbon. For example, K_H/k_D for structures below is 1.11,



compared with values of 1.14 found for a carbonium ion reaction and 1.03 for a bimolecular displacement (SN_2) in simple chemical models. The $^{16}O/^{18}O$ kinetic isotope effect for the leaving group oxygen atom indicates high C—O bond fission in the transition state. It is very difficult to distinguish whether there is a true oxocarbenium ion intermediate or there is an oxocarbenium-like transition state.

Protein stability

One of the great unsolved problems of science is the prediction of the 3-D structure of a protein from its amino acid sequence: the "folding problem." The primary objective is the prediction of the catalytic activity of an enzyme from its amino acid sequence because;

- First, the acquisition of sequence data by DNA sequencing is relatively quick, and vast quantities of data have become available through human genome project and other genome sequencing projects. But acquisition of three-dimensional data is still slow and is limited to proteins that either crystallize in a suitable form or are sufficiently small and soluble to be solved by NMR in solution. Algorithms are thus required to translate the linear information into spatial information.
- Second, we are now able to synthesize proteins by way of their genes, and so the production of new enzymes with specified catalytic activities is a challenging prospect.

Synthesizing new enzymes requires five underpinning and interrelated abilities:

1. the ability to predict the most stable fold of a particular sequence;
2. the ability to design a novel fold;
3. the ability to predict whether the desired fold is kinetically accessible;
4. the ability to design the precise features for specific binding in the fold; and
5. the ability to design the precise orientation of groups in the protein for efficient catalytic function.

A denatured protein makes many interactions with solvent water. As the protein folds up, it exchanges those noncovalent interactions with others that it makes within itself: its hydrophobic side chains tend to pack with one another, and many of its hydrogen bond donors and acceptors pair with each other, especially those in the polypeptide backbone that form the hydrogen-bonded networks in helixes and sheets. Each interaction energy is small, but because of their large numbers, the total interaction energies in the native and denatured states are huge, depending on the size of the protein.

This tiny amount of energy is the difference between the free energies in the native and denatured states, and those states are almost balanced in energy. Thus, whether a protein folds depends crucially on a balance between two large numbers, each of which is very

difficult to calculate with precision. To predict the stability of a protein, we have to calculate not only the interaction energy between any two atoms within a protein, but also this energy relative to the interactions that the individual atoms make with water in the denatured state. Current potential functions are not sufficiently accurate for this purpose. But we can use protein engineering and other experimental procedures to make changes in existing proteins.

Protein denaturation

Proteins can be denatured by changing their physical or chemical environments. The most common methods are by

- Heating;
- Adding a chemical denaturant, such as urea or guanidinium chloride (GdmCl);
- Changing the pH to acidic or alkaline;
- Applying high pressure.

Many proteins, especially small ones, denature reversibly; they regain their native structures spontaneously when returned to conditions that favor folding. Many larger proteins denature irreversibly; they often aggregate and precipitate after denaturant has been removed or hot solutions are cooled. Even small, readily renaturable proteins tend to precipitate at high concentrations of their denatured states. The reversible equilibria for denaturation of proteins can be treated by thermodynamics in the same way as simple chemical equilibria, except that there are some added twists because of the multitude of weak noncovalent interactions.

Types of denaturation

- Heat denaturation
- Solvent denaturation
- Acid-base denaturation
- Cold denaturation

Thermodynamics of protein folding

The denatured state has considerable conformational freedom: it is not a rigid structure but individual segments of polypeptide chain can move relative to one another,

and groups can rotate about single bonds. The denatured state has an inherently high configurational entropy from the simple Boltzmann formula S = k In W. The native state is very conformationally restricted and has a low entropy. Thus, as a protein folds, it loses considerable entropy, which must be balanced by a gain of enthalpy for the free energy to favor folding. The enthalpy of packing of side chains in the native state is favorable and compensates, for its low entropy. The classic thermodynamic description of protein folding is that it has large negative values of ΔS and ΔH. The contribution of solvent has two important consequences.

- Entropy of release of water on folding
- The specific heat of unfolding
- Free energy of unfolding
- Cold unfolding
- Measuring thermal denaturation
- Measurement of ΔC_p from change of $\Delta H_{D\text{-}N}$ with pH

a. Entropy of release of water on folding

The native state of a protein has many of its hydrophobie side chains shielded from water because they are packed in hydrophobie cores. Conversely, the denatured state has many of its hydrophobic side chains exposed to solvent. The water molecules stack around these in "icebergs" as they maximize their hydrogen bonds with one another. This lowers the entropy of water, because the individual molecules have less freedom of movement, and lowers the enthalpy because more hydrogen bonds are made. Similarly, the hydrogen bond donors and acceptors in the polypeptide backbone of the denatured protein are largely exposed to solvent and tie down more water molecules. These water molecules are released as the protein folds, and the gain in entropy of water compensates considerably for the loss of conformational entropy.

b. The specific heat of unfolding

The specific heat plays an important role in the thermodynamics of protein folding. The change in specific heat, ΔCp, for most simple chemical reactions is very small because the reactants and products have similar values of Cp. But, the specific heat

of denatured states is particularly high, mainly because of the icebergs of solvent around the exposed hydrophobic groups. As the temperature is raised, the icebergs melt and their latent heat of fusion contributes significantly to Cp of the denatured state. Denatured protein has weak structured regions makes a sufficient contribution to the specific heat. The value does become distinctly lower at higher temperatures, due to melting of icebergs.

The high value of ΔCp has several important consequences such as enthalpy and entropy

The enthalpy of unfolding of a protein changes significantly with temperature. Consider that the protein denatures reversibly according to a simple two-state transition in which the denatured state D is in equilibrium with the native structure N. Then, according to classical thermodynamics, the value of $\Delta H_{(D-N)}$ at a temperature T_2 $\Delta H_{(D-N)\ (T2)}$ is related to that at another is related to another temperature T_1 $\Delta H_{(D-N)\ (T1)}$.

$$\Delta H_{\mathrm{D-N}(T_2)} = \Delta H_{\mathrm{D-N}(T_1)} + \Delta C_{\mathrm{p}}(T_2 - T_1)$$

As ΔC_p is positive, the enthalpy of unfolding becomes more positive as the temperature increases. The fact that as the enthalpy of the unfolded state increases the temperature increases because the icebergs surrounding the hydrophobic side chains melt progressively and have a fewer hydrogen bonds with one another.

The entropy of unfolding becomes higher with increasing temperature because fewer water molecules are tied down in the icebergs that surround the denatured state.

c. Free energy of unfolding

As the temperature increases, the term $T\Delta S_{D-N}$ in the below equation increases and causes the protein to unfold when it becomes greater ΔH_{D-N} i.e., when ΔG_{D-N} becomes negative. Thus, thermal unfolding is caused by the entropy of the unfolded state dominating at high temperature.

$$\Delta G_{\mathrm{D-N}(T_2)} = \Delta H_{\mathrm{D-N}(T_1)} + \Delta C_{\mathrm{p}}(T_2 - T_1) - T_2(\Delta S_{\mathrm{D-N}(T_1)} + \Delta C_{\mathrm{p}} \ln (T_2/T_1))$$

d. Cold unfolding

The high value of ΔC_p leads to the phenomenon of cold denaturation; as the temperature becomes lower, the enthalpy of unfolding decreases, according to the below equation and the free energy term becomes dominated again by the entropy component.

$$\Delta H_{D-N(T_2)} = \Delta H_{D-N(T_1)} + \Delta C_p(T_2 - T_1)$$

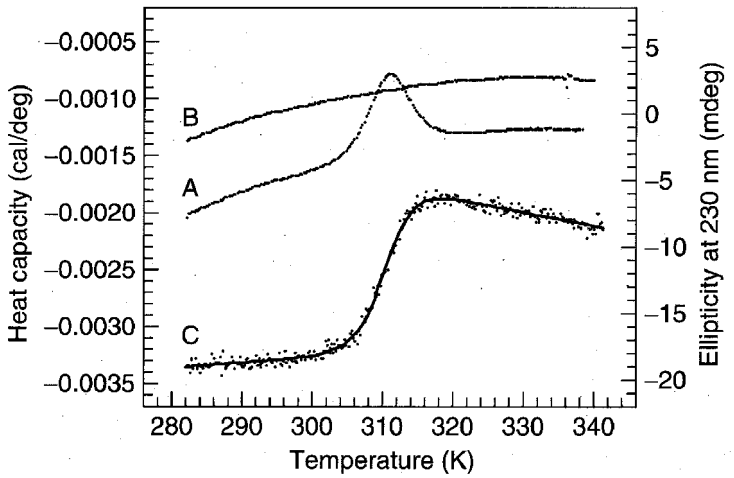
The melting temperature for cold denaturation, T_C is given by

$$T_c = \frac{T_m^2}{T_m + 2(\Delta H_{D-N}/\Delta C_p)} \approx T_m - \frac{2(\Delta H_{D-N})}{\Delta C_p}$$

Proteins with a small numerical value of ΔH_{D-N} and a large ΔCp are expected to cold unfold at higher temperatures.

e. Measuring thermal denaturation

The thermal unfolding of proteins is best measured by differential scanning calorimetry, which measures the heat absorbed by a protein as it is slowly heated through its melting transition (Refer figure below).



A solution of about 1 mg of protein in 1 mL of buffer and a separate reference sample of buffer alone are heated electrically. The additional current required to heat the protein solution is recorded. As the protein denatures, there is a large uptake of heat because the process is highly endothermic. The temperature at the maximum of the peak is the T_m, the melting temperature. The area under the curve is the enthalpy of unfolding, ΔH_{D-N}. The slope of the baseline at temperatures above the T_m is the product of the specific heat of the denatured state and its concentration. At the T_m, ΔG_{D-N} is zero

because there are equal concentrations of D and N, and ΔG_{D-N} = - RT In ([D]/[N]). Thus, the value of Tm equal to the ratio of its enthalpy and entropy of denaturation.

The equation for ΔG_{D-N} may be rearranged and ΔH_{D-N} and ΔS_{D-N} substituted to give

$$\frac{[D]}{([D] + [N])} = \frac{\exp(\{T\Delta S_{D-N} - \Delta H_{D-N}\}/RT)}{1 + \exp(\{T\Delta S_{D-N} - \Delta H_{D-N}\}/RT)}$$

The fraction unfolded [D]/([D]+[N]) is determined by spectroscopy and fitted to the above equation.

f. Measurement of ΔC_p from change of ΔH_{D-N} with pH

A convenient procedure to measure ΔC_p is to measure ΔH_{D-N} in increasingly acidic solutions. Proteins usually become progressively less stable with decreasing pH, and the T_m becomes progressively lower. The decrease in stability results from the protonation of carboxylic acid groups, and these are known to have small enthalpies of ionization. It is found that $\Delta H_{D-N(cal)}$ decreases with decreasing pH at acidic pH, and a plot of $\Delta H_{D-N(cal)}$ against Tm for each pH is linear. The slope of the plot is ΔC_p.

Solvent denaturation

These denaturants solubilize all the constituent parts of a protein, from its polypeptide backbone to its hydrophobic side chains. The free energy of transfer of the side chains and polypeptide backbone from water to solutions is linearly proportional to the concentration of denaturant. As the denatured state is more exposed to the solvent than the native state, the denatured state is stabilized preferentially by the denaturant. Thus the free energy of denaturation at any particular concentration of denaturant is given by

$$\Delta G_{D-N} = \Delta G^{H_2O}_{D-N} - m_{D-N}[\text{denaturant}]$$

Acid-or base induced denaturation

Most proteins denature when the pH is changed to very low or very high values. The primary reason for extremes of pH causing denaturation is that proteins usually have buried groups that have highly perturbed pK_a's.

Two-state versus multistate transitions

In simple two-state equilibrium of D⟷ N, protein denaturation is a cooperative transition in which denaturation occurs in a single step without any intermediates accumulating. There are many evidence to prove that protein folding takes place through intermediates and experimental methods are available to prove the presence of the intermediates.

a. Spectroscopic evidence for two state transition

A two-state transition is usually identified by all spectroscopic probes changing simultaneously as the equilibrium changes. The far ultraviolet circular dichroism signals, which is used to measure of secondary structure, whereas the near ultraviolet, is used to measure of tertiary structure. Fluorescence and near ultraviolet absorbance spectra also probe tertiary structure.

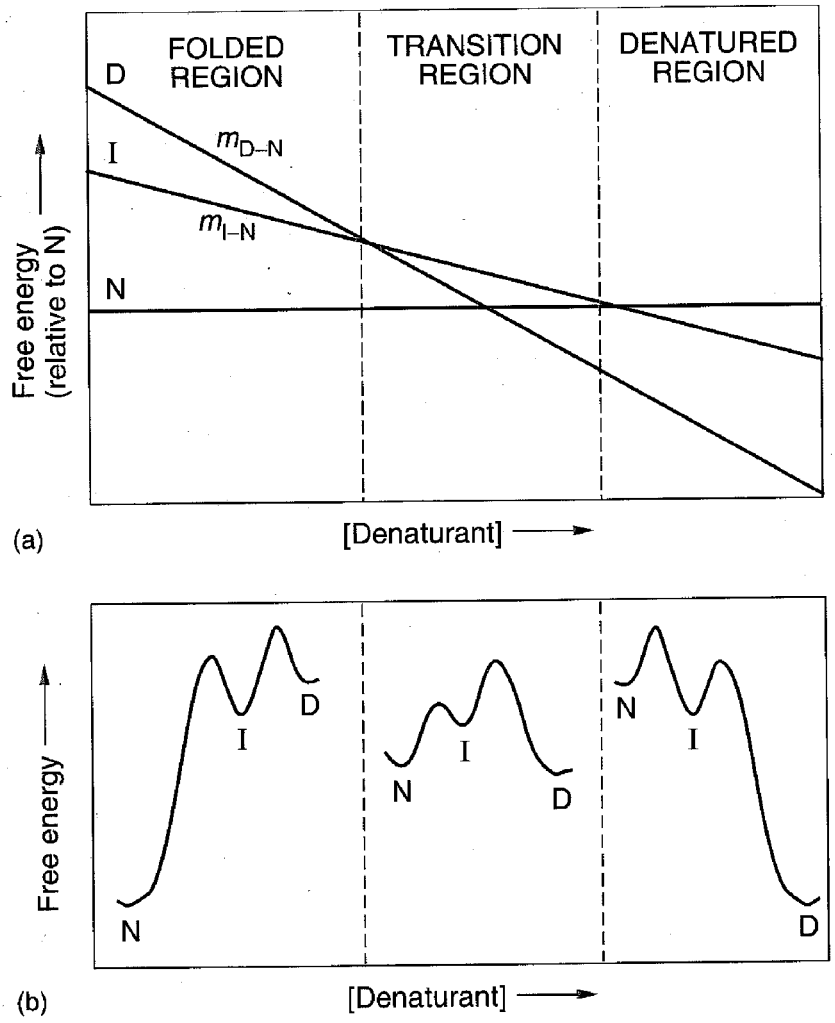
b. Enthalpy test for a two-state transition

The calorimetric methods are used to can detect additional intermediates during transitions. There are clear-cut cases where two or more phases are seen in melting curves, or thermograms, showing that intermediates are appearing. The difficult case is when only one transition is seen. Whether or not there is a single transition can be tested by measuring $\Delta H_{D\text{-}N(cal)}$ and $\Delta H_{D\text{-}N(vh)}$ for the transition. The ratio of $\Delta H_{D\text{-}N(vhl)}/\Delta H_{D\text{-}N(cal)}$ is the diagnostic test for the two-state intermediates. If the ratio is one, then no intermediates accumulate; there may be intermediates at very high energy. If the ratio is less than one, then an intermediate is present. If the ratio is greater than one, this indicates that aggregation or artifact is occurring.

d. Apparent two state transition

Intermediates that are present at equilibrium at low concentration will not be detected by most spectroscopic methods. Further, they will not be detected by calorimetry, either, if they do not build up during the thermal transition.

There are many proteins in which there are stable intermediates present in water, but these are not observed during the denaturation transition. For barnase, for example (Refer figure below), there is a folding intermediate (I) that is more stable than the highly denatured state (D) at low concentrations of denaturant.



But as the concentration of the denaturant is increased, D becomes progressively more stable than I. The stability of I varies with denaturant in a manner analogous to solvent denaturation equation, in which m_{I-N} replaces the term m_{D-N}. Because intermediates are more compact than D, $m_{D-N} > m_{I-N}$. It is found in the transition region for barnase, as for many proteins, that I is the highest energy species and so is not observed at equilibrium. The same is true during thermal denaturation.

d. Extrapolated thermodynamic quantities from apparent two state transitions

There are two important consequences from the above figure.

First, many proteins appear to follow two-state denaturation around the midpoint of folding, despite intermediates accumulating under more renaturing conditions.

Second, the measurements made under these conditions give information on the transition between D and N, and not on D and I. The extrapolation of equilibrium data from the transition region for denaturation in a solution of denaturant will give the thermodynamic properties of the denatured state. The same is true for data from a calorimetric experiment.

Stability-Activity trade off

Some enzymes from thermophiles that are very stable at normal temperatures have low activities at the lower temperatures. It has been speculated that there is a trade-off between stability and activity. But barnase that has been stabilized by 3.4 kcal/mol (4 kJ/mol) by a combination of six mutations that are external to its active site has no loss of activity.66 However, there is a compromise between stability and activity in the structure of the active site. There are several positions in the active site of barnase that can be mutated to give more stable but less active protein. Activity can even be increased further at an unacceptable expense to stability. Some of these changes can be rationalized. The active site of the ribonuclease is highly positively charged because it has to stabilize the highly negatively charged transition state generated at the pentacoordinate phosphate. The lysine, arginine, and histidine at the active site are mutually repulsive. Other regions of the active site have to be mobile, and so some hydrogen bonds are weak. Active sites of enzymes and binding sites of proteins are a general source of instability because they contain groups that are exposed to solvent in order to bind substrates and ligands and so are not paired with their normal types of part-partners. Stability-activity trade-off is also seen with residues in the natural polypeptide inhibitor of barnase, barstar, that has evolved to bind as rapidly as possible to barnase, and also in the active site of T4 lysozyme.

UNIT – IV KINETICS OF PROTEIN FOLDING

Basic methods- two state kinetics - multistate kinetics - transition states in protein folding - $^{1}H/^{2}H$ exchange methods – folding peptides

Kinetics of protein folding

Introduction

Anfinsen discovered that the small proteins ribonuclease A and staphylococcal nuclease could be reversibly denatured.1 On removal of a chemical denaturant, such as urea, they spontaneously refold to their native structures after denaturation. Similarly, they spontaneously refold on cooling after thermal denaturation. The kinetics of folding and unfolding of proteins appear to be a very complex process, but this process follows very simple rate laws, governed by a few basic principles. In the following session, the basic principles acts as a prelude to examining the mechanism of protein folding using two powerful techniques for examining detailed structural events in protein folding, Φ-value analysis and $1_H/2_H$-exchange.

Basic methods

The folding of proteins is usually studied in vitro by first denaturing them in solutions of urea, guanidinium chloride (GdmCl), or acid and then diluting the denaturant. Techniques such as

Techniques	Merits and Demerits
• Stopped-fiow methods	• Most convenient method, because they cover a useful time range and are ideal for mixing two reagents.
• Fluorimetry	• Sensitivity • This technique is generally used to monitor tertiary interactions because the fluorescence yield and emission wavelength of tryptophan are sensitive to its environment.
• Stopped-flow circular dichroism	• Detect changes in secondary structure in the far ultraviolet. But it has an inherently poor signal-to-noise ratio and requires

	considerably more protein than does fluorescence.
• Relaxation methods or flash photolysis	• Used for investigating faster reactions
• Temperature-jump procedures	• are ideal for denaturing proteins by jumping from ambient to elevated temperatures. This method has been adapted to study rapid events in folding on time scales of nanoseconds to microseconds by temperature-jumping a cold-denatured protein
• Nuclear Magnetic Resonance	• Very sensitive

Multiple phases and cw-peptidyl-prolyl bonds

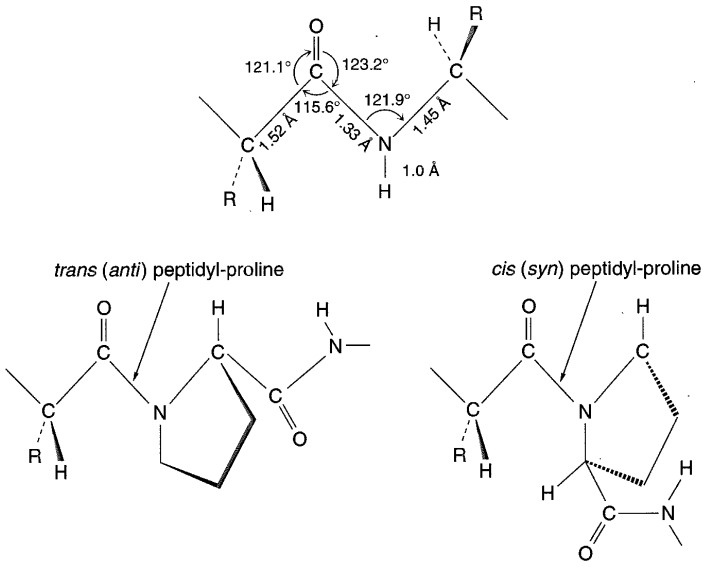
a. Effects of peptidyl-proline isomerization on kinetics

Refolding is generally found to proceed by a series of exponential phases. Many of these exponentials are a consequence of cis-trans isomerization about peptidyl-prolyl bonds. The equilibrium constant for the normal peptide bond in proteins favors the trans conformation by a factor of 103—104 or so. The peptidyl-prolyl bond is an exception that has some 2-20% of cis isomer in model peptides (Refer figure below).

It is found that the cis isomer occurs often in native structures. The interconversion of cis to trans in solution is quite slow, having half-lives of 10-100 s at room temperature and neutral pH. This has two important consequences.

- First, a protein that has several prolines all in the trans conformation in the native structure will equilibrate when denatured to give a mixture of cis and trans forms. On refolding, there will be an apparent succession of slow refolding phases as the different isomers fold in parallel. The isomerization of the proline bonds can be accelerated by being coupled withholding to have rate constants of 1 s^{-1}.
- Second, proteins that naturally have cis-peptidyl prolines equilibrate on denaturation to give denatured states that are predominantly in the wrong

conformation. The kinetics of their refolding is dominated by the isomerization events. The proline-related phases in refolding can sometimes be accelerated by peptidyl-prolyl isomerase.



That peptidy] -proline isomerization is a post-unfolding event is readily demonstrated by double-jump experiments. The protein is denatured for several milliseconds and then restored to renaturing conditions before the peptidyl prolines have had time to isomerize. The proline-related phases are then no longer seen.

b. Effects of peptidyl-proline isomerization on equilibria

Equilibrium parameters derived from kinetics will differ from those measured at equilibrium because of the effects of proline isomerization. The additional equilibria between cis- and trans-peptidyl-proline bonds in the denatured state add to its stability and so favor denaturation (Refer equation below).

$$N_{trans} \rightleftharpoons D_{trans} \rightleftharpoons D_{cis}$$

c. Effects of peptidyl-proline isomerization on unfolding kinetics

The proline isomerization phases are normally seen in the folding kinetics. Denaturation of the protein at [denaturant] >> $[denaturant]_{50\%}$ is close to 100%, and only a single exponential phase is observed in the unfolding kinetics for most small proteins. But, at denaturant concentrations of approximately $[denaturant]_{50\%}$, a protein with a trans-peptidyl-prolyl bond goes first to D_{Trans} as in the above equation, and then there is a

second minor phase as D_{Trans} —> D_{Cis}, displacing the equilibrium. This phase will be a major one if the protein has a cis bond. The two phases can, of course, become coupled and give complex kinetics.

Two-state kinetics

Chymotrypsin inhibitor 2 (CI2) folds rapidly by simple two-state kinetics; that is, $D \rightleftharpoons N$, with a $t_{1/2}$ of 13 ms^{-1}. CI2 is a small 64-residue protein that has all its peptidyl-proline bonds in the favorable trans conformation. The occurrence of two-state kinetics does not prove that there are no intermediates on the folding pathway; but there could be intermediates at high energy and are kinetically undetectable. Two-state behavior has subsequently been found for many other small proteins.

While studying the two-state kinetics, we have to consider the following parameters such as

- Effects of denaturant on unfolding and folding kinetics
- Interpretation of the rate laws for denaturation and folding: The Tanford β value
- Effects of temperature on folding
- Two-state kinetics and intermediates
- Kinetics tests for intermediates

1. Effects of denaturant on unfolding and folding kinetics

A plot of the logarithm of the rate constant for unfolding, K_u against the concentration of denaturant is found to be close to linear for many proteins at [denaturant] >> $[denaturant]_{50\%}$

$$\ln k_u = \ln k_u^{H_2O} + m_{k_u}[\text{denaturant}]$$

Where K_u^{H2O} is the value extrapolated to the absence of denaturant and m_{ku}, is a constant of proportionality which can be expressed in terms of activation energies,

$$\Delta G_{\ddagger-N} = \Delta G_{\ddagger-N}^{H_2O} - m_{\ddagger-N}[\text{denaturant}]$$

where $\Delta G_{\ddagger-N}^{H_2O}$ is the free energy of activation for unfolding in water.

For CI2 and some other small proteins, the folding rate constant, K_f, follows a similar relationship for [denaturant] >> $[denaturant]_{50\%}$

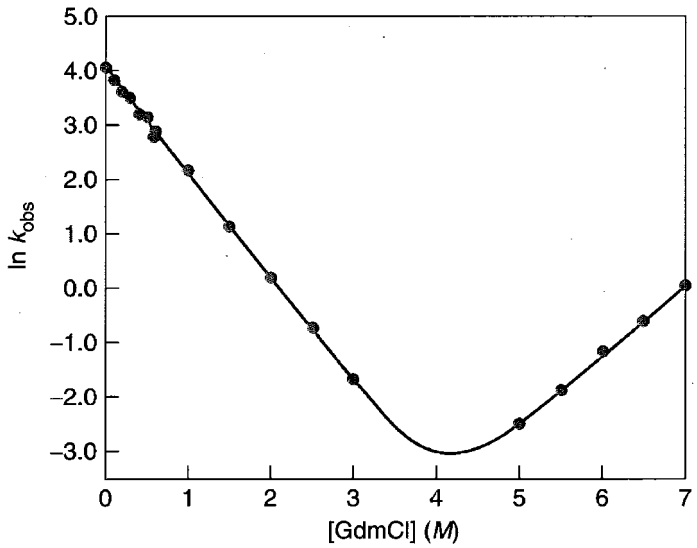
$$\ln k_f = \ln k_f^{H_2O} - m_{k_f}[\text{denaturant}]$$

which can be expressed in terms of activation energies;

$$\Delta G_{\ddagger-D} = \Delta G_{\ddagger-D}^{H_2O} + m_{\ddagger-D}[\text{denaturant}]$$

The two curves form a V-shaped kinetics curve (Refer figure below), which is sometimes called a chevron plot, constructed by combining the two rate constants:

$$\ln k_{obs} = \ln (k_f^{H_2O} \exp(-m_{k_f}[\text{denaturant}]) + k_u^{H_2O} \exp(m_{k_u}[\text{denaturant}]))$$



Chevron plot

The linearity of the equations holds very well for denaturation and folding of CI2 in GdmCI, but there is some downward curvature for the unfolding of barnase in urea, which follows the below equation:

$$\ln k_u = \ln k_u^{H_2O} + m_{k_u}[\text{urea}] - m^*[\text{urea}]^2$$

2. Interpretation of the rate laws for denaturation and folding: The Tanford β value

Proteins are denatured by using denaturants such as urea and GdmCI because all parts of them are more soluble in denaturant, and their free energy of transfer to denaturant is nearly linear with [denaturant]. The transition state for protein folding and unfolding is intermediate in structure between the denatured and native states. Thus, the transition state is stabilized by denaturant with respect to the native structure but destabilized relative to the denatured state. Accordingly, denaturant lowers the activation

energy of denaturation and raises the activation energy of folding. The stabilization energy terms are clearly linear with [denaturant].

If each group, *i*, in the protein has a free energy of transfer of δg_i cal/mol from water to a 1-M solution of denaturant and each group becomes exposed by only a factor α_i on denaturation, then the effective free energy change for the group on denaturation is $\alpha_i \delta g_i$. This leads to an m value for unfolding of $m_{D\text{-}N} = \Sigma;\alpha_i \delta g_i$. this value applies even to the formation of the transition state for unfolding:

$$m_{\ddagger\text{-}N} = \sum_i \alpha_{i\ddagger} \delta g_i$$

where $a_{i\ddagger}$ is the fractional degree of exposure in the transition state for denaturation. Then the Tanford β value, β_T is given by;

$$\beta_T = \frac{m_{\ddagger\text{-}N}}{m_{D\text{-}N}} = \frac{\sum_i \alpha_{i\ddagger} \delta g_i}{\sum_i \alpha_i \delta g_i}$$

β_T is a measure of the average degree of exposure in the transition state relative to that of the denatured state from the native state. It is a useful index of the compactness of the transition state. Because the plots against denaturant may be slightly curved, it is better to define $m_{D\text{-}N} = \delta\Delta G_{D\text{-}N}/\delta$[denaturant] and $m_{\#\text{-}N} = \delta\Delta G_{\#\text{-}N}/\delta$ [denaturant] over the ranges of [denaturant] employed in the comparison.

Accordingly, Tanford $\beta_T = \delta\Delta G_{D\text{-}N}/\delta$ [denaturant])/ $\delta\Delta G_{D\text{-}N}/\delta$ [denaturant]). The Tanford β value is often measured for unfolding kinetics. This is especially convenient for reactions in which there are folding intermediates. For two-state systems, the β values for folding and unfolding should add up to 1.

3. Effects of temperature on folding

The rate constant, k, for most elementary chemical reactions follows the Arrhenius equation, $k = A \exp(- E_A/RT)$, where A is a reaction-specific quantity and E_A the activation energy. As E_A is always positive, the rate constant increases with temperature and gives linear plots of In ***K*** versus I/T. Kinks or curvature are often found in Arrhenius plots for enzymatic reactions and are usually interpreted as resulting from complex kinetics in which there is a change in rate-determining step with temperature or

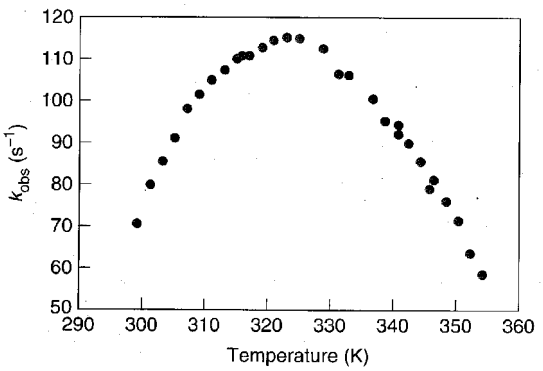
a change in the structure of the protein. The Arrhenius equation is recast by transition state theory (Refer equation below) to

$$k = k_B T \frac{\kappa}{h} \exp \frac{\Delta S^{\ddagger}}{R} \exp \frac{-\Delta H^{\ddagger}}{RT}$$

where kB is the Boltzmann constant, h is the Planck constant, k is the transmission coefficient, AS* is the entropy of activation, and A//* is the enthalpy of activation. The value of ΔH* is usually determined from the slope of a plot of In (k/T) versus 1/T, the **Eyring plot**. ΔH* is always positive for an elementary reaction step because energy is required to create the partly broken bonds of the activated complex. The denaturation of proteins appears to follow the above equation. But folding reactions are often distinctly curved, and the kinetics is said to be non-Arrhenius.

Non-Arrhenius folding kinetics-Negative enthalpies of activation

The simple two state kinetics of folding of CI2 has been followed over a wide range of temperature (Refer figure below).



The rate increases with increase in temperature reaches the maximum and finally decreases with increasing temperature. $\Delta H^{\#}$ varies with temperature, being positive at lower temperatures and negative at high temperatures.

$\Delta H^{\#}$ for protein folding has a large ΔC_p term for overall denaturation If the transition state for folding is relatively compact and shields the hydrophobic side chains, then there will be the characteristic ΔC_p term associated with the hydrophobic effect (Refer the equation below)

$$\Delta H_{\ddagger-\mathrm{D}(T_2)} = \Delta H_{\ddagger-\mathrm{D}(T_1)} + \Delta C_{\mathrm{p}(\ddagger-\mathrm{D})}(T_2 - T_1)$$

where $\Delta C_{p(\#-D)}$ is algebraically negative because it is measured from the denatured state to the transition state.

Computer simulation models predict that at high temperature, the model polymer is highly random and has a high configurational entropy that is lost in the folding reaction, and so the barrier to folding is entropic. At low temperatures, there are local minima in the model that lower the entropy of D and require enthalpic barriers to be broken. The contribution from these effects is probably much less than that from the classical hydrophobic effect. The free energy of activation, $\Delta G_{\#-D}$ is always positive. At high temperature, the barrier is entropic, with $T\Delta S_{\#-D}$ being dominant. At low temperature, the barrier is primarily enthalpic, with $\Delta H_{\#-D}$ being dominant.

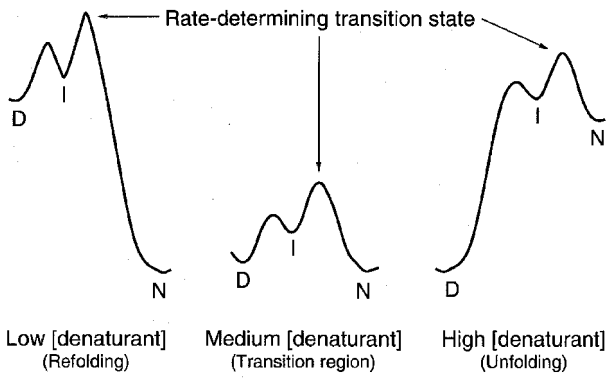
The ratio $\Delta C_{p\#-D} / \Delta C_{p(N-D)}$ is a measure of the relative burial of hydrophobic surface areas in the transition and folded states, if the hydrophobic model is correct. The ratio is 0.51 for CI2, compared with a value of β_T of 0.6, which is a measure of the overall change in surface area.

4. Two-state kinetics and intermediates

a. Apparent two-state kinetics

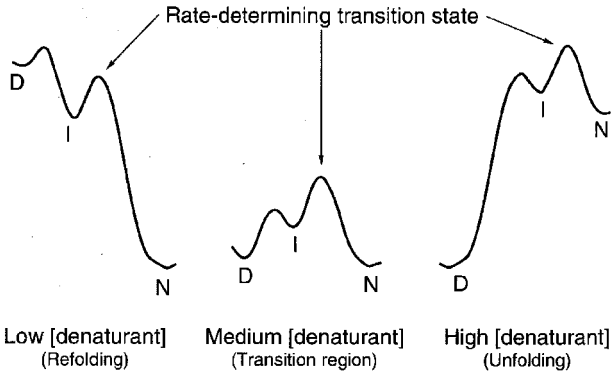
Many multistate denaturation squilibria exhibit apparent two-state behavior around the transition region because the intermediates are present at low concentrations, which applies even for folding and unfolding kinetics. The occurrence of simple two-state kinetics does not prove that there are no intermediates on the reaction pathway.

For example, the energy profile in the below figure has a high energy intermediate.



But the rate laws that describe the reaction are identical to those of a simple two-state reaction at all concentrations of denaturant. This is easily demonstrated algebraically, but it also can be seen intuitively, because there is the same transition state, and the ground states are always D and N at all concentrations of denaturant.

In the below figure is a more typical situation where the intermediate is more stable than the more highly denatured state D in the absence of denaturant



5. Kinetics tests for intermediates

Intermediates present in the reaction pathway are usually detected by the appearance of additional phases. Most small proteins have only one non-proline-related folding phase in the accessible time range. But the simple kinetics can be deceptive, and there can be hidden intermediates, besides the high energy states. Many procedures such as spectroscopic studies, isotopic studies, tryptophan fluorescence, circular dichorism for detection of secondary and tertiary structures were described to test for equilibrium two-state transitions; these procedures are also effective for detecting intermediates that may accumulate close to the transition region.

The reaction illustrated in the above figure often leads to kinetics that appears to be two-state: in the absence of denaturant, the transition from D to I is often too fast to be measured, and only the slower transition of I to N is observed. There are powerful kinetic tests for detecting intermediates under these conditions even if only a single transition is observed. There are three powerful tests to detect these intermediates such as

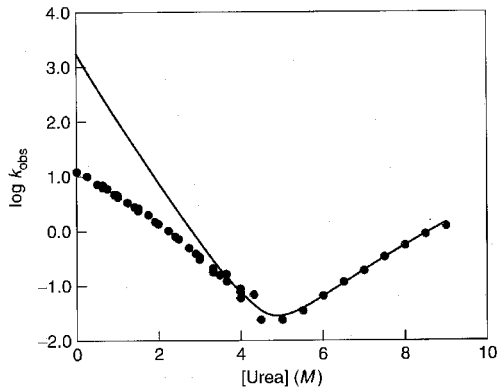
- The first test is that the ratio of the forward and reverse rate constants for a two-state transition must give the same equilibrium constant as that measured directly by solvent denaturation. The ratio of the rate constants for the apparent folding

step and for unfolding in water of the three-state reaction as in the above figure gives the equilibrium constant between I and N, and not between D and N.

- The second test is that the sum of the forward and reverse m values must equal the m value for equilibrium unfolding for a two-state transition; that is,

$$m_{\ddagger-\mathrm{N}} + m_{\ddagger-\mathrm{D}} = m_{\mathrm{D}-\mathrm{N}}$$

- The third test is deviations from the simple V-shaped curves of rate constants versus [denaturant] that were shown in chevron plot for CI2. An intermediate was detected from the denaturant dependence of the single phase for the folding of barnase (Refer blow figure) because of downward curvature. This is sometimes called "rollover."



- The rate constant for folding for the two-state transition for barnase can be calculated as a function of [urea] from the equilibrium and rate constants for denaturation. The calculated values deviate from the observed values being much higher as [urea] tends to zero. The behavior is a diagnostic test for the occurrence of an intermediate.

Folding intermediates and denatured states are usually sticky because of exposed hydrophobic surfaces and so tend to aggregate. This aggregation causes artifacts in kinetics that can easily be mistaken for additional intermediates. Aggregation can often be detected by light scattering.

Multistate kinetics

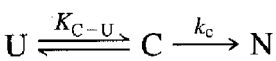
Proteins folds rapidly following certain kinetic patterns involving zero to n number of intermediates in its reaction pathway. If it is not involving any intermediates in its reaction pathway, it is assumed to follow two state kinetics. If additional kinetic phases are observed in a folding pathway, it may involve folding intermediates.

1. Are intermediates on or off pathway?

The most difficult question to answer is whether an observed "intermediate," I, is on or off the reaction pathway. That is, is there the compulsory order D—> I—> N, or is I a nonproductive side product that has to decompose back to D for folding to proceed? The problem is that although we are performing kinetic experiments, there are equilibria involved and some of the key species are effectively at equilibrium. It is a fundamental tenet of kinetic and thermodynamic analysis that measurements of the equilibrium distribution of intermediates give just the relative thermodynamic properties of the intermediates and not the pathway between them. This is a consequence of the first law of thermodynamics: one cannot derive a kinetic mechanism from an equilibrium analysis, because the free energy of a particular molecule at equilibrium depends only on the state of the molecule and is independent of the pathway by which it is formed. This concept is so important that it is a worthwhile exercise to go through particular mechanisms and discover the ambiguities. So as not to presuppose the order of events, we shall temporarily drop the terminology "I" and assume folding from a highly unfolded state, U.

a. Kinetic consequences of rapid pre-equilibria

Suppose that, as on the left of Figure 18.7, there is a compact intermediate state, C, on the pathway that is a genuine folding intermediate, and so the reaction follows the scheme:



where $K_{c\text{-}u}$ = [C]/[U] and kc is the rate constant for C reacting to give N. If U and C are in rapid equilibrium, the fraction of the denatured state, D, that is present as [C] is

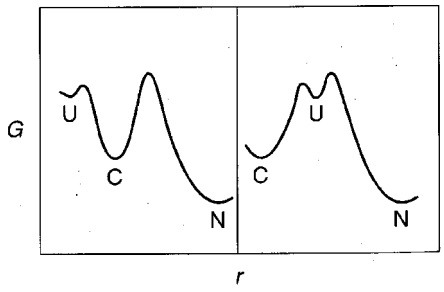
$K_{c\text{-}u}/(1+K_{c\text{-}u})$. As U and C equilibrate rapidly, we can observe only a single rate constant for folding that is given by

$$k_{obs} = \frac{k_c[D]K_{C-U}}{1 + K_{C-U}}$$

Since $K_{c\text{-}u}' >> 1$ and [C] >> [U], the observed rate of formation of N is given by

$$\frac{d[N]}{dt} = k_c[C]$$

Now suppose that C is not on the pathway but is a compact product of a side re-reaction, as on the right of Refer figure below;



$$k'_{obs} = \frac{k_f[D]}{1 + K_{C-U}}$$

Since $K_{C-U} \gg 1$ and [C] ≫ [U],

$$\frac{d[N]}{dt} = \frac{k_f[C]}{K_{C-U}}$$

that is,

$$\frac{d[N]}{dt} = k'_f[C]$$

Equations K_{obs} and K'_{obs} are kinetically equivalent: the rate of formation of N by both pathways follows a rate constant multiplied by the concentration of C, the dominant species in solution. Thus, the rate constant for formation of N reports back simply on the

difference in structural properties between the transition state for the formation of N and the ground state that is C. This means that we cannot determine whether the reaction is U->C->N or C-> U-> N by kinetics. A rapidly interconverting system of U and C that is being slowly converted to N; that is,

$$\{U \rightleftharpoons C\} \longrightarrow N$$

For example, suppose while examining the effects of increasing concentrations of denaturant, which has two effects: the intrinsic rate constants for folding, k_f and k_c, become lower, and the concentration of U increases relative to C. Initially, the change in concentration is not important while $K_{c\text{-}u}$ is greater than 1, and so both mechanisms slow down because of the intrinsic rate constants decreasing. When [C] .= [U] at higher concentrations of denaturant, k_{obs} for multistate involving an intermediate, slows down by a further factor of 2 since $K_{c\text{-}u}/(1 + K_{c\text{-}u}) = 0.5$ (in equation K_{obs}) and does for multstep reaction in which C is not on the pathway sinc $1/(1+K_{c\text{-}u})$.

b. Both kinetic steps resolvable

From the observation it is found that in the final step of folding has a single rate constant. It is possible to resolve, where a highly unfolded denatured protein in concentrated denaturant is diluted into a folding buffer and there is fast formation of C, either on or off the pathway, followed by slower folding, that two relaxation times are observed, but measurement of rate constants alone cannot give the order of events because the rate equations are symmetrical in the relaxation times.

In theory, the associated magnitudes and signs of the amplitudes of the rate constants can determine the sequence, for the following reasons.

- If the intermediate is on the pathway, there will be a slight lag in the rate of formation of N.
- Conversely, if the intermediate is off the pathway, there will be an initial rapid partitioning of U between N and C so that there is a burst of formation of N followed by slower folding as C gives N via U.

But the amplitudes of the lag or burst phases in [N] are low when the rate constant for the formation of C is much greater than for N, and so data of exquisite accuracy are required to detect a burst or lag. The detection is further complicated because the sum of all the spectroscopic signals from U, C, and N are collected, and so the absolute values of extinction coefficients, fluorescence yields, etc., must be known. In addition, heterogeneity in the denatured state, resulting naturally from isomerization of peptidyl-proline bonds or from the presence of small amounts of impurities, can further obscure the analysis.

c. Indirect evidence for sequence of events

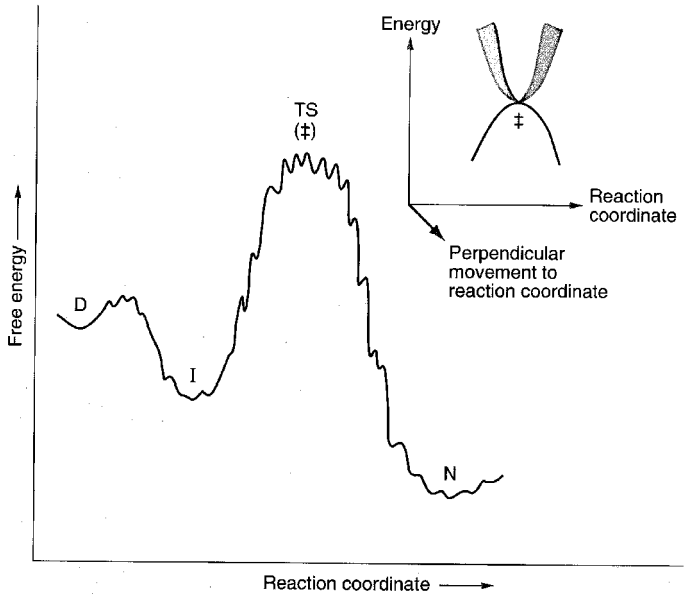
The structure of a compact state on the folding pathway of barnase has been mapped out in detail at the level of individual residues using the Φ-value analysis. Its structure is clearly between that of the unfolded state and the major transition state for folding. It has the same elements of secondary structure and the same tertiary contacts, but they are all weaker, especially in the hydrophobic core. The structure has all the properties expected of a folding intermediate that is on the pathway and has no indications of alternative structure. This does not prove that it is on the pathway, but the most likely explanation is that it is.

Transition state in protein folding

It is pertinent to ask what is the nature of a transition state in protein folding and if there is evidence that transition states do occur.

The transition state is a structure that is at a saddle point in an energy surface, at a maximum of energy along a reaction coordinate and at a minimum of energy perpendicular to it (Refer figure below).

Transition states of simple chemical reactions have only a few bonds being made and broken, and these are very strong, so that quantum theory has to be applied. But protein folding involves the making and breaking of very many weak bonds, which can be described by classic statistical mechanics and Newton's equations.



The energy surface can be calculated from these equations, and it will have a saddle point. The transition state is the ensemble of states that differ slightly from one another in energy around the saddle point. This type of energy surface should lead to transition states in protein folding being at very wide and long saddle points, with many small dips in the profile. The energy surfaces in protein folding also have very high entropy components because of the large changes in configurational entropy as a protein folds and the changes in entropies of hydration. The energy landscape around transition states can be probed by structure- reactivity relationships. These are movements of the transition state along the reaction coordinate. Protein folding studies in mutants have an advantage over those in simple organic reactions because the Tanford β_T value measures the position of the transition state along the reaction coordinate. By combining Tanford β_T values and Φ-value analysis, it has been shown that transition states in protein folding do act as if they are at the bottom of long and wide saddle points.

Φ-value analysis

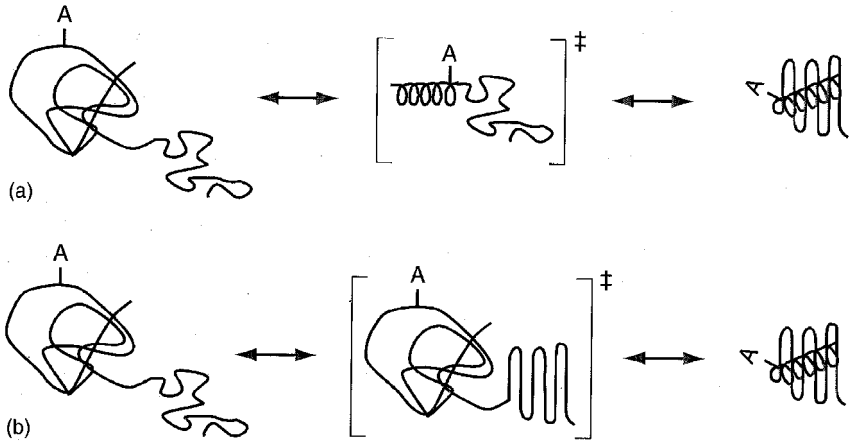
The contribution of the binding energies of different side chains of the tyrosyl-RNA synthetase to catalysis were measured by systematic protein engineering experiments. Apparent binding energies of each side chain were measured by comparing catalytic rate constants of wild-type and mutant enzymes. The structures of transition

states and intermediates in folding may be inferred from the changes in the kinetics and equilibria of folding on mutation.

The procedure is described in terms of Φ-value analysis, by analogy with classical Bronsted β-value analysis. Φ-value analysis is the only procedure for obtaining the structures of transition states (and some unstable intermediates) at the level of individual residues, at almost atomic resolution. This is because the only way of analyzing transition states experimentally is by kinetics, and the only way of analyzing detailed structure is by structure - activity relationships.

1. Changes in energy levels on mutations

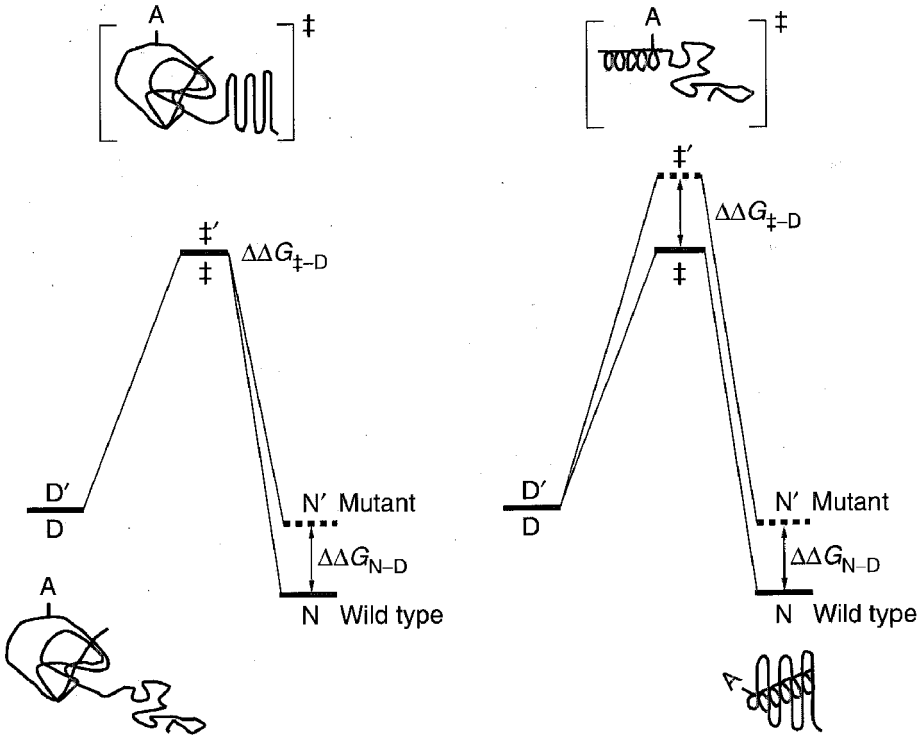
Point mutation of Ala —» Gly in the solvent-exposed face of a helix in a protein that destabilizes it by 1 kcal/mol (4 kJ/mol) relative to the denatured state (Refer figure below).



When measuring the kinetics of folding and unfolding of the protein and the transition states for folding and for a folding intermediate are also destabilized by 1 kcal/mol (4 kJ/mol) relative to the denatured state. Then it seems likely that the helix is fully formed at the site of mutation in those two states, because it responds to a destabilizing mutation in exactly the same way as the fully folded structure does. Suppose, in addition, that other mutations at the same position and at neighboring residues also destabilize the transition state and intermediate by the same amount as they do the fully folded structure. Then it is beyond reasonable doubt that the helix is fully formed in those states in the region of mutation. Suppose, conversely, that a mutation destabilizes the native state by 1

kcal/mol (4 kJ/mol) but does not affect the energy of a transition state or intermediate at all, measured relative to the denatured state. Then it seems likely that the helix is fully denatured at the site of mutation in those two states, as it responds to destabilizing mutations in the same way as the denatured state does.

Further, if the mutations at the same site or at neighboring sites also do not affect the energy of the intermediate and transition states relative to the denatured state, then it is beyond reasonable doubt that the helix is fully denatured in those states at the region of mutation. The two situations above do occur for many mutations in barnase. The detailed residue-by-residue characterization of its folding intermediate and transition state can be interpreted. Each mutation is a reporter group located at the site of mutation during the folding pathway. This analysis is formalized in terms of Φ values (Refer figure below).



Suppose that a mutation destabilizes the folded structure by $\Delta\Delta G_{N-D}$ units of energy, measured relative to the unfolded state. Then, if the free energy of a transition state, measured relative to the unfolded state, changes by $\Delta\Delta G_{+-D}$, Φ is defined by

$$\Phi_F = \frac{\Delta\Delta G_{\ddagger-D}}{\Delta\Delta G_{N-D}}$$

A Φ_F value of 0 means that the energy of the transition state is perturbed on mutation by the same amount as the denatured state is perturbed. A Φ_F value of 1.0 means that the energy of the transition state is perturbed on mutation by the same amount as the native state is perturbed. It is inferred from the changes in energy that a Φ_F value of 0 means that the structure is unfolded at the site of mutation as much as it is in the denatured state, and a OF value of 1.0 means that the structure is folded at the site of mutation as much as it is in the native state.

2. Choice of mutations: Nondisruptive deletions

The ideal mutations are those that remove moieties of side chains that make well-defined interactions. It is important to avoid mutations that add new functional groups that can make extraneous interactions within the protein, mutations that change stereochemistry or can cause stereochemical clashes, and mutations that leave charged groups without solvating partners. The best mutations are those that remove a small number of methylene groups because there is little change in the solvation energy of the denatured state. These hydrophobic mutations can be on the surface or the interior of the protein. Residues on the surface are also promising targets because they do not have to pack as well as in the interior. The ideal mutations for investigating surface and buried hydrophobic interactions are Ile ->Val, Ala -> Gly, and Thr -> Ser, followed by Val -> Ala, Ile ->Ala, and Leu ->Ala. Phe ->Ala and Tyr -> Ala are radical but still work well enough in practice.

3. Relationship between Φ and the Bronsted β value

Φ_F is similar to but not identical with the Bronsted β value. Φ_F and β are the same at the two extreme values of 0 and 1, measured during bond making ie during protein folding. Φ_F (or β) = 0 corresponds to fully denatured (no bond making), and Φ_F (or β) = 1 to fully native (complete bond formation), these values were easily interpreted for barnase. In general, there need not be a linear relationship between Φ_F and the extent of

non-covalent bond formation. The worst situation is for the formation of salt bridges, because of the complexities of the energetics. But, there should be some correlation between OF and the number of van der Waals interactions made in the packing of hydrophobic cores.

4. Fractional values of Φ_F

Fractional values of Φ_F are more difficult to interpret because they can be caused by a variety of factors such as

- First, they can result either from genuinely weakened interactions in a single species or from there being a mixture of states, some with the interactions fully formed, others with the interactions fully broken. The mixture could stem from an equilibrium between various states in an intermediate or from parallel pathways in kinetics leading to a mixture of transition states.
- Fractional values may also arise if a side chain makes one set of interactions with one element of structure and a further set with a different element, and these elements react in different ways in unfolding.
- Fractional values of Φ may arise from artifacts because, a distortion of the structure of the folded molecule on mutation that is not the same in the transition or intermediate states; that is, there are various reorganization energies introduced by mutation that do not cancel. Again, analyzing a large number of mutations in the same regions of structure allows the detection of artifacts and the identification of patterns of structure.
- The other major problem is that, in general, there is not a linear relationship between Φ and the extent of formation of structure.

$^1H/^2H$-exchange methods

1. $^1H/^2H$-exchange at equilibrium

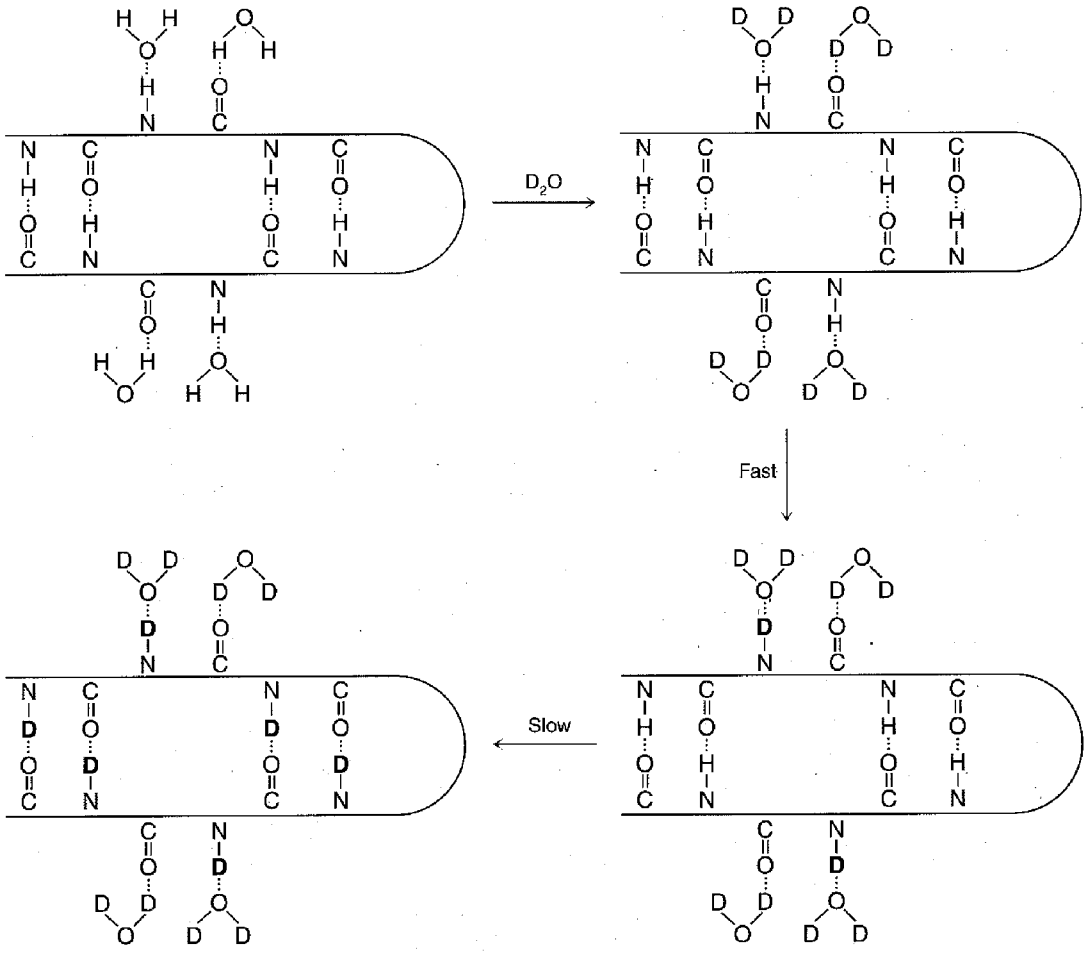
The protons of backbone >NH groups of small peptides dissolved in water are labile and exchange with those of the solvent. The exchange is readily detectable by NMR methods by studying a peptide containing the normal isotope ^1H dissolved in deuterium oxide (D_2O or ^2H_2O), or vice versa. The reaction is acid- and base-catalyzed, following the rate law

$$k_{int} = k_{H_2O} + k_{OH^-}[OH^-] + k_{H^+}[H^+]$$

where the observed rate constant for exchange is k_{int} (the "intrinsic" rate constant), the spontaneous value in water is k_{H2O}, the base-catalyzed second-order rate constant is k_{OH-}, and the acid-catalyzed second-order rate constant is k_{H+}. The absolute and relative values of these terms vary with the different amino acids and are affected by the residues immediately preceding and following in the sequence. Values of k_{int} have been tabulated for each of the individual amino acids in different sequences as a function of temperature and pH. The major term for most amino acids above pH 6 is that for the base catalysis, and the other terms may be ignored.

Backbone >NH groups of proteins may be highly protected against exchange, by factors of 104-106 or more by being in stable elements of secondary structure (for example, as in strands of β sheet in figure below) or by simply being buried in the protein. A protection factor P is defined for any proton by

$$P = \frac{k_{int}}{k_{ex}}$$



where k_{int} is the intrinsic rate constant for that proton under the reaction conditions and K_{ex} is the observed value for exchange. For hydrogen exchange of buried groups to occur, there must be either small-scale structural fluctuations, termed local unfolding or breathing, involving the breakage of hydrogen bonds and access of solvent to the exchangeable proton, or full unfolding of the protein.

$$C^{H} \underset{k_c}{\overset{k_o}{\rightleftharpoons}} O^{H} \xrightarrow[D_2O]{k_{int}} O^{D} \underset{k_o}{\overset{k_c}{\rightleftharpoons}} C^{D}$$

where C and O are either the locally "closed" and "open" conformations or globally folded or unfolded states and k_c and k_0 are the rate constants of local (or global) folding and unfolding, respectively. If D_2O is in great excess,

$$k_{ex} = \frac{k_o k_{int}}{k_c + k_o + k_{int}}$$

Under conditions that strongly favor folding, the rate constant of refolding is considerably faster than that of unfolding, so that $k_c > k_o$, and k_0 can be ignored. The above equation has two extreme limits. When $k_c > k_{int}$ as is usual for most globular proteins under most conditions, the observed rate constant of exchange reduces to

$$k_{ex} = \frac{k_o}{k_c} k_{int}$$

or

$$k_{ex} = K k_{int}$$

where $K = k_o/k_c$, is the effective equilibrium constant between open and closed forms. That is, K = [O]/[C] = 1/P. This situation is termed the EX2 limit. At the other extreme, when $k_c << k_{int}$, the rate-determining step is the open-opening, and

$$k_{ex} = k_o$$

This is termed the EX1 limit.

The mechanism of exchange may be probed by the effects of denaturant, pH, and mutation on K_{ex}. Use of mutation is particularly powerful for determining how much structural fluctuation is required for exchange to take place. Suppose, as is common, that exchange is occurring at the EX2 limit. Then we can calculate the free energy between the open and closed states from the K_{ex} equation.

$$\Delta G_{ex} = -RT \ln K = RT \ln P$$

The value of ΔG_{ex} will differ from the real value of the equilibrium free energy because of uncertainties in the value of K_{int}: the residue in state O may not be as fully

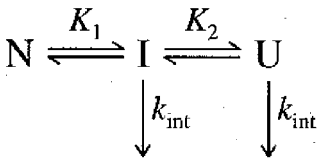
accessible as in model peptides. Suppose we make a mutation in the protein that changes its stability by $\Delta\Delta G_{D\text{-}N}$. We can then measure the change in ΔG_{ex} on mutation from the change in protection factors (P_{wt} in wild type and P_{mut} in the mutant):

$$\Delta\Delta G_{ex} = -RT \ln \frac{P_{mut}}{P_{wt}}$$

Providing that the residues examined are removed from the site of mutation and that the mutation is non disruptive, all the uncertainties in K_{int} cancel out. Mutation of many of the deeply buried residues in barnase gives exactly the same values of $\Delta\Delta G_{ex}$ as $\Delta\Delta G_{D\text{-}N}$, showing that those residues exhibit the full change in free energy of unfolding on going from G to O. Those residues thus exchange by global unfolding. Others, especially those close to the surface in helixes, have values of $\Delta\Delta G_{ex}$ that are very close to zero, and so they are unaffected by mutations that alter global stability. These residues thus exchange by local fluctuations.

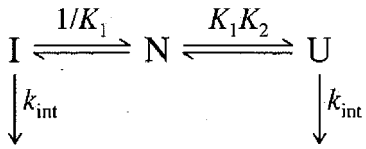
2. Exchange at equilibrium cannot be used to determine pathways

Because of the apparent power of exchange at equilibrium to measure fluctuations in structure, there have been many attempts to use the procedure to determine pathways. Equilibrium measurements just give the relative thermodynamic properties of intermediates and not the pathway between them. The value of ΔG_{ex}, the equilibrium constant between particular open and closed states of a protein derived from hydrogen exchange at equilibrium, is just such a thermodynamic measurement and does not give information about when that state is formed on a pathway. For example, consider the exchange from a reaction sequence:



where an intermediate I is formed from a native structure, N, with formation constant K1 (= [I]/[N]), and the unfolded state U has formation constant K_1K_2(= [U]/[N]), and I and U are "open" states for exchange. The rate constant for the exchange of a particular

residue i from state I is given by $K'_{ex} = K^i_{int}K_1$ and that of residue j from state U by $K^j_{int} = K^j_{int}$. Suppose, instead, that I is an off-pathway state (Refer equation below),



formed with the same equilibrium constants; $K_1 = [I]/[N]$, $K_1K_2 = [U]/[N]$. The rate constant for the exchange of a particular residue i from state I is given by $K'_{ex} = K^i_{int}K_1$and that of residue j from state U by $K^j_{int} = K^j_{int}$. Thus, identical values of ΔG_{ex} are obtained for both pathways.

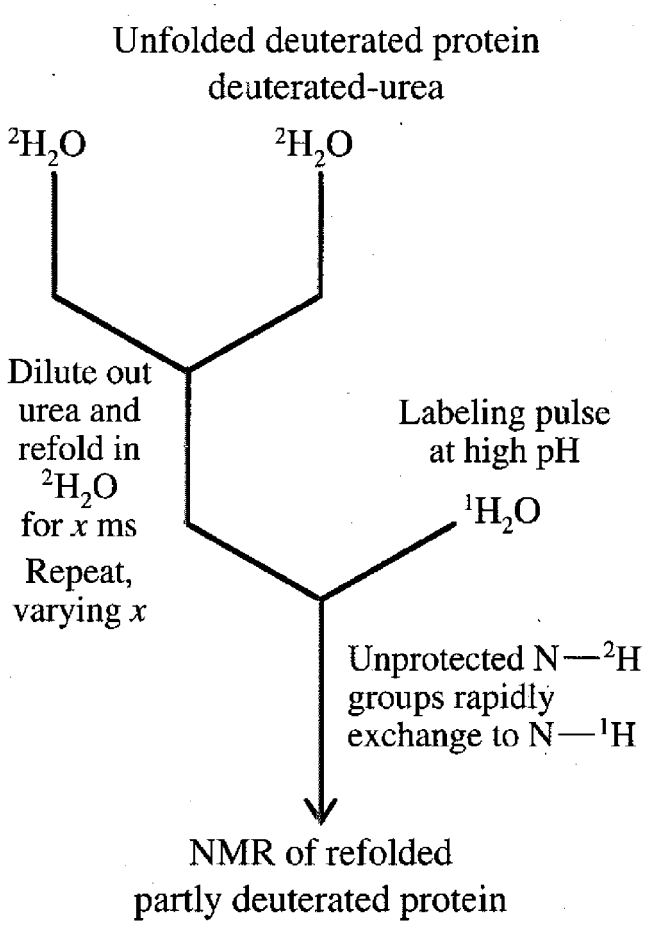
3. Uses of equilibrium 1H/2H-exchange in folding studies

$^1H/^2H$-exchange at equilibrium comes into its own in studying structure in stable intermediates. It is particularly useful for detecting regions of protected structure in partly unfolded states; regions that are protected will have significant protection factors. Suppose, for example, that a state exists in equal amounts in two rapidly inter converting conformations, one of which is highly protected and the other of which is unprotected against exchange. Then the observed protection factor is only two, no matter how high the degree of protection in the protected state, because the observed exchange is occurring from the other state. $^1H/^2H$- exchange at equilibrium is also an invaluable tool for determining the nature of the states of compounds bound to large complexes, such as molecular chaperones, that are detected by mass spectrometry as well as NMR, and for detecting small amounts of unfolded protein that would otherwise be undetectable.

4. Quenched-flow 1H/2H-exchange

The onset of protection of >NH groups as a protein folds may be measured by rapid quenching techniques. (Refer figure below). A crude form of the experiment is to denature deuterated protein in ^2H_2O containing deuterated urea. The urea is diluted out with a large volume of ^2H_2O in a quenched-flow machine, and the protein is allowed to refold for varying lengths of time (x). It is then pulsed with ^1H_2O for a defined time at a

sufficiently high pH for all unprotected deuterons to exchange nearly completely. It is then allowed to fold to completion.



The NMR spectrum is then measured for each sample to deter- determine the fraction of each >N2H group that has exchanged, and the results are plotted against x. This is a very useful procedure to determine when protected secondary structure is formed. But it is quite difficult to perform quantitatively in practice. Individual time points, each of not particularly high precision, are measured separately, and a very large number of these points is required to analyze multiphasic kinetics. Also, folding intermediates can be very labile and become unprotected in the labeling pulse, especially at high pH.

5. Φ-va!ue analysis versus quenched-flow $^{1}H/^{2}H$-exchange

Φ-Value analysis	Quenched flow exchange
Φ-values measure tertiary interactions	Quenched-flow exchange mainly detects secondary interactions.
Φ-Value analysis is by far the better technique for kinetics	Quenched flow is for intermediates
It can be used over any time scale accessible to kinetics	Quenched-flow exchange is limited to several milliseconds or more
Φ-values can be used to analyze transition states and unstable intermediates	Quenched flow exchange is limited to intermediates.
Φ-values do detect secondary structure by inference and can in principle be used to analyze the fate of virtually every side chain	Quenched flow exchange is limited to analyzing just the relatively few protons that are highly protected in secondary structure.
Φ-values give energetics as well as structure and tell whether interactions are fully or partly formed. This enables the order of events to be defined with some precision and gives evidence unavailable from any other technique.	
Φ-value analysis has provided evidence that the folding intermediate of barnase has a structure between that of the transition state for unfolding and the unfolded structure.	

Folding of peptides

1. Loops

Theory suggests that the most probable loops in proteins contain 10 residues.

Shorter loops become stiff and longer ones lose more conformational entropy.

2. α- Helixes

A simple model for the kinetics of helix formation invokes as the rate-determin-rate-determining step the formation of a nucleus of neighboring residues with Φ and Ψ angles in the a-helical region of the Ramachandran diagram. The nucleus is so improbable that there is only a single one formed in each molecule. Once the first hydrogen bond between the >C=O of residue i and the >NH of residue i + 4 is formed in the nucleus in a helix that should be stable, the subsequent growth is thermodynamically downhill because the stabilization energy of forming a single extra hydrogen bond is greater than the unfavorable energy of fixing a pair of residues in the a-helical conformation.

3. β Hairpin

The simplest β structure is the β hairpin. The β hairpin requires the pairing of hydrophobic side chains to stabilize it. The hairpin has to nucleate at its central turn, unlike the helix, which can nucleate at any residue whose >C=O can form a hydrogen bond with residue i + 4. The formation of β structure is inherently slower than the formation of α helixes because of the fewer nucleation sites and the requirements to make side-chain interactions.5 Depending on precise structure, helixes form with half-lives of a few hundred nanoseconds, and β structures form with half-lives 10 times longer.

4. Very fast folding small proteins

The fastest-folding small proteins generally fold on much slower time scales than the time scale of formation of secondary structure. The speed record is currently held by lambda(6-85), a truncated, monomeric form of the N-terminal domain of lambda repressor, which refolds with a half-life of approximately 140 μs. A thermostable lambda(6-85) variant with alanine substituted for glycine residues 46 and 48 in the third helix folds faster in dilute solutions of denaturant, with an extrapolated half-life of less than 10 μs in water. Cold-shock protein CspB from Bacillus subtilis folds in about 1 ms. Engineered mutants of the P22 Arc repressor and CI2 fold in a fraction of a millisecond.

UNIT - V FOLDING PATHWAYS AND ENERGY LANDSCAPES

Folding of CI2-nucleation condensation mechanism-folding of barnase-time resolution-insights from theory-optimization of folding rates-molecular chaperones

Folding pathways and energy landscapes

A protein folding up from its denatured state to its native structure is a challenging and far more complex than solving classical chemical mechanisms. In simple chemical mechanisms, there are usually changes in just a small number of bonds as a reaction progresses. Chemical bonds are often strong, and so stable covalent intermediates can sometimes be isolated and characterized. In protein folding, there are thousands of weak noncovalent interactions are made or broken, and it is very difficult to trap intermediates because of their unstable nature.

There is an astronomical number of conformations in the protein folding pathway. The denatured state is particularly difficult to analyze because it is an ensemble of many ill-defined and rapidly fluctuating conformations. Experimentalists must characterize all the stable and metastable states on (and off) the reaction pathway and the transition states that link them. The role of the experimentalist is to determine the sequence of intermediates and their rates of interconversion and to characterize their structures and energetics as far as possible to provide the essential basis for calculations and simulations by theoreticians.

The essence of modern work on structure and activity of proteins is analysis at the level of individual residues and atoms. Atomic and residual studies is restricted to the native state only whereas NMR is the only spectroscopic technique that can be applied in solution with sufficient resolution. New techniques are required for the study of unstable intermediates and transition states in folding.

One of the most productive techniques for analyzing simple chemical mechanisms has been the use of structure - activity relationships, in which small changes are made in the reagents and correlations made with changes in activity.

Techniques such as protein engineering, Φ value analysis are employed where there are many differences between noncovalent and covalent chemistry, the fundamental ideas are the same, and so we can adapt the classical methods. The crucial importance of Φ-value analysis is that it is the only method for studying side-chain interactions in transition states, which can then be used to infer secondary and tertiary structural information.

Other methods include computer simulation procedure using molecular dynamics. Molecular dynamics applies Newton's laws of motion to every atom in a protein and to the surrounding solvent.

Hurdles to molecular dynamics

- The computer must handle many conformations of the denatured and partly denatured states and then must calculate how they change over a relatively long period of time.
- The limited time scale accessible to simulation ie reactions must be speeded up.
- There are the same uncertainties in the potential functions that are used to calculate the energetics as there are in calculating the stabilities of the native state.

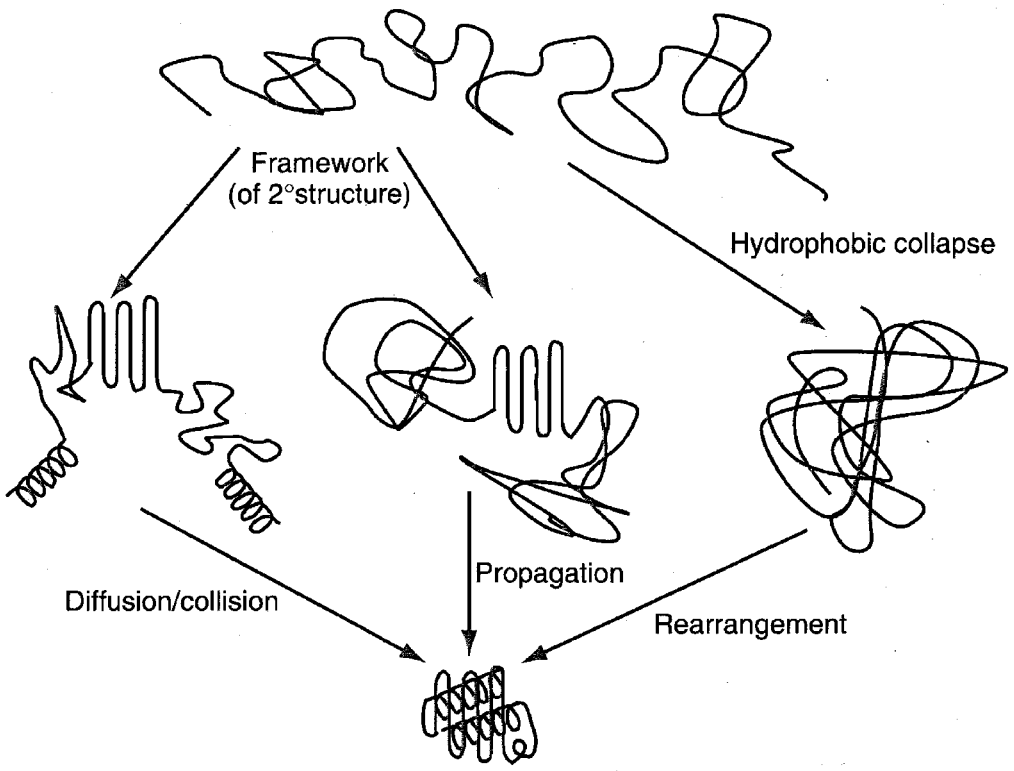
Molecular dynamics is very well suited for analyzing protein unfolding, because the calculations start from well-defined crystal and solution structures, solved by x-ray crystallography and NMR. Also unfolding can be speeded up by simulating denaturing conditions, such as high temperature or the presence of simulated denaturants. The folding pathway is then simulated by running unfolding in reverse. The molecular dynamics simulations have the potential of describing the whole pathway of folding, but they require experimental validation.

The combination of Φ-value analysis, NMR, and molecular dynamics simulation has allowed the description of folding pathways at atomic resolution.

Levinthal' paradox

Levinthal argued that because there is an astronomical number of conformations open to the denatured state of a protein, an unbiased search through these conformations would take too long for a protein to fold. In short "there must be defined pathways to simplify the choices in folding".

Three mechanisms emerged to provide these routes which simplified the search for the folded state by uncoupling the formation of secondary structure from tertiary structure, thus removing the stringent requirement of simultaneous formation of secondary and tertiary structure.



- The frame work model(Diffusion-collision model)
- Nucleation-condensation model
- Hydrophobic-collapse model
- Jigsaw model

The framework model

The framework model proposed that local elements of native local secondary structure could form independently of tertiary structure. These elements would diffuse until they collided, successfully adhered, and coalesced to give the tertiary structure (diffusion-collision model).

The classical nucleation model

The classical nucleation model postulated that some neighboring residues of sequence would form native secondary structure that would act as a nucleus from which the native structure would propagate, in a stepwise manner. Thus, tertiary structure would form as a necessary consequence of the secondary structure.

The hydrophobic-collapse model

The hydrophobic collapse model hypothesized that a protein would collapse rapidly around its hydrophobic side chains and then rearrange from the restricted conformational space occupied by the intermediate.The secondary structure would be directed by native-like tertiary structures

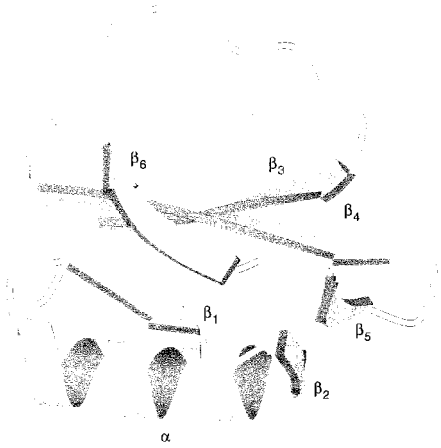
Jigsaw model

Jigsaw model is a contrary model where each molecule of protein folds by a different distinct path.

Folding of CI2

1. Structure of the native protein

- CI2 (Chymotrypsin Inhibitor 2) (Refer figure below) is a 64-residue polypeptide inhibitor of serine proteases.
- It has a binding loop, a single a helix running from residues 12 to 24, and a mixed parallel and antiparallel β sheet. The strands and the amphipathic helix interact to form the small hydrophobic core.
- CI2 is an α/β structure.
- It has four peptidyl-proline bonds are all in the trans conformation.
- CI2 is a single model structure;
- CI2 is a basic folding unit or foldon.



2. Folding kinetics

CI2 folds according to first order kinetics from a relatively open denatured state, with a half life of 10 ms because CI2 is a single module structure with interatomic interactions are uniform over the structure.

3. Structures of peptide fragments

Fragments of CI2 of increasing length from the N-terminus have been constructed and examined for residual structure in aqueous buffers by NMR and CD spectroscopy. The amphipathic helix has to make hydrophobic interactions, especially from Ala-16, with residues in the core that are located around positions 50-60, at the other end of the protein, to be stable. There are two important lessons from these and other experiments on peptide fragments.

- In general, isolated elements of secondary structure are relatively unstable without the rest of the protein. Thus, short-range interactions must be stabilized by long-range interactions. This emphasizes the necessary balance between local and long-range interactions in stabilizing proteins.
- Although the native-like secondary structure interactions are weak in the absence of tertiary interactions, residual native structure does exist in some small peptide fragments and can help guide the protein on its way to folding.

4. Structure of the denatured protein

Advance NMR methods are used to study the structure of the denatured protein but any residual native structure in CI2 under denaturing conditions is below the limits of direct structural detection by NMR. The pKa values of the 10 acidic residues in the native state of CI2 have been measured by NMR to give indirect information. Most of them are 2-3 units lower than they are in model compounds. The pKa values in the acid-denatured state are, on average, 0.3 unit lower than the model compound values in pure water, but the difference disappears as ionic strength increases. This indicates some compactness in the denatured state, but it could be induced by electrostatic interactions. By all criteria, the denatured state of CI2 is a relatively expanded state.

5. Structure of the transition state

The structure of the transition state for the folding of CI2 has been mapped by two methods using more than 100 mutants under a variety of conditions such as

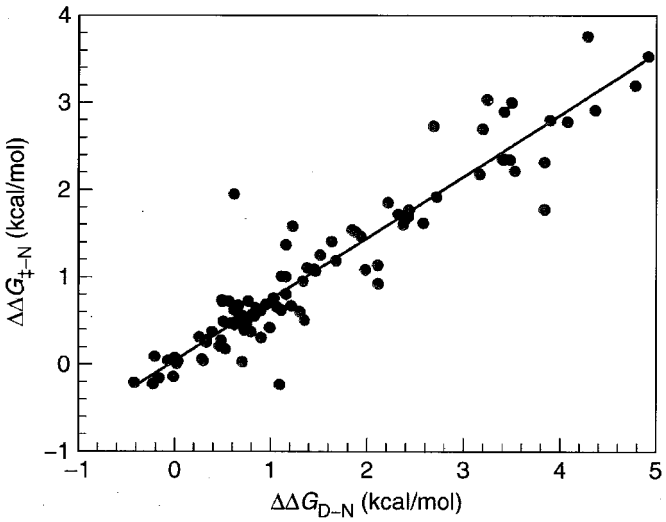
- Kinetic measurements
- Φ-value analysis

The properties were measured in water from folding kinetics and extrapolated to water from unfolding kinetics as a function of [GdmCl].

a. Gross features

The average properties of the transition state may be estimated from a variety of criteria. The Tanford β_T value, measured from the relative sensitivities of the folding kinetics and equilibrium constants to [GdmCl] is 0.6, indicating that about 60% of the surface area of the protein is buried in the transition state for folding, relative to that buried in the native structure.

The change in heat capacity of the transition state relative to that of the folded state ($\Delta C_{P(\#-D)}/\Delta C_{p\ (N-D)}$) is about 0.5, indicating that about 50% of the hydrophobic side chains become buried. A plot of changes in activation energy of unfolding ($\Delta\Delta G_{\#-N}$) against the change of free energy ($\Delta\Delta G_{D-N}$) on mutation fits well to a Bronsted plot of slope 0.7 (Refer figure below).



such that on average, about 30% of the net stabilization energy of the side chains is present in the transition state. This relatively good fit to the Br0nsted equation shows that CI2 unfolds and folds as a single cooperative unit.

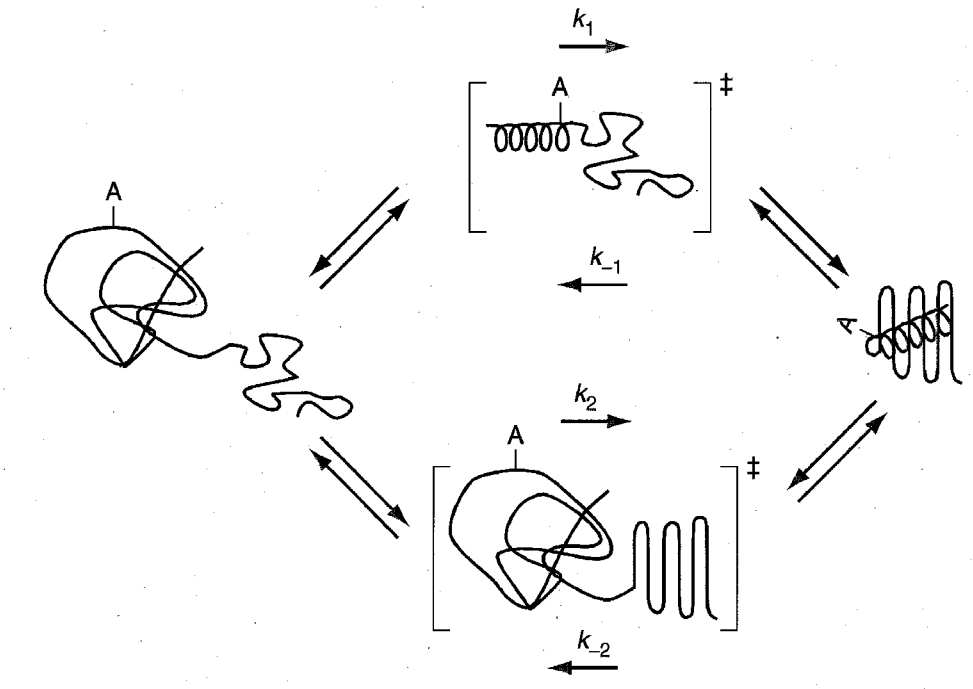
b. Fine details from ^-value analysis

The Bronsted plot represents the average change in energy of activation on mutation. The variations are seen from the individual Φ_F values, most of which are between 0.2 and 0.5.

- The higher values tend to be for the formation of the a helix, specially at its N-terminus.
- The energetics imply a structure for the transition state that looks like an expanded version of the native structure. But the expansion is not uniform. It is as if there is a center located in the N-terminal part of the helix, and the structure becomes more expanded the farther away it is from the center.

c. Fractional Φ values = fractional bond formation

A protein with an α helix and β helix has two folding pathways as shown below in the figure.

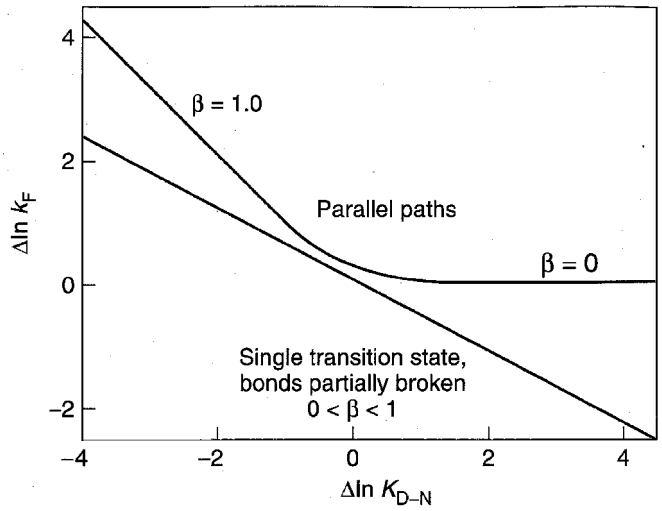


- In one hypothetical pathway (the upper pathway), the transition state has the helix fully formed($\Phi_{F(helix)} = 1$) and the sheet unfolded ($\Phi_{F(Sheet)} = 0$)
- In the other pathway, the helix is fully unfolded($\Phi_{F(helix)} = 0$) and the sheet is fully formed ($\Phi_{F(Sheet)} = 1$)

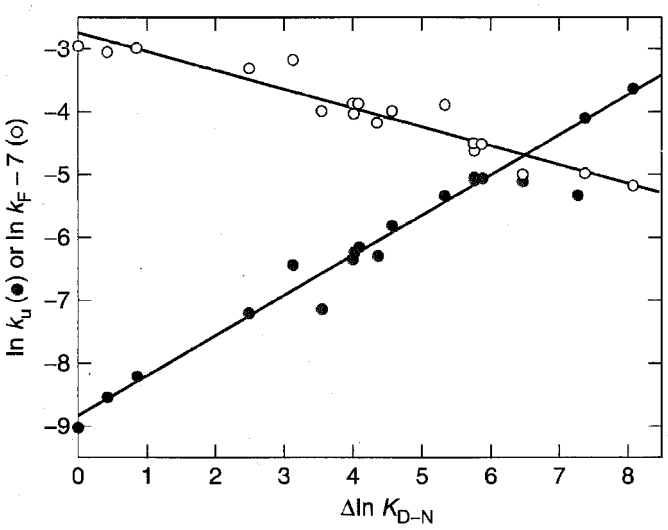
The average structure of the two transition states will thus appear to have partially formed helix and sheet and so have apparent fractional values of Φ_F.

We can detect this mechanism from the effects of mutations. If there are two distinct pathways, we can reduce the importance of the one with $\Phi_F = 1$ by destabilizing mutations.

- The first few destabilizing mutations will lower the importance of this route until it is negligible, then subsequent mutations will have no further effects. This situation would not lead to a linear Bronsted plot but would give a plot of changing slope (Refer figure below).



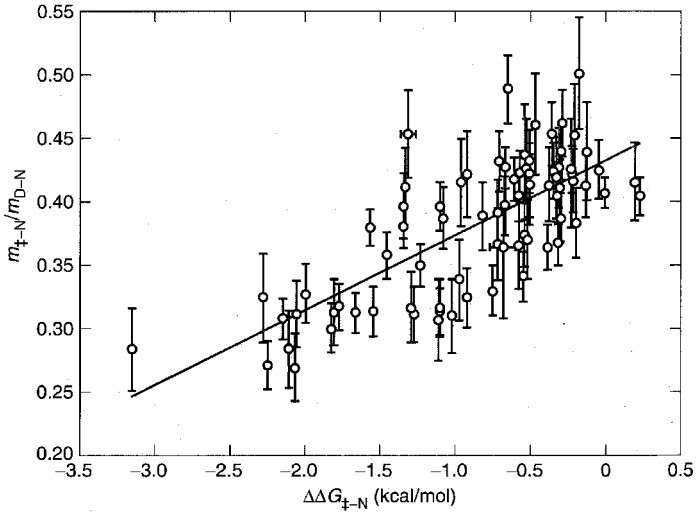
- Multiple mutations can be made in various parts of CI2 that lead to good Bronsted plots over a wide range of changes of energy.
- For example, the minicore of CI2 (three residues: Leu-32, Val-38, and Phe-50) has been extensively probed by mutation to other hydrophobic residues; a plot of In k_u versus In K_{D-N} (Refer figure below) fits nicely to a Bronsted equation with $\beta_u = 0.64 \pm 0.02$ over a range of nearly 5 kcal/mol in $\Delta\Delta G_{D-N}$ and a factor of 300 in rate constants (correlation coefficient, r = 0.99).



Thus, the data do not fit a model in which there are parallel pathways with largely different transition states but are consistent with a narrow ensemble of structures whose average resembles an expanded transition state.

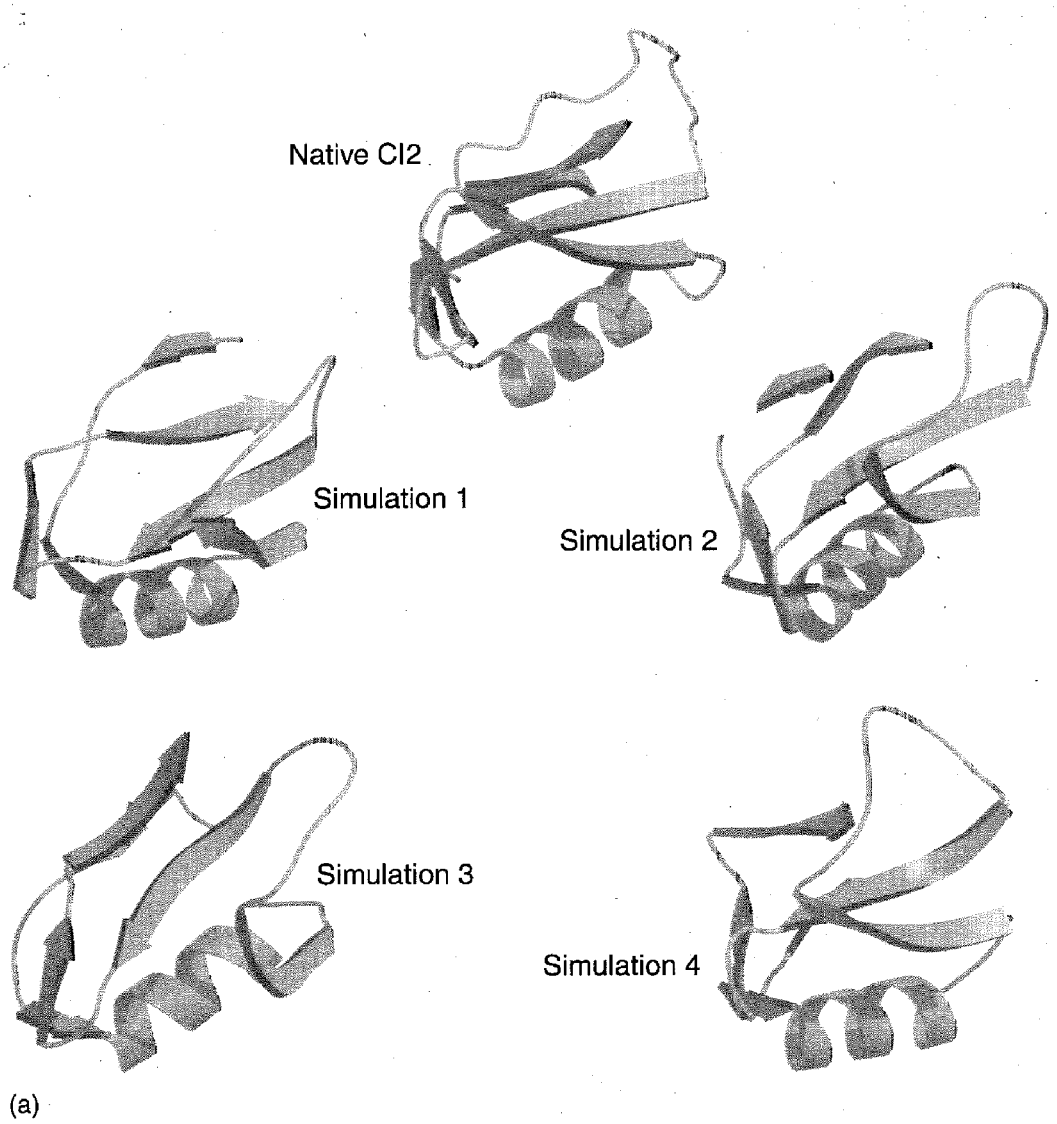
d. Hammond behavior

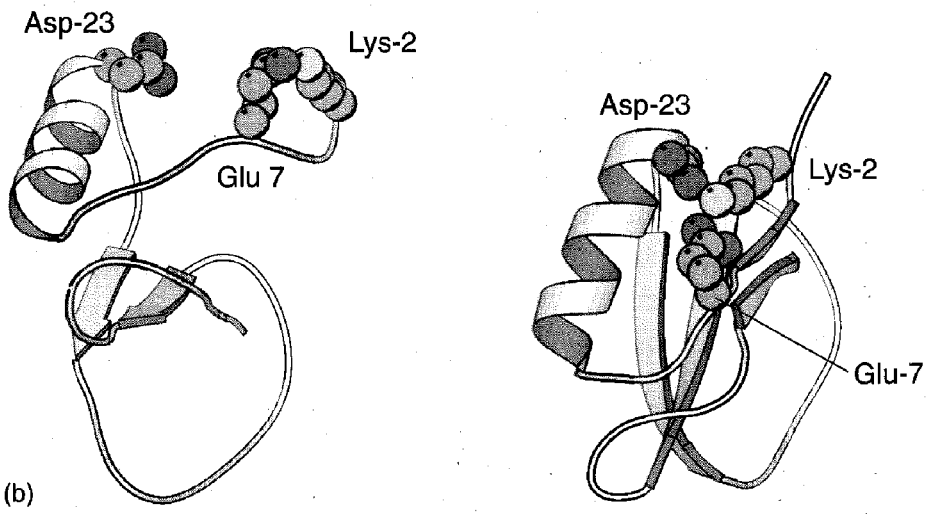
The transition state composed of closely spaced energy levels is the finding of Hammond behavior: as a reagent is destabilized, the transition state for its reaction moves along the reaction coordinate to become closer in structure to the destabilized state. Plotted in Refer figure below is the Tanford β_T value for unfolding ($m_{\#-N}/m_{D-N}$) against the change in the free energy of activation of unfolding. Wild-type protein has a β_T value for unfolding of 0.4, which drops progressively as the protein becomes destabilized to about 0.25 when the free energy of activation falls by 3 kcal/mol. This shows a smooth movement in structure in the transition state for folding, from about 60% burial of surface area in wild type to 75% in the most destabilized mutants.



6. Molecular dynamics simulations of the transition state

Application of molecular dynamics to the unfolding of CI2 gives us fair interpretation. Experimentalist looks at individual molecules. There are some variations in individual unfolding simulations. By performing a large number of simulations, the energy landscape can be determined by applying statistical mechanics to the distribution of structures. Twenty-four simulations of CI2 unfolding produced a statistical cluster of structures around the transition state that was determined experimentally. A common feature of the different simulations is that the helix is well-formed in the distribution of transition states of CI2. This shows that the Φ_F values of 0.5 to 0.6 in the a helix mean that the structure is basically formed, although the interactions are somewhat weakened. The cluster of structures explains the Hammond data: mutation or change of conditions leads to the transition state moving smoothly around the energy landscape. The simulations make predictions that can be tested experimentally. For example, as illustrated in the below figure, the C-terminal region of the helix moves away from first β strand in the transition state and breaks a salt bridge between Asp-23 and Lys-2. Now, an Asp is unfavorable energetically in this position in an isolated helix relative to an Ala. Mutation of Asp-23 to Ala destabilizes the native protein, as expected from the disruption of the Asp-23 •••Lys-25 salt bridge, but speeds up folding because it stabilizes the transition state in which the C-terminal region resembles an isolated helix. The success of the testing of the simulations lends confidence to their use.


Native CI2
Simulation 1
Simulation 2
Simulation 3
Simulation 4
(a)


Asp-23
Lys-2
Glu 7
Asp-23
Lys-2
Glu-7
(b)

The nucleation-condensation mechanism

1. The lessons from CI2 folding

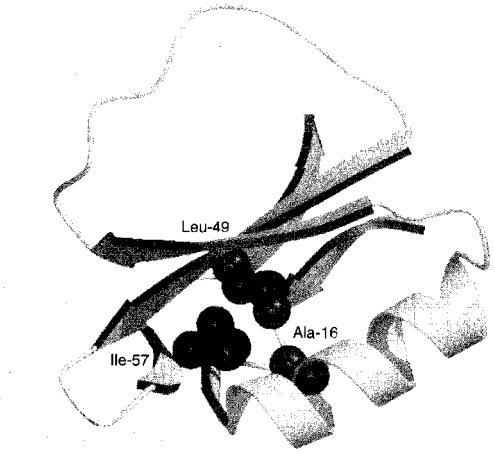
The kinetics of folding of CI2 and the Bronsted and Φ-value analysis of its transition state established important principles.

- Small proteins can fold rapidly in a first-order reaction from a relatively expanded denatured state.
- The transition state of CI2 is an expanded structure in which secondary and tertiary structures are formed in parallel.
- There are no completely fully formed elements of secondary structure in the transition state of folding of CI2; all elements are in the actual process of being formed.

All three of the classical mechanisms of folding are ruled out by these observations, as is the jigsaw model. The framework and nucleation models imply that there are fully formed elements of secondary structure in the transition state of folding. Regions within the classical nucleus or the preformed secondary structure would have Φ_F values of 1.0. The hydrophobic-collapse model is ruled out because there is not a folding intermediate and tertiary and secondary interactions are formed in parallel. Jigsaw and parallel-path mechanisms are ruled out by the Bronsted analysis.

2. The nucleation-condensation mechanism

The TS is constructed around an extended, delocalized nucleus consisting of the helix residues and the interactions it makes with hydrophobic residues in the core that are located around the position 50 specifically around Ala-16.



A mechanism called nucleation-condensation describes the folding involves the following steps

- The nucleus consists of primarily of adjacent residues.
- The nucleus does not form stable structure without assistance from interactions (Short range and long range interactions) made with residues that are distant in sequence.
- Formation of the small nucleus isn't the rate-determining because a significant fraction of the overall structure must be in approximately correct conformation to provide long-range interactions to stabilize the nucleus

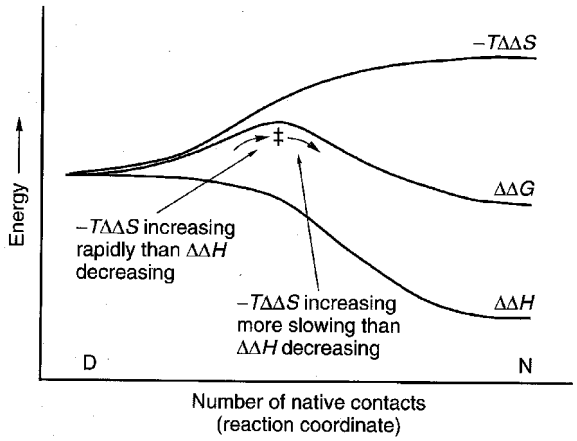
In CI2, the α-helix in the remains nascent embryonic stage until sufficient long-range interactions are built up that it becomes stable. The rest of the protein then condenses around the helix and the other native interactions that are developed during the stabilization of the helix. The onset of stability is so rapid that the helix is still in the process of being consolidated as the transition state is reached. Thus, formation of the nucleus is coupled with more general formation of structure, and so nucleation is just part of the rate-determining step. The coupling of nucleation and condensation leads to a relatively compact transition state. Thus, the whole molecule is involved in forming the transition state; the nucleus is simply the best formed part of the structure in the transition state.

This nucleation-condensation mechanism has certain characteristics.

- The nucleation site does not need to be extensively preformed in the denatured state. It may or may not be detectable in isolated peptides, but it is essential only in that it is extensively formed in the transition state. Further, it need not be formed completely in the transition state; because of the onrush of cooperative stabilizing interactions, however, it may be in the process of being formed in the transition state.
- Much of the structure is from local, contiguous residues, there are important stabilizing contributions from long-range interactions; i.e., from contacts with residues that are far removed in sequence.

Residues with high Φ values constitute the nucleus. Thus, destabilization of the nucleus does not prevent folding, but it is the most sensitive region to destabilizing mutations.

The physicochemical basis of the nucleation-condensation mechanism is that there must be a critical number of interactions made in the transition state for folding. As a protein folds there is an unfavorable large loss of entropy that is compensated by the favorable decrease in enthalpy of the interactions that are formed. After a critical number of interactions are made, the lowering of enthalpy is more rapid than the loss of entropy as further interactions are made (Refer figure below).



The reason the enthalpy is lowered more rapidly is that the stabilizing interactions cluster and so are formed cooperatively. The cluster of favorable interactions constitutes the folding nucleus.

3. Direct evidence for nucleation-condensation in assembly of protein fragments

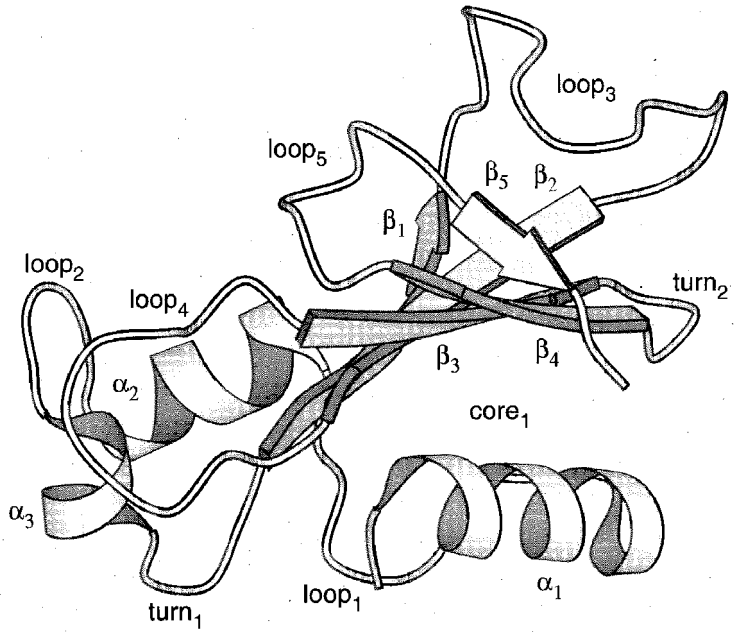
It is argued that the nucleus is formed in the denatured state of the protein immediately on its transfer from denaturing to renaturing conditions. The structures of the starting materials, transition state for folding, and products may all be analyzed under the same set of conditions that favor folding can be determined by various experimental techniques such as NMR and CD spectroscopy. CI2 can be cleaved in two by the action of CNBr on Met-40 in the binding loop, to give the peptides CI2 (l-40) and CI2 (1~64). NMR and CD spectroscopy show that they are largely of random structure in water. The few specific interactions that are present are non-native. Yet the two peptides, when

added together, associate and fold with a second-order rate constant of 4 X 10^3 s^{-1} M^{-1} to give a product that has the wild-type structure, apart from the region immediately around the cleaved bond. The transition state for the association/folding step, analyzed by Φ_F values has the same nucleus as in the folding of intact protein.

Folding of Barnase

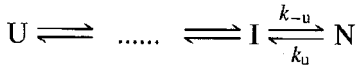
1. Structure of the native protein

- Barnase is a 110-residue RNase that is secreted from Bacillus amyloliquefaciens.
- It is a α + β structure.
- It has a major a helix running from residues 6 to 18 (α $helix_1$), two smaller helixes in the first half of the sequence, and five strands of antiparallel β sheet in the second half.
- There is a well-packed hydrophobic core formed by $helix_1$ packing onto one face of the sheet.
- All the peptidyl-proline bonds are in the trans conformation.
- Barnase is a small multifoldon protein



2. Folding of kinetics

Barnase has only one observable step for refolding, consisting of first order kinetics from a folding intermediate that is the major stable species under conditions that favor folding equilibrium.



The kinetics of formation of the intermediate from its more unfolded state is unknown. The small fraction that folds slowly because of cis —» trans peptidyl-proline isomerization.

3. Structures of peptide fragments

- The region corresponding to helix$_1$ of barnase has only about 6% or less helical structure in small fragments.
- The peptide corresponding to β hairpin at the centre of the barnase has no detectable native and non-native structure.
- The fragment 23-110 has found to have weak catalytic activity that is enhanced on the addition of the fragment 1-22 implying that a small fraction of the larger fragment does not exist in the native conformation.
- The small (1-22) and large (23-110) fragments bind to each other very rapidly and tightly to give a native-like structures which is a primary evidence that barnase can fold in separate parts.

4. Structure of the denatured protein

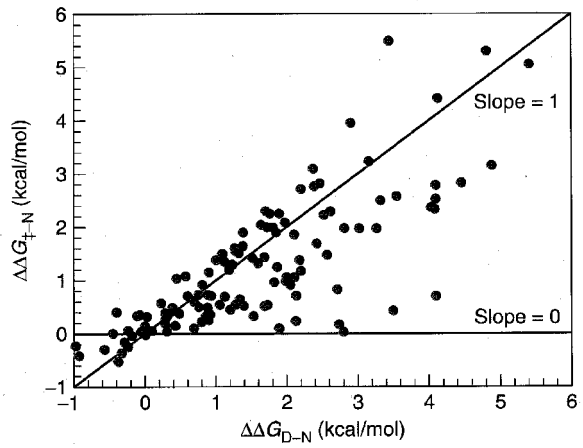
Advanced NMR methods detect weak residual native structure in helix$_1$ and helix$_2$ of barnase under denaturing conditions. There is also some local structure in the central β hairpin of barnase, although it maybe non-native. The denatured state ensemble of barnase has been analyzed by molecular dynamics, using statistical mechanics. According to the ergodic principle, the behavior of one molecule over a period of time gives the same variations in energies and structures as does the simultaneous behavior of many molecules. The denatured state of barnase, generated by a long molecular dynamics simulation of unfolding, was followed for 10,000 time steps. The ensemble is heterogeneous, but it contains non-random, residual structure with persistent interactions.

There is helical structure in the C-terminal portion of the major helix (residues 13-17) and non-native hydrophobic clustering between β strands 3 and 4, consistent with NMR data. In addition, there were tertiary contacts between residues in α l and the C-terminal portion of the β sheet which requires long range interactions to stabilize the helix.

5. Structures of the intermediate and the TS for the unfolding

Barnase was extensively analyzed by measurements on more than 130 mutants. In addition to the Φ_F values for single mutants, the tertiary structure was probed by double mutant cycles that check whether different elements are in contact in the intermediate and transition state. The transition state for unfolding has some regions with Φ_F values of 0, others with values close to 1.0, and a few regions with intermediate values.

This heterogeneity is seen in the Bronsted plot for unfolding (Refer figure below), where many points fit the lines of slopes 0 and 1.



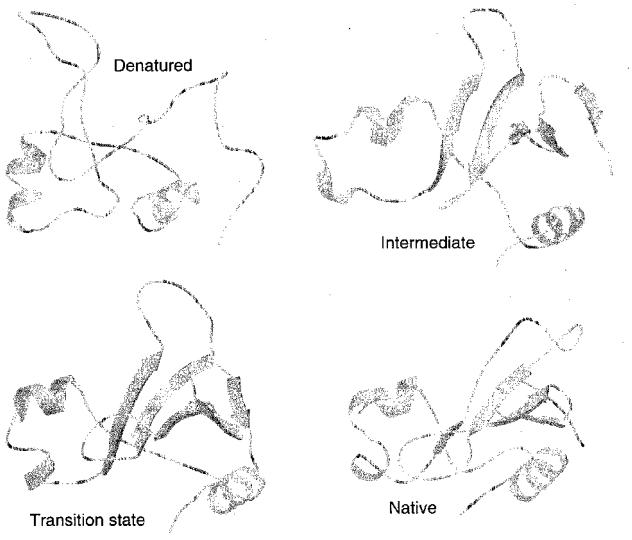
This is in contrast to the plot for CI2 (Refer figure below), which folds and unfolds as a single cooperative unit. Barnase clearly unfolds and folds in parts. The main parts of barnase that are nearly completely folded in the transition state are the β sheet, with Φ_F values of 1.0 at the center and slightly decreasing at the edges, and the a helix i, apart from the first turn. The hydrophobic core, which is formed by the docking of helix$_1$ on the sheet, is weak, with low Φ_F values. The folding intermediate is of similar structure; the Φ_F values that are 1.0 in the transition state are still 1 or only slightly lower. The major difference is in the hydrophobic core, which is very weak. Molecular dynamics simulations are in excellent agreement with these findings. The Φ_F values show clearly

that the rate-determining step in folding of barnase involves the docking of the preformed a helix$_1$ on the preformed β sheet, with the formation of the hydrophobic core being significant energetic part of the process.

The structures of the intermediate and transition state can be perturbed by radical mutations that cause significant decreases in stability. Molecular dynamics simulations are consistent with the experimental data and show that radical mutations in the major helix cause it to be more unfolded in the transition state for unfolding (the anti-Hammond effect) but the β sheet is less unfolded.

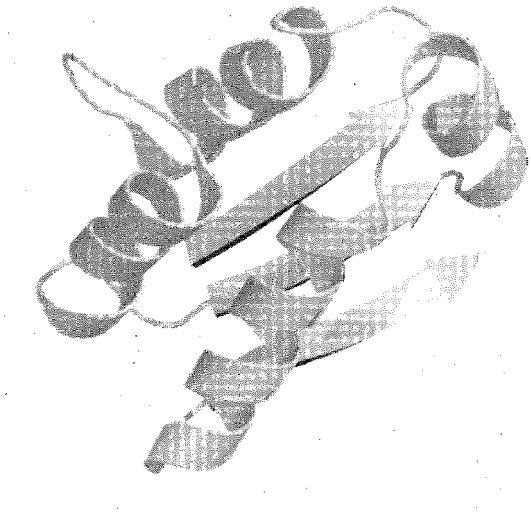
6. Molecular dynamics, Φ values and NMR studies to describe the folding pathway

The Φ values describe the energetics of interactions of side chains and implicate the presence of secondary structure in transition states and intermediates. NMR studies on denatured states give rudimentary information on persistent structure. This is precisely the type of information required to benchmark molecular dynamics simulations. The consistency between simulation and experiment in describing the denatured state of barnase and the transition states for unfolding of barnase and CI2 inspires confidence in the molecular dynamics simulations. There is excellent agreement between simulated and experimental Φ values. A description of the folding pathway of barnase from the denatured to the native state can be constructed by combining the simulation with experimental data from Φ-value analysis and NMR measurements on the denatured state (Refer figure below).



Folding pathway of barstar at microsecond resolution

- Barstar is an 89-residue polypeptide inhibitor of barnase that is produced by B. amyloliquefaciens to prevent its RNA from being destroyed by any barnase that is not secreted.
- It consists of three strands of parallel β sheet, three large helixes, and two small helixes.
- The distinction between single- and multifoldon proteins can be subjective: CI2 is clearly single and barnase multi, but barstar is borderline between the two.
- One peptide bond, that to Pro-48, is in the cis conformation in the native state, The trans —» cis peptidyl-proline isomerization event dominates the folding kinetics of barstar because the trans is the major form in the denatured state.



The early stages of folding of barstar have been measured on the microsecond time scale by temperature jumping its cold-denatured state from 2 to 10°C.There is the fast formation of a folding intermediate ($t_{1/2}$ ~ μs) with the peptidyl-proline 48 bond trans, followed by the formation with hri ~ 60 ms of a second intermediate that is highly native-like because it binds to and inhibits barnase. The native-like intermediate then undergoes trans —> cis peptidyl-proline isomerization on the time scale of minutes to give the final native structure (Refer equation below).

$$D \rightleftharpoons I_{trans} \rightleftharpoons N'_{trans} \rightleftharpoons N_{cis}$$

The fragment corresponding to the $helix_1$-loop-$helix_2$ region of barstar has barely detectable helical structure in water. NMR methods detect weak residual native structure in $helix_1$ and $helix_2$ of barstar under denaturing conditions, indicating that long range interactions stabilize secondary structure in the denatured state. Φ_F-value analysis shows that the first helix becomes substantially consolidated as the intermediate is formed in a few hundred microseconds, as does the second to a lesser extent. The conversion of D to l_{trans} fits the nucleation-condensation mechanism.

Insights from theory

There is an arsenal of theoretical methods for the analysis of folding, ranging from precise atomic analysis by molecular dynamics simulation, through models involving simplified polymer chains, to completely abstract procedures. The most precise procedure of molecular dynamics simulation applies readily to unfolding and can take a protein to the denatured state in solution. Molecular dynamics simulation has not been applied to the folding of a completely random chain because of the plethora of possible conformations. In comparison, the more abstract procedures allow simple models for folding to be analyzed directly, and they start from a state prior to the partly collapsed state in solution. These procedures tend to stress general principles. All the methods have given insights into the principles of folding that can be tested against experiment, make predictions, and fill in the entire energy landscape. Some of the models are

- Lattice simulations
- Spin glass theory and other abstract methods
- The folding funnel

1. Lattice simulation

A step down in detail from molecular dynamics simulation is to simplify the protein into a string of beads that is arranged on a cubic lattice; e.g., a 27-mer on a cube of 3 X 3 X 3 points, or a 64-mer on a 4 X 4 X 4 cube. In its simplest form, each bead of the string is spherical, has no side chains, and is assigned a characteristic of being hydrophobic or polar. A simple potential function is assigned to the interaction of the various beads, defining them as being polar or nonpolar. The folding of the random chain

can be simulated by Monte Carlo methods. Each simulation gives a different pathway that is recorded and analyzed. In effect, a folding experiment is performed in the computer. Once the computer experiment is benchmarked to fit in with reality, then the parameters in the program can be systematically changed to monitor how folding changes. The modeler can "mutate" the protein and watch its properties evolve with changing structure. Insights that have come from this approach include the following:

- For a protein to be able to fold, there must be a series of structures with decreasing energy levels so that the final state is of distinctly lower energy than structurally unrelated states (an "energy gap").
- Strings of beads that are modeled on real proteins fold much better than random sequences in the computer.
- Optimal rates of folding occur when there is not an initial collapse of structure, hydrophobic or otherwise, but when there is a specific and extended nucleus in the transition state; model proteins evolve toward nucleation mechanisms with extended nuclei.
- The folding of larger proteins splits up into domains.

2. Spin glass theory and other abstract models

Spin glass theory was originally formulated to analyze the orientations of the spins of ferromagnetics that are frozen in a diamagnetic material (e.g., manganese dissolved in molten copper and frozen). The spins want to be antiparallel to each other, but they cannot all be and so are "frustrated." By analogy, a residue in a protein is frustrated if it wants to be in more than one conformation; for example, to be both in a helix and in a sheet.The most stable structure is "minimally frustrated," compared with misfolded states. Another approach is to use "capillarity," a combination of considering the protein as a droplet, with a surface tension, and using ideas of nucleation. From these and other methods of physics, including classical polymer theory, it is possible to describe protein folding in terms of energy landscapes. A "smooth" energy landscape allows rapid transitions. A "rough" energy landscape has kinetic traps that slow down folding and so is to be avoided. These methods make predictions analogous to those of

the more discrete lattice simulations, such as the size of the nucleus, non-Arrhenius kinetics of folding, and the transition from single- to multinucleation.

3. The folding funnel

a. The golf course analogy

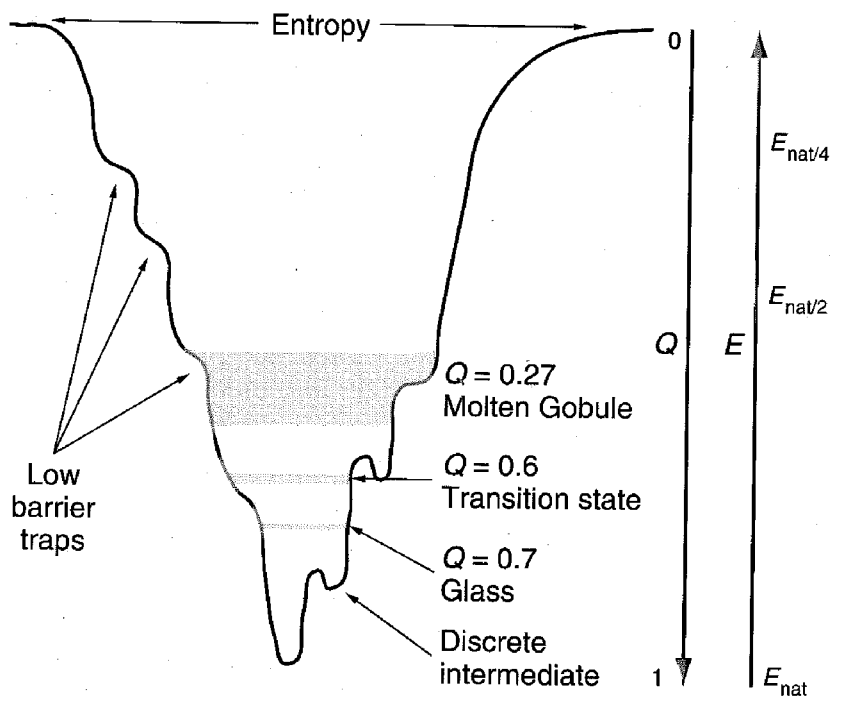
If a golfer was blindfolded, stood at the edge of a very large golf course, and was told to hit the golfball in a random direction, then the chances of sinking the ball in a single hole are infinitesimally small. This is analogous to a protein existing in a large number of non-native states of equal energy and having just one native state of much lower energy. There is no driving force to push the protein in the direction of folding; this is the Levinthal paradox. Suppose, instead, that the golf course sloped down from all directions to the hole. Then gravity would funnel the ball to the hole, and the golfer would always score a hole-in-one. Theoreticians have likened the process of a protein overcoming the Levinthal paradox to a progression down a funnel.

b. The funnel

A cross section through a funnel is shown in figure below. The funnel is a conceptual mechanism for understanding the self-organization of a protein and how it avoids the Levinthal paradox.

At the top of the funnel, the protein exists in a large number of random states that have relatively high enthalpy and high entropy. There is a fight between the maximization of entropy keeping the protein as random as possible at the top of the funnel and the minimization of enthalpy trying to drag the protein down the funnel as it folds.

Progress down the funnel is accompanied by an increase in native-like structure as folding proceeds. Proteins fold from the random state by collapsing and reconfiguring. The reconfiguring occurs by a Brownian type of motion between conformations that are geometrically similar and follows a general drift from higher energy to lower energy conformations.



The funnel is thus a progressive collection of geometrically similar collapsed structures, one of which is more thermodynamically favorable than the rest. Progress down the funnel can be described by a parameter Q, the fraction of correct contacts made in a state i.

$$Q_i = \frac{(\text{number of native pairwise contacts})_{\text{in state i}}}{(\text{total number of pairwise contacts in native state})}$$

Folding is driven by an increase in Q as the collection of states rapidly interconverts; i.e., by the acquisition of native-like interactions.

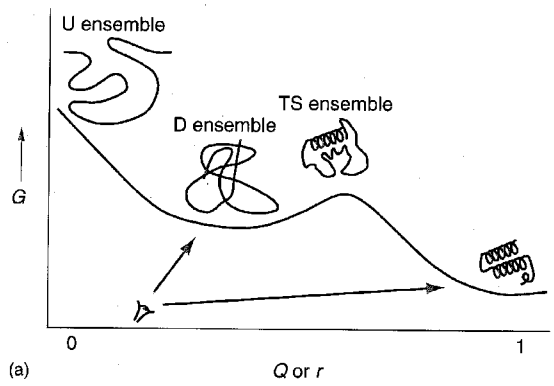
The width of the funnel in the above figure represents the entropy of the system. The depth represents the energy, and also Q. The molten globule state (a loosely collapsed state with fluctuating tertiary interactions and very weak secondary structure) is formed with Q around 0.27. The transition state occurs around Q = 0.6. There is a single unique state of low energy for the protein to fold; otherwise, it gets trapped in the "glass" region with Q = 0.7. If the increase in Q is not uniform, but some regions increase in Q more rapidly than others, then there is nucleation. There is an explicit assumption in the funnel mechanism that there is a kinetic flow down through a series of states that are progressively lower in energy. The assumption that kinetic pathways follow

thermodynamic energies downhill is normally unwarranted, because there is not a formal link between kinetics and thermodynamics.

The assumption is justified when the barriers between states are very low (~k_BT) so that they are all in rapid thermal equilibrium, as in the early stages of folding, but it is not justified at the later stages when there are high energy barriers (>k_BT). The increase in Q in the early stages of folding is the downward slope in the golf course that solves the Levinthal paradox.

c. Two-dimensional view of the funnel

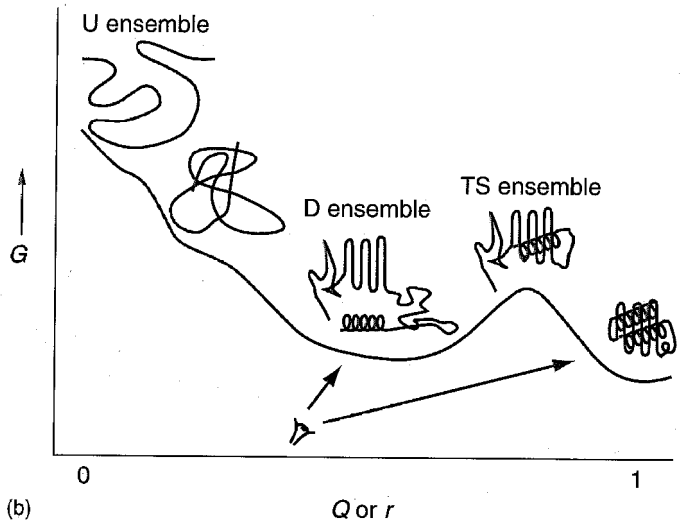
The energy landscape funnel can be viewed or converted to the free energy-reaction coordinate diagram of the experimentalist by lumping together the various states as "ensembles" (Refer figure below).



It is perfectly acceptable to marry the energy landscapes of theory and the folding pathways of experiment by using the idea of ensembles. Conventional two-state folding for a small protein such as CI2 (Refer the above figure) has energetically "downhill" folding to the denatured state ensemble that consists of a large number of conformations that have flickering native interactions. These then converge to the "transition state ensemble," which is a much more restricted set of states. A more complex protein has downhill folding from the U ensemble via more compact states to a D ensemble that has well-defined elements of structure (Refer figure below).

The energy landscape is bumpier. If it is sufficiently bumpy, an early molten globule will form that can be isolated. The earliest state that is seen by an experimentalist

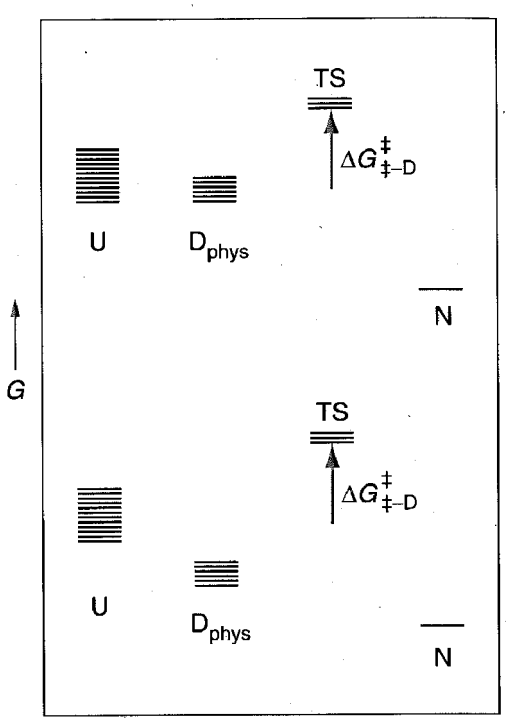
is the D ensemble. This varies from relatively expanded states of very small proteins to well-structured folding intermediates of larger proteins.



Optimization of folding rates

Computer simulation using lattice models and energy landscape theory using -abstract models predict that the fastest folding of small proteins should occur without intermediates and by an extended nucleation process. Stable intermediates slow down folding. This can be seen by analogy with enzyme catalysis. The fastest reaction occurs when the reagents are at the highest permissible energy level and the transition state at the lowest. The fastest mechanism is nucleation-condensation because it has a starting state with the least possible structure (Refer figure below).

There is a difference between the evolutionary pressure on rates of folding and on enzyme catalysis. There is a constant evolutionary pressure to increase the efficiency of enzymes because less protein has to be produced and so energy is saved. But a protein has to fold only fast enough to be produced. Any further increase in folding rate is of no apparent advantage to the cell, and so evolutionary pressure has a cutoff. Thus, proteins need not have evolved to the most efficient folding mechanisms. Indeed, intermediates are frequently found in the folding of small proteins. There will not be a unique mechanism of folding, but a range in which mechanisms will blur into one another.



1. Factors that determine rate constants for two state folding

There is a severe practical problem in looking for correlations between the rate constants for folding of small proteins and their structural or thermodynamic properties- specific structural features can dominate the rate of folding.

For example, we know from the protein engineering studies on barnase and CI2 that specific mutations can slow down the rate of folding by several orders of magnitude and some mutations even speed up folding by more than an order of magnitude. Thus, when comparing different classes of proteins, the "noise" from specific structural features is likely to obscure any "signal" from underlying trends in folding rates. We can eliminate some of the noise by looking for correlations in families of mutant proteins derived from mutagenesis experiments on a single protein. Clearly, a mutation in a region that is unstructured in the transition state will not affect the rate of folding even if it affects stability. For mutations in the single folden protein CI2 that folds by nucleation condensation, there is a Bronsted plot between rate and equilibrium constants for the folding, which shows that there is a relationship between stability and folding rate in this limited system. But, for the folding of the multimodular barnase that does have unstructured regions in the transition state, there is not a simple Bronsted plot.

There appears a statistically significant correlation between the rate constants for folding of single domain proteins and the average sequence separation between contacting residues in the native state. Proteins that have primarily local contacts (i.e., have a low contact order) tend to fold more rapidly than those that have more non-local interactions (i.e., have a high contact order).

Molecular chaperones

Proteins face obstacle on their way to successful folding. At sufficiently high concentrations, the denatured states of even the small, fast-folding proteins aggregate and precipitate instead of folding. Denatured states and folding intermediates are generally "sticky" because they have exposed hydrophobic surfaces. These states can also polymerize by forming extended β-sheet structures. Aggregation processes are of second or higher order in concentration and so eventually compete with first-order folding kinetics as concentrations increase. Even if aggregation does not occur rapidly, misfolded states or even very slowly reacting intermediates can form and be kinetically trapped so that they aggregate or become degraded in the cell. Proteins with multiple disulfide linkages can have problems in forming the correct ones spontaneously. The biochemist can perform tricks to encourage proteins to fold in vitro. Cycles of addition and removal of denaturant help unfold misfolded states, thus allowing further attempts at successful folding.

This approach can be combined with redox buffers that allow rapid equilibration of disulfide linkages. The means that nature uses to overcome the problems of folding of real proteins gives insights into mechanistic problems of folding.

The cell uses a series of accessory proteins, collectively termed molecular chaperones, to help some proteins (perhaps ~10%) fold correctly. Some of their functions are obvious. There are protein disulfide isomerases that rapidly catalyze the shuffling of disulfide linkages. Peptidyl-prolyl isomerases (e.g., cyclophilin, FK506 binding protein) have a weak activity in equilibrating the cis-trans isomers. The cell does not allow unfolded polypeptide chains of any significant length to be free in solution because of problems of aggregation or proteolysis. There are many chaperones that have evolved to bind denatured or unfolded states at various stages in their development. Trigger factor,

for example, binds to nascent polypeptide chains as they are formed on the ribosome. Many of the chaperones double as heat shock-proteins (Hsp). When a cell is put under stress that can cause proteins to denature, such as too high a temperature, it produces heat-shock proteins(Hsp). Hsp70, is a ubiquitous heat-shock protein in eukaryotes. It is known in E. coli as DnaK for historical reasons because it was first discovered from a supposed role in DNA replication.

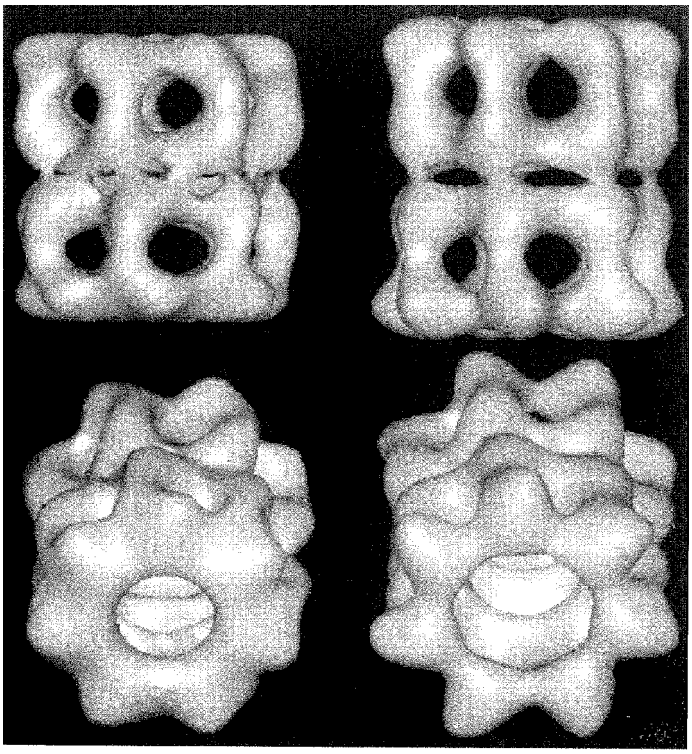
Hsp70 is also important in protein trafficking and the conveying of proteins across membranes, because the denatured state is important in these processes. In protein biosynthesis, the unfolded state of the nascent polypeptide chain is passed on to DnaK, which maintains it in an extended form. The chain, under the influence of ATP and co-chaperones such as DnaJ and GrpE, is handed over to GroEL

2. GroEL (

GroEL is the most interesting molecular chaperone. GroEL from E. coli is a typical member of the Hsp60 class of molecular chaperones, also known as chaperonins; GroEL is Cpn60. It works with a co-chaperone, GroES, that is a typical member of the HsplO or Cpn 10 class of chaperonins. Successful folding of protein substrates is coupled in vivo to the hydrolysis of ATP, but some substrates work in vitro with GroEL in the absence of ATP.

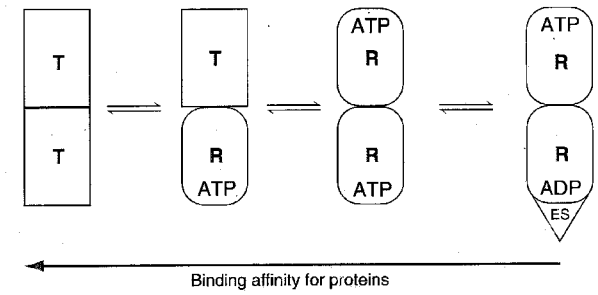
GroEL consists of 14 identical subunits, each of 57 kDa, that form a cylinder containing a central cavity. Each subunit consists of three domains. The equatorial domain (residues 6-133 and 409-523) has the binding site for ATP and forms most of the intersubunit contacts. The intermediate domain (residues 134-190 and 377-408) connects the equatorial domain to the apical domain (residues 191-376), which binds polypeptides.

a. GroEL is an allosteric protein that has weak and tight binding states GroEL undergoes a series of conformational changes on the binding of its ligands, revealed at low resolution by electron microscopy and at high resolution by x-ray crystallography. Free GroEL is a short, squat, symmetrical molecule. On the binding of ATP, it elongates (Refer figure below).



b. GroES binds as a heptamer to the apex of GroEL in the presence of ATP to give a bullet-shaped molecule. In doing so, it causes the cavity in the ring to which it is attached (the cis) to become larger, while the other end of the cis ring is sealed off. Electron microscopy studies on a GroEL GroES complex that had been frozen in the presence of denatured substrates found that a proportion bind in the cis "cage," although a higher proportion is found bound to the apex of the opposite {trans) ring. At higher ratios of GroES to GroEL, symmetrical GroEL. (GroES) complexes are formed.

c. The hydrolysis of ATP follows classical sigmoid kinetics, characteristic of an allosteric protein. GroEL undergoes a switch from a tense T state in the absence of ligands to the relaxed R state on the binding of ATP (Refer figure below). The T state has a predominantly hydrophobic cavity. The conformational changes on the binding of ATP and GroES cause the hydrophobic surfaces to become buried and new hydrophilic ones to be exposed.Most important, the change in state is accompanied by a change in the affinity toward denatured proteins; GroEL binds them very tightly, GroEL 14-GroES7 much more weakly.



b. GroEL slows down individual steps in protein folding

Individual steps in the GroEL mechanism were measured from its effects on the refolding of mutants of barnase. GroEL binds barnase at diffusion-controlled rates. Barnase refolds when complexed to GroEL, but it does so 400 times more slowly than when it is free in solution. The GroEL.GroES.nucleotide complexes retard the rate of refolding only fourfold, paralleling the loss of affinity of GroEL for denatured states on the binding of ATP and GroES. GroEL also retards the rate of refolding of CI2. Truncation of any of the hydrophobic side chains of CI2 by mutation leads to both weaker binding and less retardation of folding.

These experiments show two key features of the GroEL mechanism:

- weaker binding of denatured states leads to faster folding and
- GroEL is targeted against denatured and partly denatured states of proteins in general, because it binds hydrophobic side chains, which are the signatures of such states.

Folding is slowed down because the bonds between the denatured state and GroEL are broken during folding. Thus, the T state of GroEL is an enzyme that is complementary in structure to the denatured states of proteins. Its role is not to catalyze the steps of folding per se, but to act in the direction of unfolding to increase the yield of products.

c. GroEL has an unfolding activity

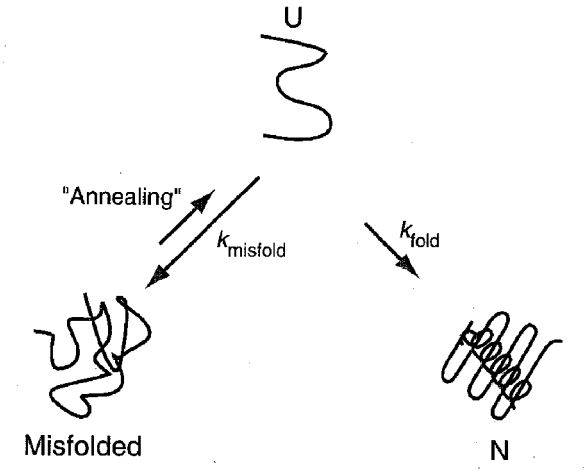
Under physiological conditions, the T state of GroEL catalyzes the $^{1}H/^{2}H$- exchange of buried >NH groups of baraase that must be fully unfolded for exchange to happen. GroEL can thus cause stable states to become transiently unfolded. The weak binding R state has much weaker activity.

d. The active site: Minichaperones show the basic activity of GroEL

A peptide of some 150 residues of GroEL (e.g., residues 191-345) folds stably into a monomer that is functionally active as a chaperone in vitro. Further, it can be covalently attached to agarose and other solid supports where the monodispersed fragment is extremely active as a chaperone. Crystal structures of recombinant "minichaperones" reveal that the active site is a flexible hydrophobic patch." It fits best to extended β strands. The basic function of GroEL is to provide a surface for binding exposed hydrophobic patches of denatured states and folding intermediates, which prevents self-aggregation. The process of sequestration is then enhanced by the binding site providing a "strain" mechanism for disrupting patches into separate β strands.

e. The annealing mechanism

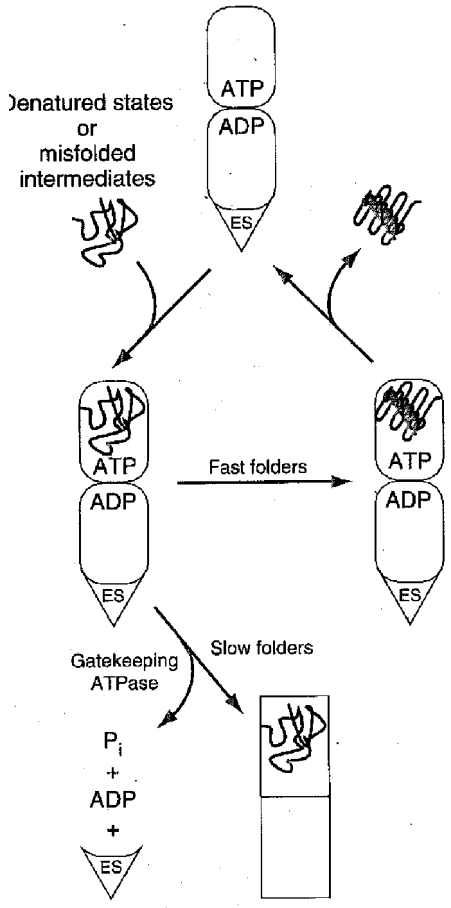
The essence of the mechanism of GroEL is first to prevent hydrophobic folding intermediates from sticking to one another and then, when necessary, to allow a misfolded state to unfold and have a fresh start for productive folding (Refer figure below).This is analogous to an annealing activity, whereby flaws in a metal, or mispairing in DNA, can be corrected by annealing: successive cycles of heating and cooling provide the activation energy for escaping from a kinetic trap to reach thermodynamic equilibrium. Every time the unfolded state in the below figure reacts, it partitions to give a proportion $K_{fold}/K_{misfold}$ of correctly folded native state. Successive rounds of annealing and refolding decrease the amount of misfolded product by a fraction of $K_{fold}/K_{misfold}$ at eacn step



f. Role of ATP hydrolysis: Energy transduction and the annealing mechanism

GroEL is essential for the growth of E. coli cells. There are temperature sensitive mutants of E. coli that are viable at 37°C but do not grow at 43°C because they have a mutation in GroEL that causes it to be nonfunctional at the higher temperatures. Viability can be restored by expressing a small minichaperone from a plasmid introduced into the cell. But, the complete absence of GroEL cannot be replaced. Single-ring constructs are also functional in vivo.The basic chaperone activity of the 150-residue active-site peptide of GroEL is thus enhanced by the allosteric properties of the large GroEL-GroES complex, which is regulated by the hydrolysis of ATP.

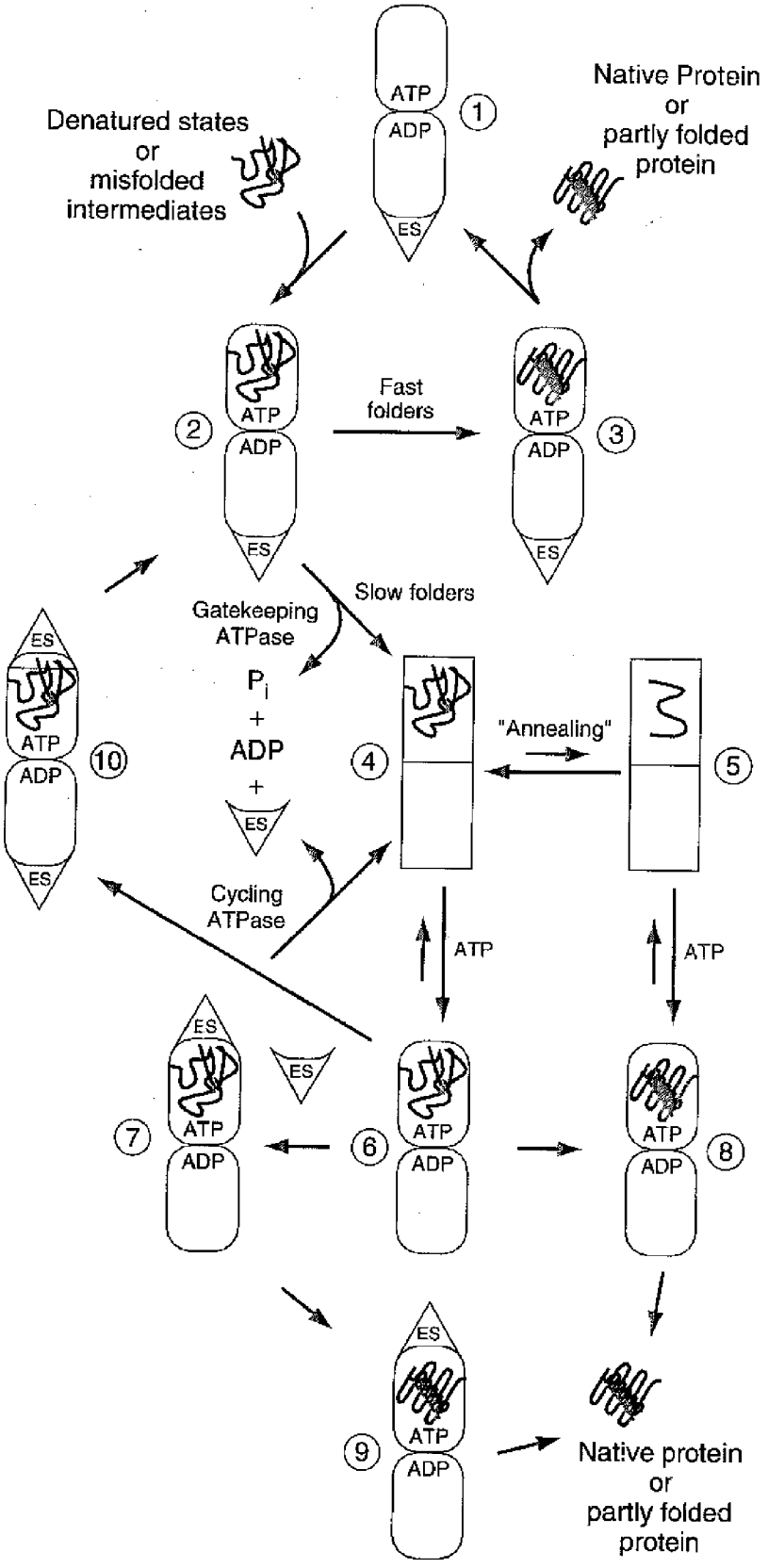
The first role of ATP hydrolysis is to keep out unwanted proteins. Every cycle of the GroEL reaction consumes 7 mol of ATP. In order not to waste ATP or saturate GroEL with all the nascent chains in the cell, nature uses the time constant for ATP hydrolysis to act as a filter or gatekeeper (Refer figure below).



The half-life for the hydrolysis of ATP in the GroEL-GroES nucleotide complex is some 20 s. This is 60 times longer than that for the refolding of barnase when bound. Proteins that fold as fast as barnase and do not need a chaperone bind to the GroEL-GroES-nucleotide holoenzyme and refold before significant ATP is hydrolyzed. But slower folding proteins that require help reside long enough to be channeled into the annealing process.

The second role of ATP hydrolysis in the GroEL mechanism is to regulate the affinity toward different substrates and to shuttle the protein between the tight-binding/slow-folding T state and the weak-binding fast-folding R state. The natural resting state of GroEL in the cell is the GroEL-GroES-nucleotide holoenzyme R state. To generate the T state, ATP must be hydrolyzed, which leads to the concomitant expulsion of GroES. The shuttling of a bound protein between the R and T states would put it in a hydrophobic unfolding environment at one extreme and a hydrophilic refolding environment at the other. This accomplishes two results: it enhances the annealing mechanism, and it allows tightly bound states to escape from the R state.

The complex mechanism shown in figure below summarizes these steps. In addition, some proteins may need to be sequestered inside the cage-like structure of GroEL, which can probably accommodate proteins up to 70 kDa. The minichaperones or GroEL by itself are capable of effecting the refolding of many proteins, but the full machinery of the holoenzyme complex is required for those proteins that are particularly difficult to refold.


Denatured states
or
misfolded
intermediates
ATP
ADP
ES
1
Native Protein
or
partly folded
protein
2
Fast
folders
3
Gatekeeping
ATPase
Slow folders
P_i
+
ADP
+
ES
10
4
"Annealing"
5
Cycling
ATPase
ATP
ATP
7
6
8
9
Native protein
or
partly folded
protein